Monetary Authority
of Singapore

FINANCIAL STABILITY REVIEW

Macroprudential Surveillance Department
Economic Policy Group
November 2024

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Definitions and Conventions

As used in this report, the term "country" does not in all cases refer to a territorial entity that is a state as understood by international law and practice. As used here, the term also covers some territorial entities that are not states but for which statistical data are maintained on a separate and independent basis.

In this report, the following groupings are used:

- "ASEAN-5" comprises Indonesia, Malaysia, Philippines, Singapore and Thailand.
- "AE" refers to advanced economies comprising Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom and United States.
- "EM Asia" comprises Asian economies such as China, Hong Kong, India, Indonesia, Malaysia, Singapore, South Korea and Thailand.
- "EM" refers to emerging economies comprising EM Asia economies as well as Argentina, Brazil, Chile, Colombia, Czech Republic, Hungary, Mexico, Poland, Russia, Saudi Arabia, South Africa and Türkiye.
- "Eurozone" comprises Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Portugal, Slovakia, Slovenia and Spain.
- "G3" comprises the Eurozone, Japan, and the United States.

Abbreviations used for financial data are as follows:

- Currencies: Australian Dollar (AUD), British Pound (GBP), Canadian Dollar (CAD), Chinese Yuan (CNY), Euro (EUR), Hong Kong Dollar (HKD), Indian Rupee (INR), Indonesian Rupiah (IDR), Japanese Yen (JPY), Korean Won (KRW), Malaysian Ringgit (MYR), New Taiwan Dollar (TWD), New Zealand Dollar (NZD), Philippine Peso (PHP), Singapore Dollar (SGD), Thai Baht (THB), US Dollar (USD).

Other abbreviations:

ABSD	Additional Buyer's Stamp Duty
ADV	Average Market Daily Trading Volume
AE	Advanced Economy
AFC	Asian Financial Crisis
AUM	Assets under Management
BIS	Bank for International Settlements
BoE	Bank of England
CAR	Capital Adequacy Ratio
CBS	Credit Bureau Singapore
CCB	Capital Conservation Buffer
CCyB	Countercyclical Capital Buffer
CCP	Central Counterparty
CCR	Core Central Region
CET1	Common Equity Tier 1
CFD	Contract for Difference

CFMs	Capital Flow Management Measures
CIS	Collective Investment Scheme
CISS	Composite Indicator of Systemic Stress
COVID-19	Coronavirus Disease 2019
CPF	Central Provident Fund
CO	Commodity
CR	Credit
CRE	Commercial Real Estate
DDRS	Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd
DOS	Singapore Department of Statistics
DSGE	Dynamic Stochastic General Equilibrium
D-SIB	Domestic Systemically Important Bank
ECB	European Central Bank
EDB	Economic Development Board
EM	Emerging Markets
EME	Emerging Market Economy
EPG	Economic Policy Group
EQ	Equity
ESG	Enterprise Singapore
FCI	Financial Conditions Index
FDI	Foreign Direct Investment
FI	Financial Institution
FCY	Foreign Currency
FSI	Financial Stress Index
FSR	Financial Stability Review
FVI	Financial Vulnerability Index
FX	Foreign Exchange
GDP	Gross Domestic Product
GFC	Global Financial Crisis
GFCy	Global Financial Cycle
GLS	Government Land Sales
HHI	Herfindahl-Hirschman Index
HKMA	Hong Kong Monetary Authority
HQLA	High-quality Liquid Assets
HY	High-yield
ICR	Interest Coverage Ratio
IMF	International Monetary Fund
IOSCO	International Organization of Securities Commissions
IR	Interest Rate
IRS	Interest Rate Swaps
IWST	Industry-wide Stress Test
LCR	Liquidity Coverage Ratio
LGD	Loss Given Default
LTD	Loan-to-deposit Ratio
MAS	Monetary Authority of Singapore
MPMs	Macroprudential Measures

MSD	Macroprudential Surveillance Department
MTM	Mark-to-Market
NAV	Net Asset Value
NBFI	Non-bank Financial Institution
NFP	Non-farm Payroll
NIM	Net Interest Margin
NPA	Non-performing Assets
NPL	Non-performing Loan
NSFR	Net Stable Funding Ratio
NUS-CRI	National University of Singapore Credit Research Initiative
OA	Ordinary Account
OCR	Outside of Central Region
OECD	Organisation for Economic Co-operation and Development
OIS	Overnight-index Swaps
OLS	Ordinary Least Squares
OTC	Over-the-counter
PCA	Principal Component Analysis
PCR	Principal Component Regression
PD	Probability of Default
PDI	Personal Disposable Income
PSTASSA	Professional, Scientific, Technical, Administrative, Support Service Activities
q-o-q	Quarter-on-quarter
RBC 2	Revised Risk Based Capital Framework
RCR	Rest of Central Region
RHS	Right Hand Side
RWA	Risk-weighted Assets
SFA	Securities and Futures Act
SGS	Singapore Government Securities
SGX	Singapore Exchange
SME	Small and Medium-sized Enterprise
SORA	Singapore Overnight Rate Average
STI	Straits Times Index
TDSR	Total Debt Servicing Ratio
TNA	Total Net Assets
UIP	Uncovered Interest Parity
URA	Urban Redevelopment Authority
VARs	Vector Autoregressions
y-o-y	Year-on-year
ZIRP	Zero Interest Rate Policy

Preface

The Monetary Authority of Singapore (MAS) conducts regular assessments of Singapore's financial system to identify potential vulnerabilities and review its resilience to potential shocks and risks. The analyses and results are published in the annual Financial Stability Review (FSR), which aims to contribute to a better understanding of issues affecting Singapore's financial system among market participants, analysts and the public.

Section 1 of the FSR provides a discussion of the risks in the external environment. This is followed by an analysis of the Singapore corporate and household sectors in Sections 2 and 3 respectively. A review of the financial sector is then provided in Section 4. The final section features specific topics on financial stability.

The production of the FSR is coordinated and prepared by the Macroprudential Surveillance Department (MSD) of the Economic Policy Group (EPG) that comprises Ang Wei Han, Adrian Chiu, Cheryl Ho, Huang Zhengjie, Koh Zhi Xing, Sherwin Lau, Eugene Lee, Wendy Lee, Leou Jie Dong, Aloysius Lim, Kalena Lim, Janice Loh, Low Yen Ling, Lye Royii, Ng Heng Tiong, Desmond Ong, Alex Phua, Edward Robinson, Celine Sia, Glenda Soh, Tan Shu Yan, Tan Siang Meng, Tan Wei Ting, Teoh Shi-Ying, Tu Chang, Christopher Walker, Xiong Wei, Yeo Han Sia and Davis Yep.

The FSR also includes contributions from other MAS departments including Banking Departments I, II & III, Corporate Finance & Disclosures Department, Economic Analysis Department, Economic Surveillance & Forecasting Department, Enterprise Knowledge Department, Insurance Department, Investment Intermediaries Department, Markets, Infrastructures & Intermediaries Department and Prudential Policy Department.

The FSR may be accessed in PDF format on the MAS website:

https://www.mas.gov.sg/publications/financial-stability-review

Overview

Elevated uncertainty and latent financial imbalances pose risks to the global financial system

Global economic activity has remained resilient, while disinflation progressed in 2024. However, the global economy confronts heightened uncertainty, trade tensions and geopolitical conflicts that raise the probability of adverse shocks. Trade-dependent small economies could face a confluence of risks, including potential terms-of trade shocks, slower global growth, higher-for-longer rates, and renewed dollar strength.

The build-up of financial imbalances, including fiscal and credit risks, stretched asset valuations and leveraged positions, may amplify these risks. Sudden spikes in global financial market volatility, as evidenced in August, could lead to the disorderly unwinding of leveraged positions, sharp asset repricing or a sudden retrenchment in cross-border financial flows.

Against this backdrop, emerging markets (EMs) could be vulnerable to currency and capital flow volatility, though EM Asia is in a better position to manage such risks, with stronger buffers and expanded policy toolkits. Even so, policymakers must be vigilant to possible shocks and preserve sufficient policy space for nimble responses.

Strong balance sheets helped Singapore corporates, households and financial institutions to cope with the higher borrowing costs and market volatility

Corporates, households and banks in Singapore have remained resilient. Overall bank credit quality is strong as most corporates have low debt levels, whilst households have benefitted from continued wage growth and strong balance sheets.

- Singapore's corporates have maintained debt levels significantly below pre-COVID averages. While leverage risks increased due to weaker corporate earnings and still-elevated borrowing costs, firms continue to have healthy debt maturity profiles and adequate liquidity buffers.

- The household sector in Singapore has a strong balance sheet and good credit quality despite the higher interest rates over the past year. Leverage risk has moderated with domestic mortgage rates recently beginning to come down. Debt servicing ability remains healthy, supported by firm wage growth. Liquidity risk is low as the excess of cash and deposits over household liabilities has continued to grow.

- The banking sector has maintained strong capital and liquidity positions. For the non-bank sector, insurers remain well-capitalised and most investment funds have managed liquidity risks well.

The economy's overall resident credit-to-gross domestic product (GDP) ratio declined from 143% in Q3 2023 to 135% in Q3 2024, driven by nominal GDP growth. Accordingly, the resident credit-to-GDP gap remained negative in Q3 2024. MAS will maintain the Countercyclical Capital Buffer (CCyB) at 0% for 2025.

Singapore corporates, households, and financial institutions have adequate buffers but need to be prepared for volatility in the external environment

MAS' stress tests show that the corporate and household sectors have adequate capacity to manage shocks to incomes and financing costs. Most firms and households have strong balance sheets, while macroprudential measures to discourage overleveraging have contributed to households' resilience. However, a small proportion of highly leveraged corporates and households could face debt-servicing pressures, and they should manage their finances prudently.

The results of the Industry-wide Stress Test (IWST) 2024 exercise affirm that banks would be able to withstand a range of severe macrofinancial stresses. Nevertheless, a sharp deterioration in the global outlook could pressure banks' profitability and capital positions, even as the sector continues to maintain sound underwriting standards and adequate loan provisions. Banks and non-bank financial institutions (NBFIs) should also remain vigilant against potential liquidity risks.

Macroprudential Surveillance Department, Economic Policy Group
Monetary Authority of Singapore
27 November 2024

1 Global Macrofinancial Environment

Heightened uncertainty and latent financial imbalances pose a moderate risk to the global financial system

Global economic activity has remained resilient in the face of higher interest rates, while inflation pressures have eased further, bolstering market expectations for a soft landing. With inflation retreating, many central banks have begun to reduce policy rates, while others have signalled easing ahead.

However, the global economy confronts heightened policy uncertainty, trade tensions, and geopolitical conflicts that increase the probability of adverse shocks. For trade-dependent small economies, a negative scenario could include terms-of-trade shocks, compounded by a global growth slowdown, higher-for-longer rates, and renewed dollar strength. Latent financial vulnerabilities, including persistent, elevated debt levels, stretched asset valuations and highly leveraged positions, may exacerbate the resulting market volatility. This could lead to the disorderly unwinding of leveraged positions, sharp asset repricing or a sudden retrenchment in cross-border credit and investment flows.

Against this backdrop of uncertainty and risk, EMs could be vulnerable to currency and capital flow volatility, though EM Asia could rely on buffers built up over the past decade as a source of resilience. Policymakers will need to remain vigilant and preserve sufficient policy space to ensure a nimble response.

Macroeconomic, monetary policy and geopolitical uncertainty are elevated

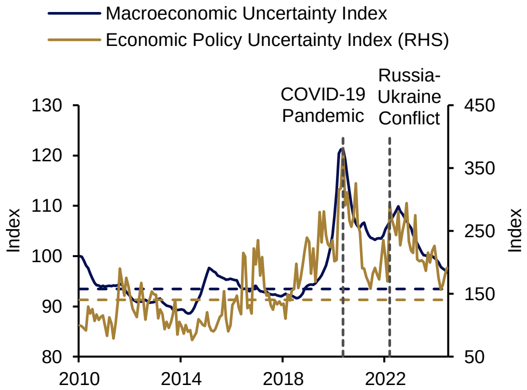
Global macroeconomic and monetary policy uncertainty are elevated relative to pre-pandemic levels, underscoring concerns about potential downside risks to growth and upside risks to inflation (**Chart 1.1**). As the effects of the shared global post-pandemic inflation shock fade, monetary policy paths could diverge based on differing domestic conditions. Desynchronisation of monetary policy could in turn increase currency volatility, including in EM Asia.

At the same time, deepening geopolitical and financial fragmentation pose another set of challenges to the macrofinancial environment (**Chart 1.2**). Rising protectionism and trade tensions, along with continuing conflicts in the Middle East and Ukraine, may lead to larger and more frequent supply shocks. These could renew price pressures, particularly as the post-pandemic inflation surge has not fully receded. More broadly, rising protectionism may weaken the capacity for international trade to cushion domestic inflationary pressures. The resultant dynamics could induce a pause in monetary easing or even a pivot to interest rate hikes. Restricted financial flows could also impede economic activity, with a disproportionate impact on emerging market economies (EMEs).[1]

1 See International Monetary Fund (2023), "Chapter 3: Geopolitics and Financial Fragmentation: Implications for Macro-Financial Stability", *Global Financial Stability Report,* April.

Chart 1.1 Uncertainty around the macroeconomic outlook remains elevated…

Macroeconomic and economic policy uncertainty indices[2]



Source: Jurado, Ludvigson and Ng (2015); and Davis (2016)

Note: Dashed lines indicate the average between 2010–19.

Chart 1.2 … alongside rising geopolitical tensions and trade uncertainty

Geopolitical risk and trade policy uncertain indices[3]

Source: Caldara and Iacoviello (2022); and Caldara, Iacoviello, Molligo, Prestipino and Raffo (2020)

Note: Dashed lines indicate the average between 2010–19.

Fiscal and credit risks add to financial imbalances

Fiscal risks in many countries are elevated. Worsened by pandemic-era fiscal deficits, average debt-to-GDP ratios increased from 60% before the pandemic to 68% most recently (**Chart 1.3**). Driven by the conjunction of higher debt levels with elevated interest rates, interest payments now represent an increased proportion of government revenues, rising from 7.7% in 2018 to 9.1% in 2023 (**Chart 1.4**).

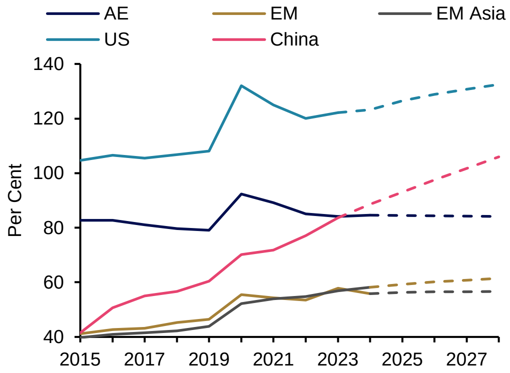
In many cases, fiscal deficits are expected to widen further, given limited fiscal consolidation efforts and, in some cases, expansionary fiscal policies of newly elected governments. Governments are mostly focused on near-term risks when managing public finances, reducing fiscal space for initiatives that enhance longer-term productivity and economic growth.

2 Jurado, K, Ludvigson, S and Ng, S (2015), "Measuring Uncertainty", *American Economic Review*, Vol 105(3), pp.1177-1216; Davis, S (2016), "An Index of Global Economic Policy Uncertainty", *NBER Working Paper*, No.22740.

3 Caldara, D and Iacoviello, M (2022), "Measuring Geopolitical Risk", *American Economic Review*, Vol 112(4), pp.1194-1225; Caldara, D, Iacoviello, M, Molligo, P, Prestipino, A and Raffo, A (2020), "The Economic Effects of Trade Policy Uncertainty", *Journal of Monetary Economics*, Vol 109, pp. 38-59.

Chart 1.3 Public debt-to-GDP ratios are elevated

General government debt-to-GDP

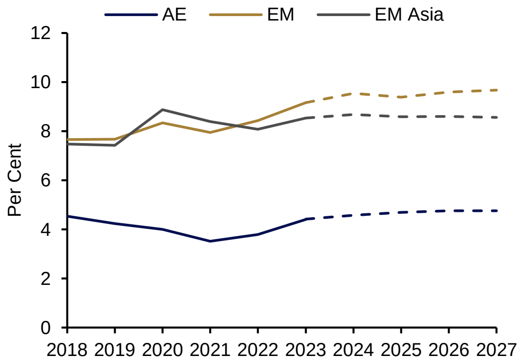


Source: International Monetary Fund (IMF), MAS estimates

Note: Dashed lines are IMF forecasts under current conditions. Under IMF's adverse scenario involving weaker growth, tighter financing conditions and fiscal slippages, future global debt-to-GDP is estimated to reach 115% of GDP.

Chart 1.4 Shares of government interest expenditures are rising

Government interest expenditure as a percentage of revenue



Source: S&P, MAS estimates

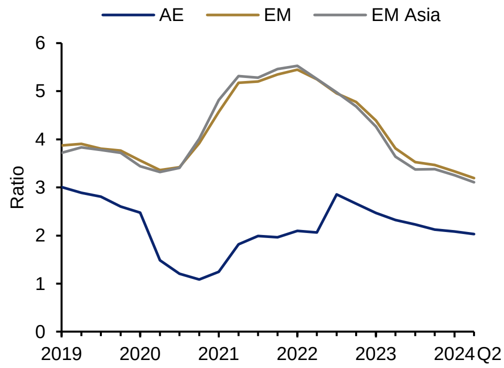
Corporate debt sustainability also poses some concerns, as many firms have yet to bear the full costs of the post-pandemic rise in interest rates. More corporates are facing debt servicing pressure (**Chart 1.5**), while refinancing risks remain elevated. Much of the USD13 trillion in corporate debt coming due in the next two years will need to be refinanced at higher rates (**Chart 1.6**). Corporate spreads now at historically tight levels could also be vulnerable to a sharp repricing in response to shocks (**Chart 1.7**).

While households generally maintained stable debt-to-personal disposable income (PDI) ratios (**Chart 1.8**), there are disparities across segments. Banks reported vulnerabilities among lower income groups and higher delinquencies for their auto and credit card loan portfolios.[4]

[4] See Bank of England (2024), *Financial Stability Report*, June; European Central Bank (2024), *Financial Stability Review*, November; Board of Governors of the Federal Reserve System (2024), *Financial Stability Report*, November.

Chart 1.5 Debt servicing abilities of firms weakened globally

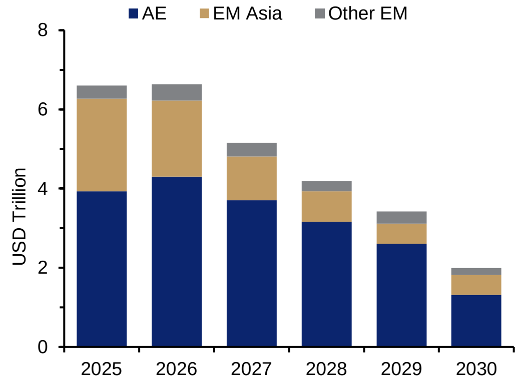
Median interest coverage ratio of firms



Source: Refinitiv, MAS estimates

Chart 1.6 About 30% of corporate debt will come due in the next two years

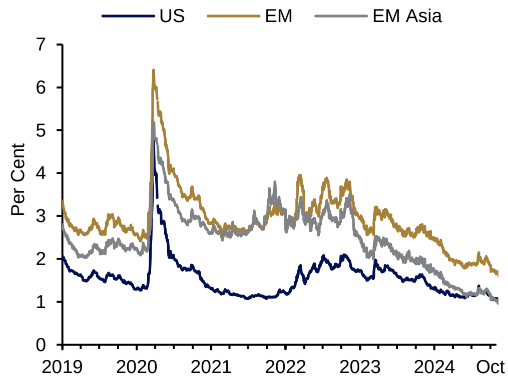
Global corporate debt maturity profile



Source: Dealogic

Chart 1.7 Tight corporate spreads could be vulnerable to shocks

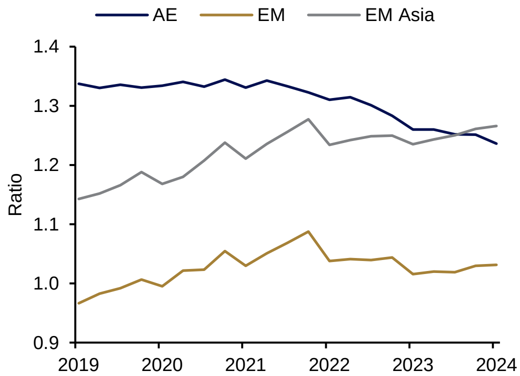
USD corporate bond option adjusted spreads



Source: ICE Data Indices, LLC, retrieved from Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis.

Chart 1.8 Household debt-to-PDI ratios remained stable, with disparities across segments

Household debt-to-PDI



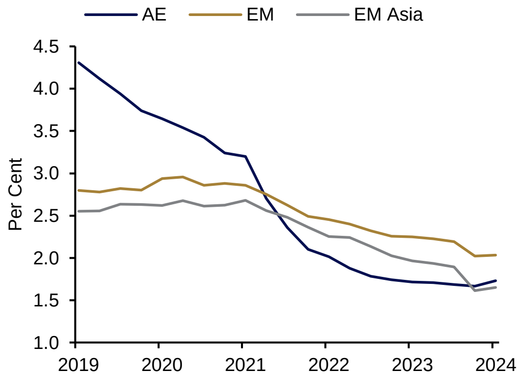
Source: Bank of International Settlements (BIS), Haver Analytics, MAS estimates

Note: Chile, Colombia, Indonesia, Malaysia, Mexico, Russia and Turkiye are excluded due to data unavailability.

Banks may see some deterioration in asset and credit quality in the medium term. Although aggregate non-performing loan (NPL) ratios remained subdued (**Chart 1.9**), banks are vulnerable to losses from weaker corporates and households, deteriorating commercial real estate (CRE) assets, and leveraged investors including NBFIs. NBFIs also pose higher risks to banks through increasing funding and liquidity interdependence, and growing linkages to opaque private equity and credit markets. In general, AE banks' provisioning is lower than that of EME banks, which may reflect differences in their exposures to vulnerable borrower and loan segments, as well as in supervisory regimes (**Chart 1.10**).

Chart 1.9 NPL ratios remained subdued, though asset quality varies across banks and loan types

NPLs to total gross loans ratios

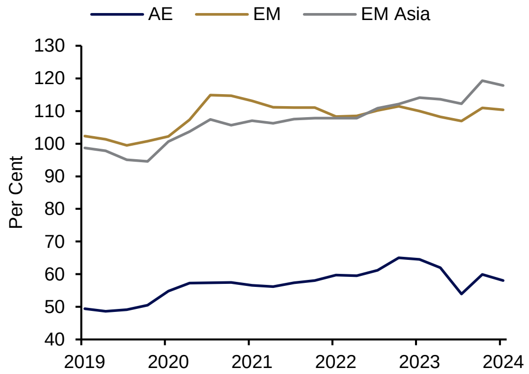


Source: IMF, Haver Analytics, MAS estimates

Note: Aggregate NPL ratios may not reflect disparities in banks' asset quality, as well as extension of loan repayment assistance which includes deferment of NPL classification.

Chart 1.10 Provisioning in AEs is lower than in EMEs

Provisions to NPLs



Source: IMF, Haver Analytics, WIND, MAS estimates

Note: Chile, Russia, South Africa and South Korea are excluded due to data unavailability.

Global markets remain vulnerable to unanticipated shocks

Markets may overreact to unexpected data releases and other surprises, as occurred following the release of a weaker-than-expected US non-farm payroll (NFP) report in early August. Movements in fixed income, equity, and other markets were disproportionate to the scale of the data surprise (difference between actual and forecast data), which was close to the historical median. Such reactions to US Federal Reserve decisions and major data releases have become more common over the past 12 months (**Chart 1.11**).[5]

While the August episode was short-lived, it reflects underlying financial market vulnerabilities that could amplify stresses. These include elevated asset prices, particularly for large capitalisation technology stocks (as proxied by share prices of the "Magnificent 7" companies)[6] and leveraged positions built up in the preceding period of low volatility and optimistic investor expectations (**Chart 1.12**). Hedge funds have also become more reliant on carry trades, raising the risk of rapid and correlated trade unwinding during periods of market stress.[7]

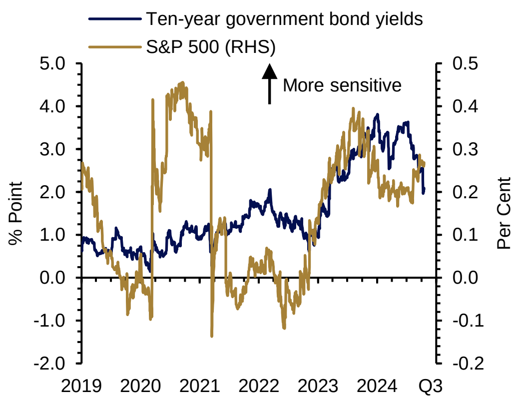
5 See Buttonwood (2024), "Why have markets grown more captivated by data releases?", *The Economist*, October 10.

6 Alphabet (Google), Amazon, Apple, Meta Platforms, Microsoft, Nvidia and Tesla

7 See International Monetary Fund (2024), "Chapter 1: Steadying the Course: Financial Markets Navigate Uncertainty", *Global Financial Stability Report,* October; Packer, F, Schrimpf, A, Sushklo, V and Zarra, N (2024), "Box C: Hedge fund exposure to the carry trade", *BIS Quarterly Review*, September.

Chart 1.11 Markets are hypersensitive to Fed decisions and major data releases

Difference in daily equity and bond market movements on days with US Federal Reserve decisions or major data release and days without (12-month moving average)

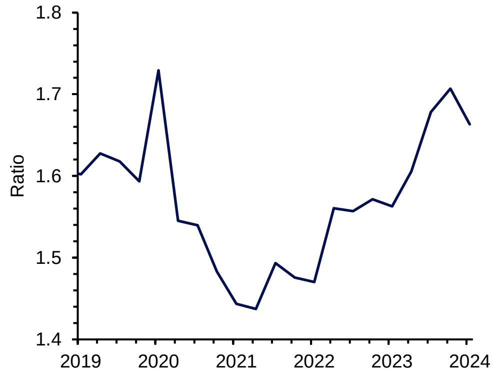


Source: Bloomberg

Note: Data releases refer to US inflation and labour data.

Chart 1.12 Aggregate financial leverage of US hedge funds grew in recent years

Hedge fund leverage ratio



Source: Board of Governors of the Federal Reserve System

Looking forward, potential triggers for sudden market movements include more hawkish-than-expected stances by the Bank of Japan, data releases that challenge market expectations of a soft landing in the US, fiscal sustainability concerns in both AEs and EMs, and ratcheting up of geopolitical and trade tensions. Sell-offs would likely involve unwinding of correlated and leveraged trades involving multiple counterparties, accompanied by spikes in long-term bond yields. A global flight-to-safety could also provoke indiscriminate sell-offs of EM assets, tighter financial conditions through higher borrowing costs, and departures from the economic recovery path.

EM Asia is in a better position to manage potential exchange rate and capital flow volatility

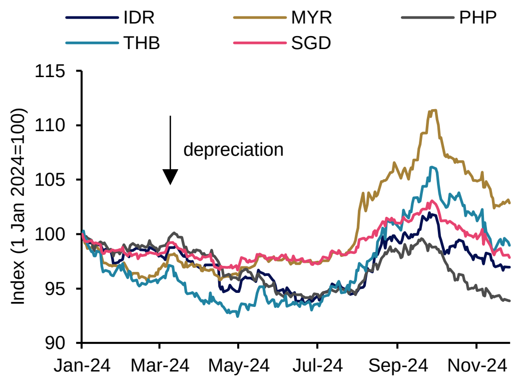
Regional economies weathered the post-pandemic round of interest rate hikes well, avoiding the capital surges and sudden stops they had suffered through earlier shocks (see Special Feature 1 “Asian Capital Flows and the Global Financial Cycle”). Nevertheless, global factors are still important determinants of domestic prices and activity in EM Asia. Notably, the influence of the external cycle on GDP growth and domestic interest rates has not diminished post-global financial crisis (GFC).

Broad-based USD strength led to bouts of depreciation pressures on EM Asia currencies in 2024 (**Chart 1.13**). Through the episodes of currency volatility, EM Asia economies managed external vulnerabilities by maintaining stronger buffers against outflows and deploying expanded policy toolkits to mitigate financial stability risks (**Chart 1.14**). EM Asia central banks engaged in discretionary policy management, accepted some currency flexibility, and used targeted foreign exchange interventions and administrative measures to temper currency volatility and capital outflows.

EM Asia economies could be susceptible to the potential supply shocks noted above and could see a resurgence in currency volatility. Central banks should prepare for a combination of shocks, by adopting credible and sustainable policies to support economic and financial resilience, and maintaining adequate levels of foreign exchange reserves.

Chart 1.13 ASEAN-5 currencies faced bouts of depreciation pressures in 2024

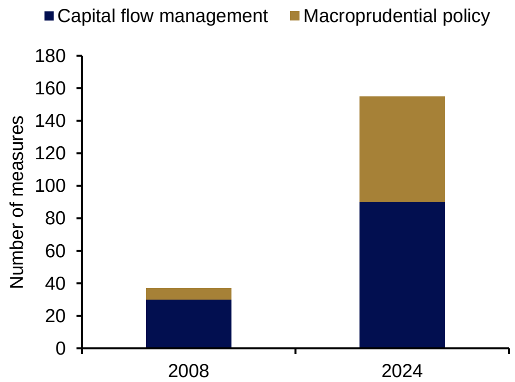
Performance of ASEAN-5 currencies vs USD



Source: Bloomberg

Chart 1.14 EM Asia central banks deployed an expanded toolkit to manage volatility in 2024

Number of capital flow management and macroprudential policy measures in ASEAN-4 pre-GFC vs 2024



Source: AMRO, MAS estimates

Singapore's financial institutions flagged macroeconomic uncertainty and geopolitical risk as top risks

In the most recent MAS Systemic Risk Survey, Singapore's financial institutions (FIs) were most concerned about macroeconomic uncertainty (growth slowdown and inflation resurgence), and geopolitical risk induced by escalation of conflicts, as well as trade tensions and uncertain policies of newly elected governments. FIs also assessed the macrofinancial implications of the recent US presidential election for the region. The findings are discussed in more detail in Box A "Systemic Risk Survey of Singapore's Financial Sector".

Box A
Systemic Risk Survey of Singapore's Financial Sector

In the April and October 2024 Systemic Risk Surveys, Chief Risk Officers of Singapore financial institutions identified macroeconomic uncertainty and geopolitical risk as two of the most impactful risks to Singapore's financial system. Details are in **Table A1** below.

Table A1 FIs' perceptions of risks to Singapore's financial system

Risk Categories cited by at least 10% of respondents	% of citations		Impact ranking	
	April 2024	October 2024 (Δ)	April 2024	October 2024 (Δ)
Macroeconomic uncertainty (growth slowdown and inflation resurgence)	67	62 (-5)	1	1
Geopolitical risk induced by escalation of ongoing conflicts, as well as trade tensions and uncertain policies of newly elected governments	51	54 (+3)	2	2
Technology and cyber risk	51	46 (-5)	3	3
Money laundering and terrorism financing risks	25	27 (+2)	4	4
Transition and physical risks due to climate change	16	16	5	6 (↓1)
AI developments	13	11 (-2)	6	5 (↑1)

Sources: MAS estimates.

Notes: (i) Impact ranking is the aggregate of all respondents' ranking of each risk by potential impact. (ii) Both runs of the survey had response rates of 98% from 57 FIs.

Uncertainty over growth and inflation remains top of mind

Uncertainty regarding the global macroeconomic outlook remains the most-cited risk, though by a slightly smaller margin than in the previous survey (April 2024). Respondents noted that increases in macroeconomic, geopolitical and monetary policy uncertainties could induce a growth slowdown and an inflation resurgence. A slower than expected growth outlook for China could have implications for corporates, funds, and loan portfolios with significant exposures to the economy.

More respondents cited geopolitical risk as a concern this time, noting the intensification of conflicts in the Middle East and Ukraine, as well as rising protectionism. Respondents were concerned that disruptions to global supply chains and commodity markets could lead to inflationary supply shocks, with negative impact on growth and financing conditions. Respondents noted that the US-China strategic rivalry has already led to supply chain reconfigurations in the region. 30% of the respondents who cited geopolitical risk (or 16% of all respondents) were concerned about the trend of rising trade frictions. Singapore banks and fund managers also highlighted the possibility of credit and market losses through their exposures to externally oriented firms that are vulnerable to supply chain disruptions.

Respondents also highlighted currency and capital flow volatility, as well as credit risk, as key concerns. An inflation resurgence could prompt a sharp pivot away from monetary easing, leading to broad-based USD strengthening and a rise in currency and capital flow volatility in the region. Geopolitical risk could increase market uncertainty and undermine investor confidence in EM Asia assets. Consequently, EM Asia assets and currencies could come under pressure from multiple fronts. Given reduced buffers and elevated debt, more households and corporates could face debt-servicing pressures under a higher-for-longer interest rate environment.

2 Singapore Corporate Sector

Singapore corporates have benefitted from the improving growth momentum in Q3, particularly in manufacturing and export activities. Corporates have also gained support from financial conditions that have turned less restrictive.

Over the past year, corporate balance sheets have remained generally stable in the face of market volatility. Defaults, including those of small and medium enterprises (SMEs), have been low despite a moderate rise in the corporate sector FVI, which reflects the cumulative effects of monetary tightening. Weaker earnings and still-high borrowing costs had contributed to an increase in leverage risk. Despite this, corporate debt levels have remained significantly below pre-COVID levels and firms have maintained healthy debt maturity profiles. There has been some pick-up in firms' foreign currency (FCY) borrowings but the risks are contained, as such firms tend to be naturally hedged. Liquidity buffers have also remained adequate despite some drawdown of excess reserves built up during the pandemic.

In the year ahead, corporate debt servicing capabilities are expected to remain resilient with less restrictive domestic financial conditions. Stress tests show that most firms still have adequate buffers to manage unfavourable earnings and interest rate shifts that could arise from inflationary shocks, negative growth surprises, trade frictions or an escalation in geopolitical tensions.

Corporate sector FVI[8] level	Q2 2023	Q2 2024
Overall corporate FVI		
Leverage risk		
Liquidity risk		
Maturity risk		
Foreign currency risk		

8 The corporate sector FVI consists of four risk indicators: (i) leverage risk measures firms' debt levels and debt repayment capabilities, (ii) liquidity risk assesses the adequacy of firms' liquid assets to cover short-term liabilities, (iii) maturity risk gauges firms' ability to rollover short-term debt, and (iv) foreign currency risk is proxied by the share of FCY debt to total debt.

2.1 Financial conditions in Singapore

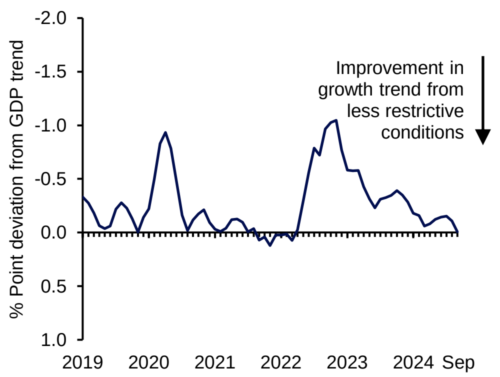
Domestic financial conditions have turned less restrictive

Financial conditions in Singapore, as measured by the FCI,[9] have become less restrictive as global interest rates have started to ease (**Chart 2.1**). Domestic SGD interest rates have moderated from the highs seen last year. In the equity market, the Straits Times Index (STI) was up by 6.4% as of Q3 2024 compared to a year ago. Spreads of Singapore investment grade corporate bonds also narrowed significantly in Q3 2024, falling below pre-COVID levels (**Table 2.1**).

Bank credit to resident corporates has started to expand following the prolonged contraction during 2023. In the SME segment, the number of borrowers rose in 2024 (see Chart Panel 2A "Small and Medium-sized Enterprise Financing Conditions", Chart 2A2). Looking ahead, banks expect firms' loan demand to increase alongside stabilising lending terms and conditions.[10]

Chart 2.1 Domestic financial conditions have been less restrictive over the past year

Domestic FCI (monthly)



Source: Bloomberg, DOS, Urban Redevelopment Authority (URA), MAS estimates

Table 2.1 Indicators of domestic financial conditions

	Q4 2019	Q3 2023	Q3 2024
Domestic FCI (% Point deviation from GDP trend)	−0.09	−0.33	−0.06
3-month compounded SORA (%)	1.33	3.69	3.59
10-year SGS yields (%)	1.74	3.17	2.82
STI (points)	3188.7	3238.1	3443.8
CEMBI Investment Grade+ SG Spread* (bps)	113.0	133.6	78.8

Sources: Bloomberg, J.P. Morgan Markets, MAS
Note: Figures are quarterly averages.

* This measure is computed by J.P. Morgan Markets as the yield spread of USD corporate bonds over US Treasury bonds.

9 For more details on the FCI, please see Special Feature 2 "Expanding the Toolkit for Macroprudential Surveillance".

10 These observations were based on MAS' survey of banks conducted in September 2024, which included realised credit conditions over the past six months and expectations for Q4 2024 and Q1 2025.

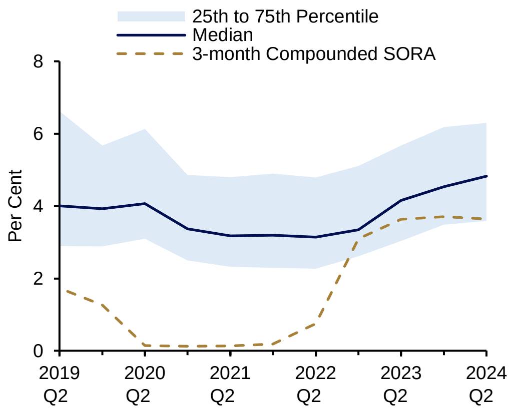
2.2 Assessment of vulnerabilities

Firms' low overall debt levels helped to cushion the impact of rising borrowing costs

Borrowing costs for firms rose further in 2024 as they refinanced debt amid still-high interest rates (**Chart 2.2**). That said, the impact was cushioned by debt levels that remained significantly below pre-COVID levels (**Chart 2.3**). Going forward, borrowing costs are expected to stabilise in the coming months following the peaking of the global rate cycle.

Chart 2.2 Borrowing costs rose in 2024 with the continued passthrough of elevated interest rates

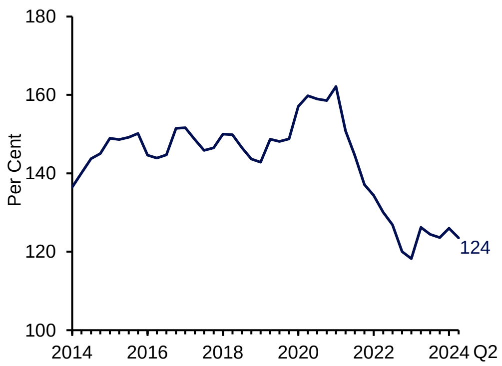
Interest expense as a share of total debt of SGX-listed firms



Source: Refinitiv Eikon, MAS estimates

Chart 2.3 Corporate debt-to-GDP remained at relatively low levels

Singapore's corporate debt, % of GDP



Source: BIS, Dealogic, DOS, MAS estimates

Firms' debt servicing has remained manageable, and the earnings outlook is expected to improve next year

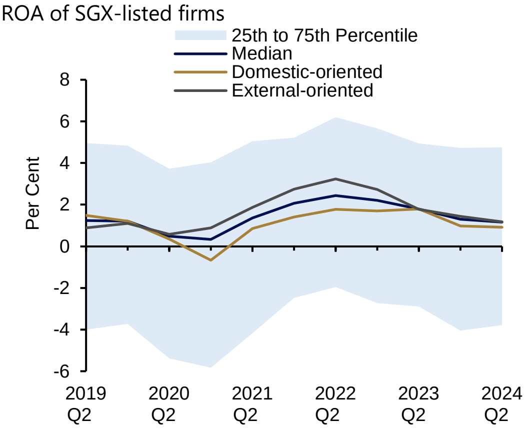
Most firms in Singapore have continued to generate sufficient earnings to service their debt despite higher interest expenses. While the median interest coverage ratio (ICR) of SGX-listed firms fell moderately alongside weaker corporate earnings in H1 (**Chart 2.4**), it remained relatively robust at above two in Q2 2024 (**Chart 2.5**).

Earnings are expected to register an improvement in H2 2024, reflecting the pick-up in economic growth in Q3 2024,[11] driven by external-oriented firms. Looking ahead, the strength of the tech cycle and less restrictive global financial conditions are projected to contribute further to improving earnings and stabilising of domestic financing costs. Affirming this outlook, the Q4 2024 Business Expectation surveys by the Economic Development Board (EDB) and the Department of Statistics (DOS) show manufacturers and services firms anticipating

[11] For more details, please see the October 2024 Issue of MAS' Macroeconomic Review.

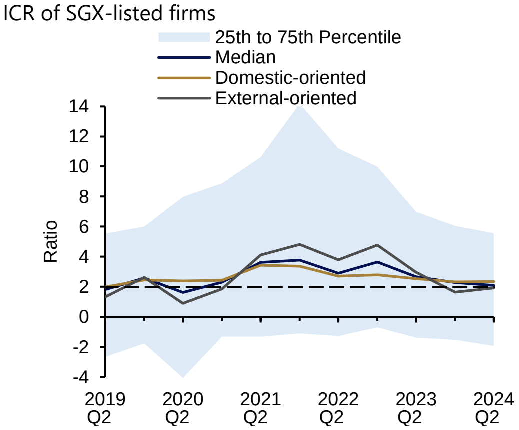
improved business prospects for the period October 2024 to March 2025, compared to the third quarter of 2024.

Chart 2.4 Corporate profitability fell in 2024



Source: Refinitiv Eikon, MAS estimates

Chart 2.5 Firms' debt servicing abilities moderated in 2024 but remained healthy



Source: Refinitiv Eikon, MAS estimates

Refinancing risk is mitigated by firms' healthy liquidity buffers and debt maturity profiles

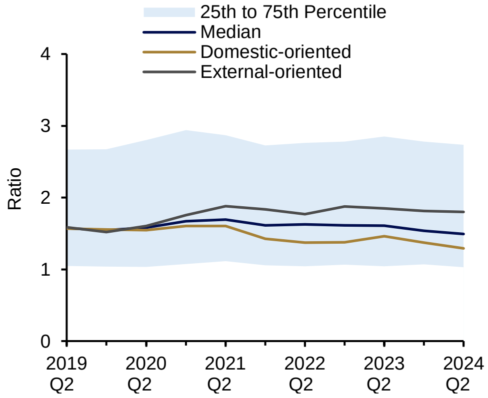
Refinancing risk has remained manageable, reflecting healthy liquidity buffers and favourable debt profiles (**Charts 2.6, 2.7 and 2.8**). Debt maturity profiles were stable, with the proportion of short-term debt coming due relatively unchanged at 43% in Q2 2024, and the share of bonds due by 2025 comprising a manageable 14% share of total outstanding bonds as of Q3 2024.

External-oriented firms have continued to maintain larger precautionary buffers due to global uncertainty and risks (**Charts 2.6 and 2.7**). SMEs, which tend to have weaker balance sheets and smaller buffers, have maintained a healthy median current ratio of above one,[12] even as higher borrowing costs had led to a decline in their cash buffers (**Chart 2.9**). With most SME loans collateralised by property, SMEs continued to have access to financing as property valuations have held up (see Chart Panel 2A, Chart 2A4).

[12] Based on latest available financials of SME customers collected in MAS' survey of banks, the median current ratio (which measures current assets as a proportion of current liabilities) of SMEs in 2023–2024 was healthy at around 1.2x.

Chart 2.6 Firms held adequate current assets to cover their current liabilities

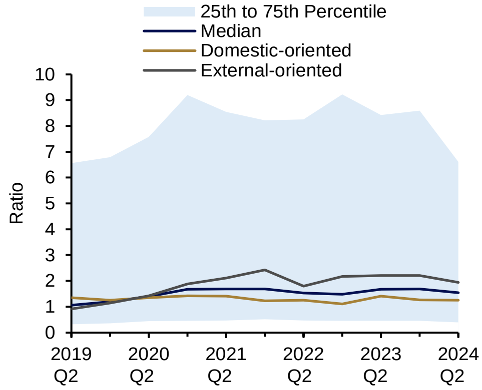
Current assets to current liabilities of SGX-listed firms



Source: Refinitiv Eikon, MAS estimates

Chart 2.7 Firms maintained sufficient cash to repay short-term debt at short notice

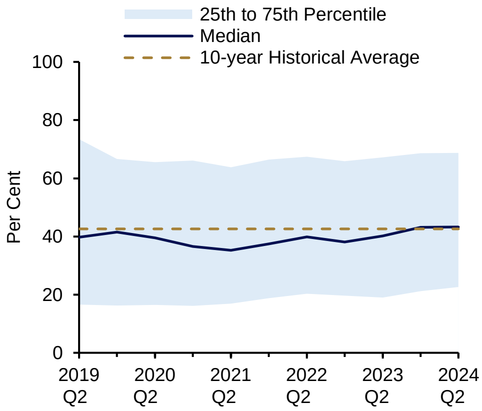
Cash to short-term debt of SGX-listed firms



Source: Refinitiv Eikon, MAS estimates

Chart 2.8 Short-term debt to total debt ratios have remained stable

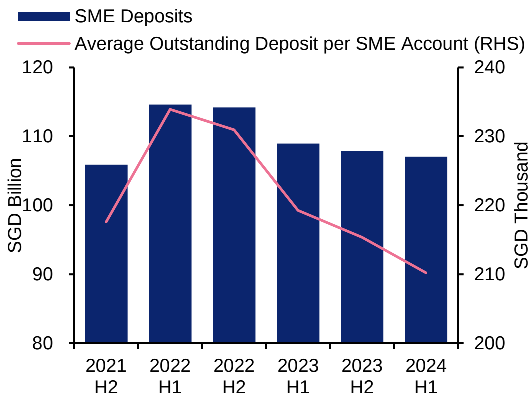
Short-term debt to total debt ratios of SGX-listed firms



Source: Refinitiv Eikon, MAS estimates

Chart 2.9 SME cash buffers have fallen

Overall and average outstanding SME deposits



Source: MAS survey

Hedging strategies helped to limit the foreign currency mismatch risk

The share of FCY-denominated bonds as a proportion of total outstanding bonds issued by firms in Singapore ticked up slightly from 58% in Q3 2023 to 62% in Q3 2024. Nevertheless, financial disclosures lodged with SGX revealed that firms that borrow in foreign currencies tend to be naturally hedged, with revenues in currencies matching their liabilities. Such firms also rely on derivatives to manage their FCY exposure,[13] which would not have been reflected in the FCY debt measure.

[13] An update of MAS' earlier study on the hedging practices of firms in Singapore (which used text-mining of financial disclosures in firms' annual reports), suggests that firms with larger FCY debt continue to rely on natural hedging and active hedging with derivatives to manage their exposures.

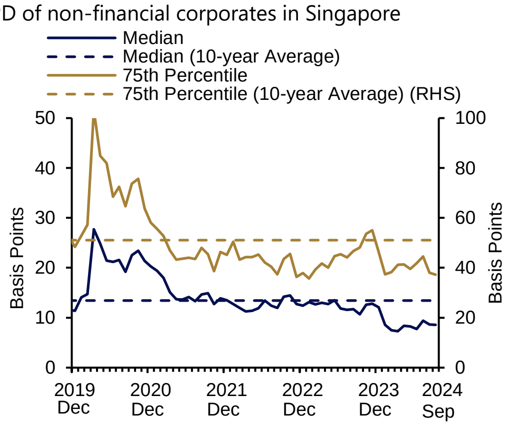
Forward-looking probability of default and corporate NPL ratios have stayed low

National University of Singapore Credit Research Initiative (NUS-CRI)'s Probability of Default (PD)[14] indicator shows the forward-looking PDs of listed firms has remained low and stable in 2024, suggesting that the vulnerability of Singapore corporates has not risen through the recent period of slower earnings growth and higher interest rates (**Chart 2.10**).

Corporate NPL ratios, including those for SMEs, fell in 2024 (**Chart 2.11** and Chart Panel 2A, Chart 2A5), indicating that SMEs' financial health has remained relatively stable, a view supported by a sounding of bank lending officers. A bank survey of SMEs[15] reported a relatively positive business outlook, with about half of SMEs surveyed expecting better performance over the next six months, 40% assessed the outlook to remain stable, and only 11% took a negative view of their business prospects.

Chart 2.10 Probability of default of Singapore firms has remained below historical averages

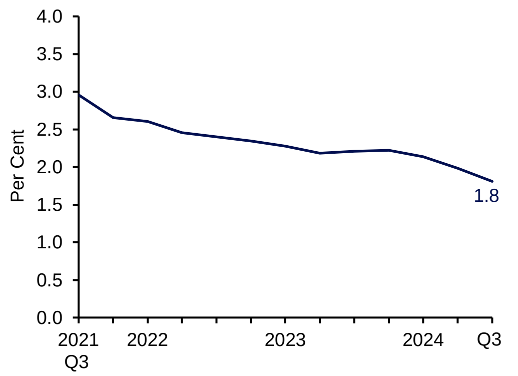
PD of non-financial corporates in Singapore



Source: NUS-CRI, MAS estimates

Chart 2.11 Singapore banking system's corporate NPL ratio continued to fall in 2024

Singapore banking system's overall corporate NPL ratio



Source: MAS

2.3 Outlook

MAS' stress test suggests that Singapore's corporate sector will continue to be resilient against further shocks

Rising geopolitical and trade tensions could dampen growth and disrupt supply chains. Supply shocks could renew inflationary pressures, prompting central banks to raise interest rates again. To evaluate these risks, MAS conducted a stress test on SGX-listed corporates to assess how the sector would respond to a joint shock from lower demand and higher interest

[14] This indicator serves as a timely, market-relevant metric to enhance corporate surveillance as it is a forward-looking measure of listed corporates' vulnerability. It assesses a firm's default risk over the next 12 months using firm-specific accounting and market-based inputs, as well as macrofinancial factors. For more details, see "Special Feature 3: Enhancing Corporate Surveillance with Probability of Default Model" in MAS' 2021 Financial Stability Review.

[15] The OCBC SME Business Outlook poll was conducted between 2 September 2024 and 30 September 2024, collecting about 1,100 responses from SME business owners in Singapore.

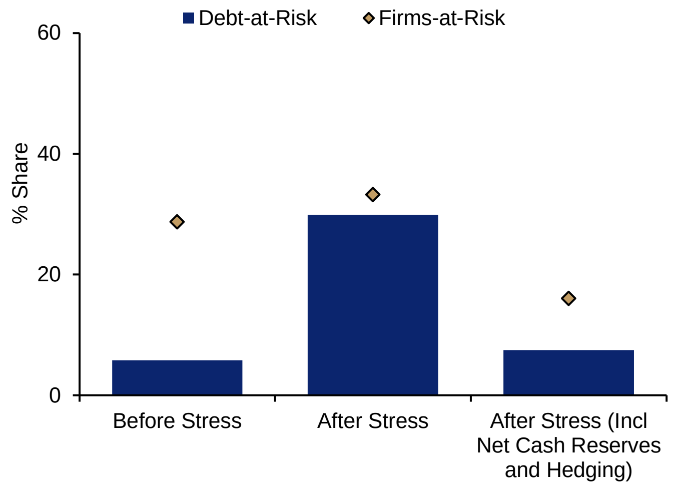
rates, i.e. a 10% decline in earnings[16] and a 300 bps rise in interest rates[17] from already elevated Q2 2024 levels.

The results suggest that most corporates are resilient to interest rate and earnings shocks, with cash reserves providing buffers. Based on their Q2 2024 financial position (including net cash reserves and hedging), the percentage of firms-at-risk (defined as having ICR less than one) stood at 16% after stress, comparable to the 15% from the Q2 2023 stress test. Debt-at-risk (defined as debt held by firms-at-risk) came in lower at 7% of overall corporate debt (**Chart 2.12**), compared to 8% in Q2 2023.

Corporates that are less resilient with more debt-at-risk are pre-dominantly external-oriented firms whose margins were compressed by the challenging global environment, as well as those with capital-intensive business models and higher operating leverage.

Chart 2.12 Most Singapore corporates would remain resilient under simultaneous adverse shocks to interest rate and earnings

Listed firms with ICR less than 1 (as a share of total number of listed firms, and as a share of total listed firms' debt)



Source: Refinitiv Eikon, MAS estimates

Firms should maintain sufficient financial buffers to mitigate downside risks

Corporates have weathered the challenging macrofinancial environment well over the past year, with default risks continuing to fall. Amid the macroeconomic and financial uncertainties, stress test results suggest that most listed firms should remain resilient, though external-oriented and highly leveraged firms could be more vulnerable. Against this backdrop, firms should provision adequate buffers to mitigate such risks, including by managing the maturity of their debt obligations and increasing liquidity buffers, as a way to further reduce their financial vulnerabilities.

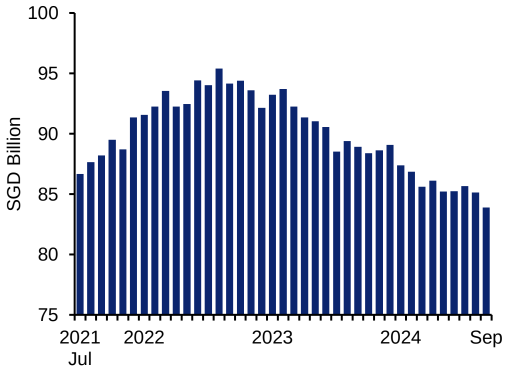
16 The assumption references the decline of the corporate sector's gross operating surplus in past crises.

17 A simulation of a larger premium charged by banks on top of a further pass-through of interest rate rises.

Chart Panel 2A Small and Medium-sized Enterprise Financing Conditions

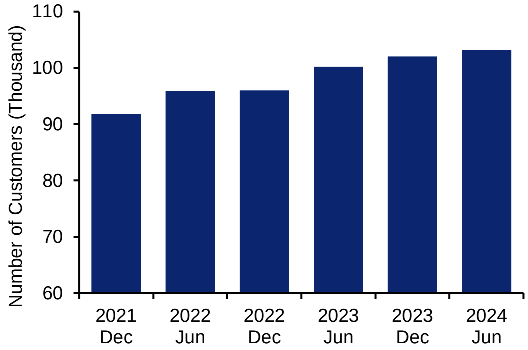
Aggregate SME loan growth was weak in 2024, as declines in outstanding loans under expired government financing schemes outpaced new loans. However, the number of SME customers increased.

Chart 2A1 SME loans outstanding



Source: MAS

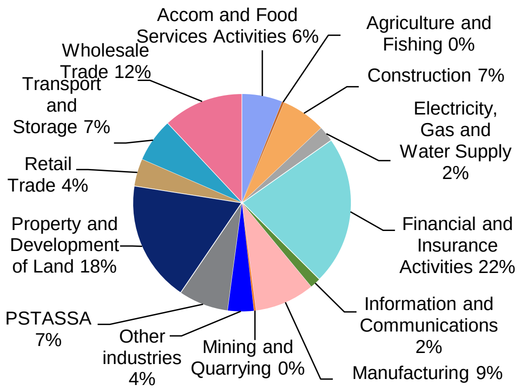
Chart 2A2 SME customers



Source: MAS survey

The Property and Development of Land sector accounted for the largest share of non-financial SME loans. Property remained the predominant form of collateral for SME loans.

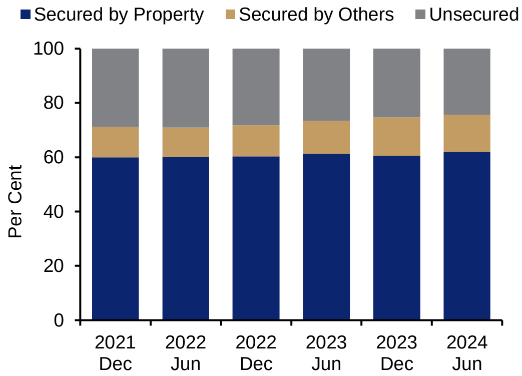
Chart 2A3 SME loans by sector (as of September 2024)



Source: MAS

Note: PSTASSA refers to Professional, Scientific, Technical, Administrative, Support Service Activities.

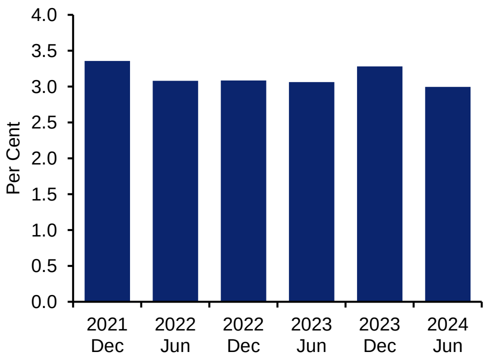
Chart 2A4 Outstanding SME loans by type of collateralisation



Source: MAS survey

SME NPL ratio fell slightly over the past year.

Chart 2A5 SME NPL ratio



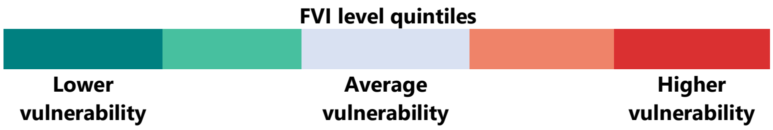
Source: MAS survey

3 Singapore Household Sector

The household sector in Singapore has remained financially resilient, with a strong balance sheet and good credit quality as reflected in the improvement of the household sector FVI. Leverage risk has moderated as mortgage rates have eased, while debt servicing ability is healthy alongside stable income growth. Liquidity risk has remained low as the growth in cash and deposits has outpaced that of total household liabilities. Asset price volatility risk is assessed to be contained as the private housing market has been stabilising, with a slowing pace of price increases through the first three quarters of the year.

Overall, the risks to the household sector from the uncertain global macroeconomic outlook are expected to be contained, given strong existing buffers. MAS' stress tests of households affirm that most borrowers have the capacity to manage higher debt servicing costs and lower income, reflecting prudent credit practices by banks and sound macroprudential policies. Nevertheless, households should consider the heightened uncertainty in the macroeconomic environment before taking on more leverage.

Household sector FVI[18] level	Q3 2023	Q3 2024
Overall household FVI		
Leverage risk		
Liquidity risk		
Asset price volatility risk		



3.1 Assessment of leverage risk

Growth in household debt was outpaced by asset growth and supported by healthy income growth

While overall household debt increased in the past year (**Chart 3.1**), financial assets grew faster.[19] As healthy income growth outpaced debt accumulation, household debt as a share

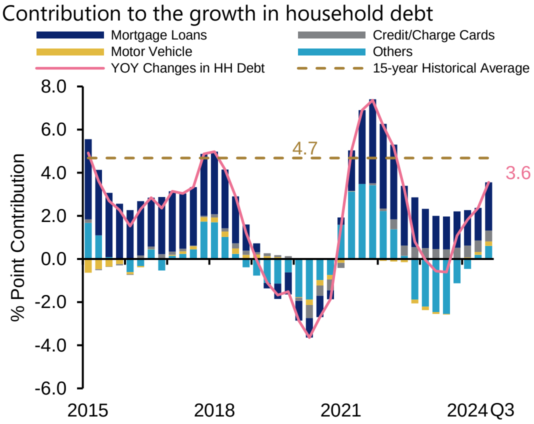
[18] The household sector FVI consists of three indicators: (i) leverage risk assesses household mortgage debt levels alongside debt repayment risk; (ii) liquidity risk assesses household vulnerability to short-term debt such as credit card borrowings compared to income and liquidity buffers; (iii) asset price volatility risk assesses households' exposures to market risks in the private residential property market.

[19] In Q3 2024, households' financial assets grew 8.0% y-o-y, higher than the growth in overall household debt.

of PDI has been on a downward trend since Q4 2021. Most recently, it has appeared to stabilise at a multiple of 1.1 as of Q3 2024 (**Chart 3.2**).

Chart 3.1 Gradual increase in household debt due to growth in mortgages

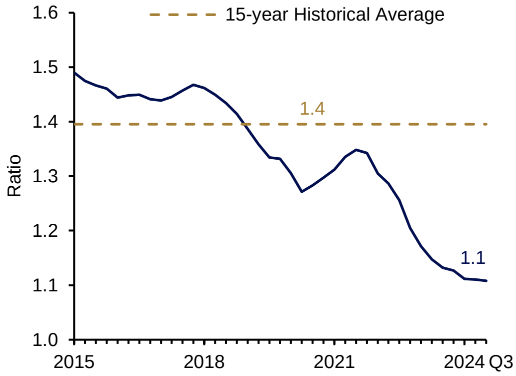
Contribution to the growth in household debt



Sources: DOS, MAS estimates

Chart 3.2 Household debt-to-PDI declined for the 12th consecutive quarter

Household debt-to-PDI

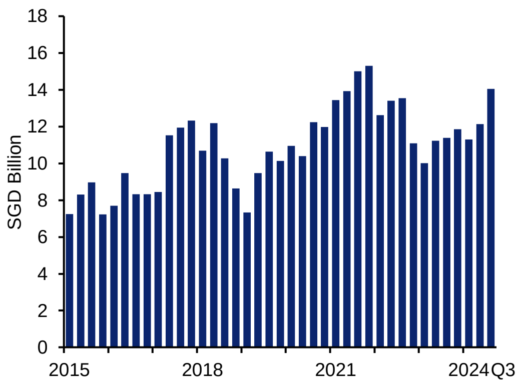


Sources: DOS, MAS estimates

About three-quarters of total household debt comprises housing loans that are backed by property collateral. Outstanding housing loans increased by a modest 1.6% year-on-year (y-o-y) in Q3 2024, as new loans were largely offset by borrowers paying down existing mortgages amid elevated interest rates (**Charts 3.3 and 3.4**).

Chart 3.3 New housing loans increased…

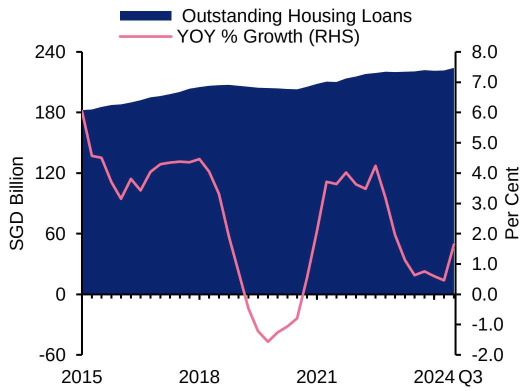
New housing loans



Source: DOS, MAS estimates

Chart 3.4 … but the stock of outstanding housing loans rose only moderately, as borrowers paid down existing mortgages

Outstanding housing loans



Source: DOS, MAS estimates

Households' debt servicing ability remains healthy, supported by lower mortgage rates and continued income growth

Mortgage rates moderated over the past year. Following market expectations of a peak in the global interest rate cycle in H1 2024 and monetary policy easing by major central banks,

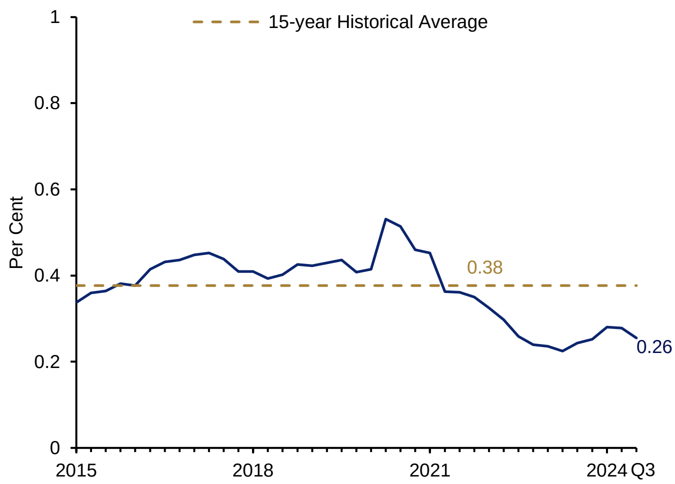
banks adjusted their mortgage rates lower. Fixed rate mortgage packages offered by banks peaked at 4.5% in end-2022 and have fallen to below 3% in H1 2024.[20]

At the same time, the risk of a delayed upward adjustment in average mortgage rates on debt servicing has diminished. As many as 90% of borrowers now bear the post-pandemic upswing in interest rates, including borrowers who have refinanced or taken up new loans since major central banks began hiking rates in H2 2022.[21] Credit quality of housing loans remains strong, with housing NPL ratios staying low at 0.3% as of Q3 2024 (**Chart 3.5**).

A stress test on households, assuming an immediate increase in mortgage rates to 5.5%, together with a simultaneous income loss of 10%, affirms that debt servicing capacity is sufficient to survive adverse shocks. More than 90% of households would be able to continue servicing their mortgages in this scenario. However, a small segment of highly leveraged borrowers, accounting for less than 5% of total mortgage loans, could be more vulnerable to repayment risk. The stress test may be regarded as identifying the upper bound of potentially distressed borrowers as households could have other sources of funds which could be used for debt servicing such as cash buffers and Central Provident Fund-Ordinary Account (CPF-OA) savings; these were excluded from the simulations.

Chart 3.5 NPL ratio rose slightly over the past year but has remained below the historical average

Non-performing loans (mortgages)



Source: MAS estimates

20 Close to half of borrowers in H1 2024 have taken up mortgage packages at relatively lower mortgage rates of less than 3%.

21 Based on MAS' survey of housing loans for selected financial institutions, which account for over 90% of total outstanding housing loans extended by the industry.

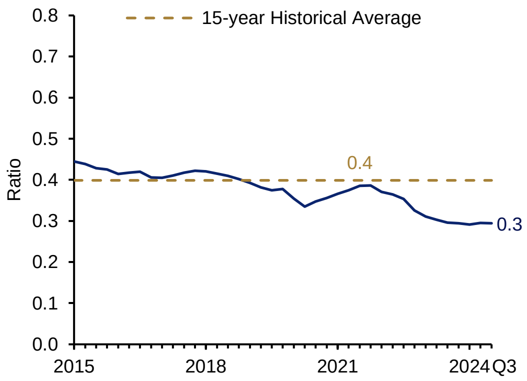
3.2 Assessment of liquidity risk

The ratio of personal loans to PDI has trended down and stands below its historical average

The ratio of personal loans to PDI, which has been on a downward trend since Q4 2021, continued to decline (**Chart 3.6**). While personal loans increased by 5.1% y-o-y in Q3 2024, the growth rate has remained below the historical average pace (**Chart 3.7**).

Chart 3.6 Personal loans to PDI moderated since Q4 2021

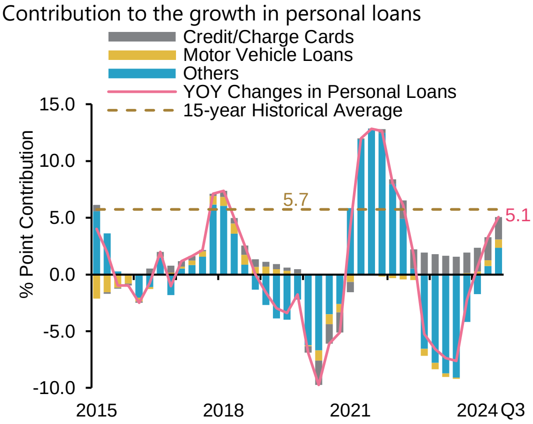
Personal loans to PDI



Source: DOS, MAS estimates

Chart 3.7 Growth in personal loans stayed below its historical average

Contribution to the growth in personal loans

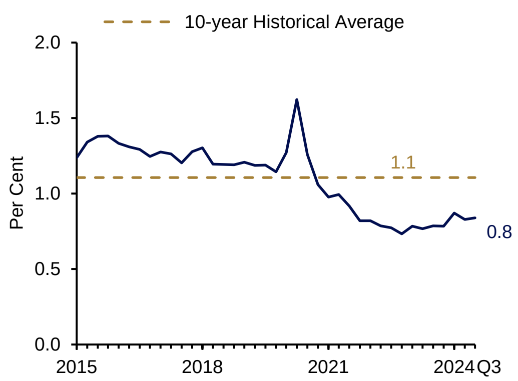


Source: DOS, MAS estimates

Almost all borrowers have been able to service their credit/charge card debt, despite the rising balances which reflected robust growth in resident outbound travel and domestic retail sales. Credit card delinquency rate has remained low at less than 1% (**Chart 3.8**). Rollover balances as a share of PDI edged up to 2.4% in Q3 2024 normalising to historical averages but below pre-pandemic levels (**Chart 3.9**). Revolvers as a share of total cardholders also remained stable at around 18% over the past year, significantly lower than pre-pandemic levels (**Chart 3.10**).

Chart 3.8 Credit card delinquencies has remained low

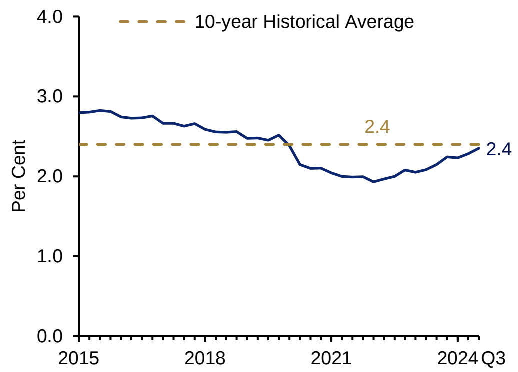
Credit card delinquency rate



Source: MAS estimates

Chart 3.9 Rollover balance to PDI normalised but is still below pre-COVID levels

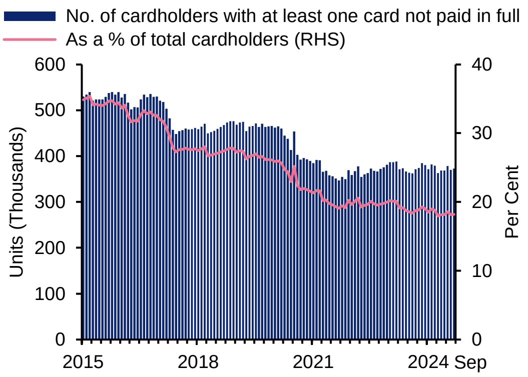
Rollover balance to PDI



Source: DOS, MAS estimates

Chart 3.10 Revolvers as a share of total cardholders have remained stable

Number of revolvers



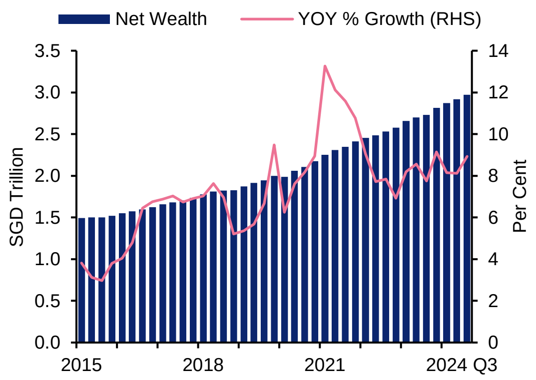
Source: Credit Bureau Singapore (CBS), MAS estimates

Sustained growth in liquid assets buffers against repayment risks from the growth in household liabilities

Aggregate household net wealth (i.e. total assets less liabilities) grew in the first half of 2024, rising by about 9% y-o-y to almost SGD3 trillion as of Q3 2024 (**Chart 3.11**). In particular, liquid assets such as cash and deposits—comprising about 20% of total assets as of Q3 2024—continued to exceed total liabilities. The growth of the former outpaced that of the latter since Q1 2022 (**Chart 3.12**).

Chart 3.11 Households' net wealth recorded strong gains in Q3 2024

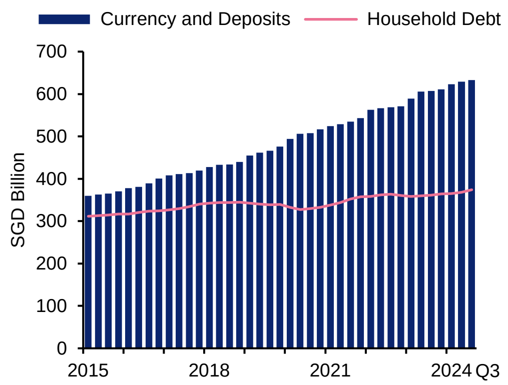
Net wealth



Source: DOS, MAS estimates

Chart 3.12 Liquid assets has continued to exceed total liabilities

Liquid assets and debt



Source: DOS, MAS estimates

3.3 Private residential property market

The private residential market has shown signs of stabilisation

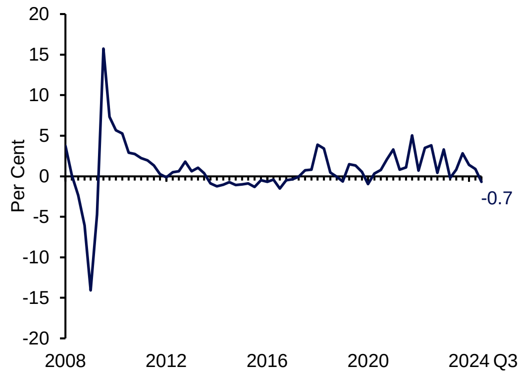
Price increases in the private residential market slowed consecutively over the first two quarters of 2024 before turning negative in Q3 2024, falling by 0.7% quarter-on-quarter (q-o-q) and marking the first decline since Q2 2023 (**Chart 3.13**). In aggregate, prices rose by 1.6% in the first three quarters this year, significantly slower than the 3.9% increase over the same period in 2023.

Transaction volumes[22] remained moderate, with average quarterly transaction volumes in the first three quarters of 2024 similar to that in 2023, which is still about 12% below 2022 levels. New sales fell by about 43% in the first three quarters of 2024 compared to the same period last year as developers held back major project launches, although this was partially offset by the increase in resale transactions of about 22% in the same period (**Chart 3.14**). More recently, over the past six months, transaction volumes have been gradually picking up.

[22] Transaction volume figures exclude Executive Condominiums (ECs), unless indicated otherwise.

Chart 3.13 Property price growth momentum has eased in recent quarters

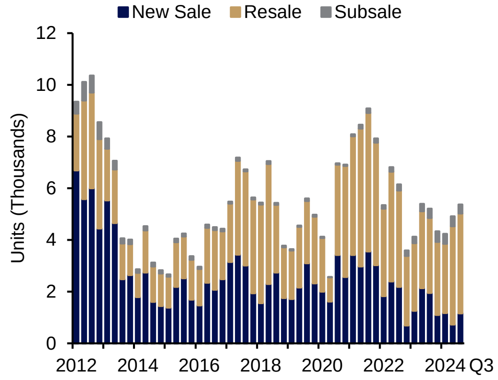
Private residential property price index (q-o-q change)



Source: URA

Chart 3.14 New sale transactions in 2024 have remained low

Number of private residential property transactions



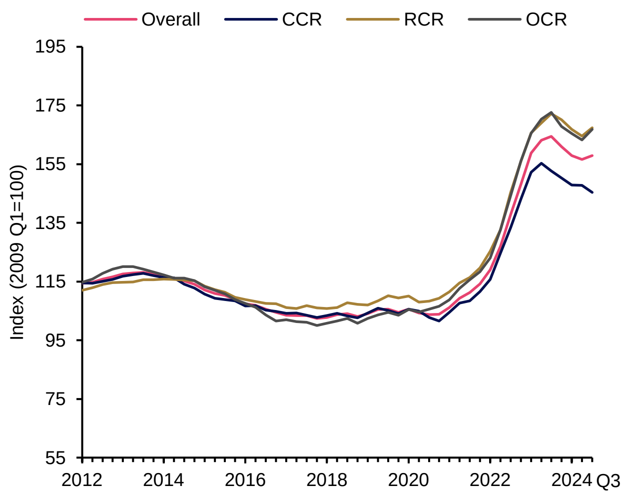
Source: URA

The private rental market has moderated from its peak

After increasing over three years to a peak in Q3 2023, private residential rents have since fallen by a cumulative 4% with the moderation observed across all market segments (**Chart 3.15**). The tight rental market has eased with the large supply of private housing completions in 2023 after construction resumed post-pandemic. Since Q1 2023, quarterly private housing completions have averaged 3,600 units, about 50% higher than in 2022. Looking ahead, the rental market is expected to be broadly stable, with a steady supply of units coming on stream and demand pressures easing in the coming quarters.

Chart 3.15 Rents moderated from its peak in Q3 2023, declines were broad-based

Private property rental price indices by region



Source: URA

Note: "CCR" refers to Core Central Region, "RCR" refers to Rest of Central Region, "OCR" refers to Outside of Central Region

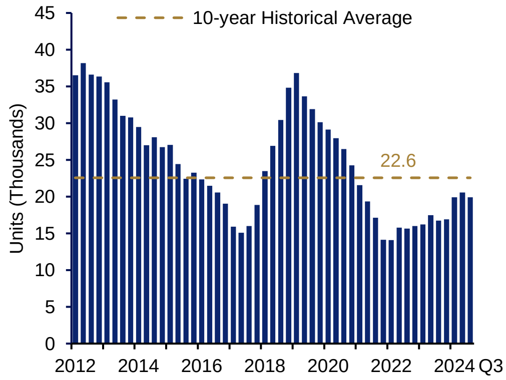
Supply and demand have been shifting into better balance

Supply-demand conditions in the private residential property market have been coming into better balance. With the ramp-up in private housing supply via the Government Land Sales (GLS) Programme, the inventory of unsold units in the supply pipeline has been correspondingly replenished (**Charts 3.16 and 3.17**), trending closer to historical averages. This has helped to meet demand for private housing.

While the recent easing of domestic interest rates may have boosted market sentiments as developers resumed project launches, the macrofinancial outlook is subject to significant uncertainties arising from geopolitical conflicts and trade tensions. Existing property market and macroprudential measures have restrained excessive demand and ensured household financial prudence. The government will remain vigilant to market developments, with the objective of promoting a stable and sustainable private housing market.

Chart 3.16 Unsold inventory continued to edge up...

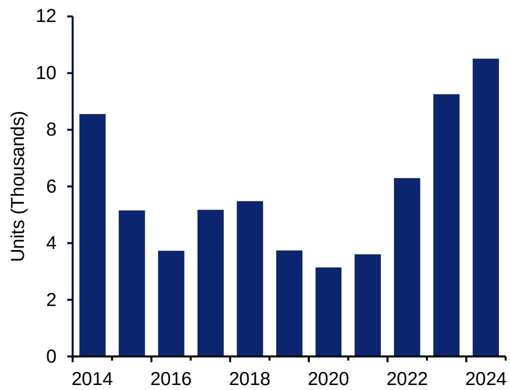
Total number of unsold private residential units from projects with planning approvals



Source: URA

Chart 3.17 ... as the Government continued to ramp up GLS supply

GLS confirmed list supply



Source: URA

3.4 Outlook

Households should exercise prudence amid macroeconomic uncertainties

Overall, the risks to the household sector are expected to be contained, given strong financial buffers. The easing of mortgage rates in H1 2024 accompanied by stable income growth has bolstered households' debt servicing ability, while the liquidity position of households has further improved. However, given the heightened geopolitical uncertainties and trade tensions in the macrofinancial environment, households should continue to exercise prudence in their financial management.

4 Singapore Financial Sector

Singapore's financial sector has remained resilient over the past year. Banks and insurers have maintained healthy capital and liquidity positions, while investment funds have managed liquidity risks well. Looking ahead, the uncertain global macroeconomic outlook with the risk of weaker growth, could adversely impact banks' and insurers' profitability and capital positions. Market participants may be exposed to bouts of market volatility that may lead to a disorderly unwinding of overvalued or leveraged positions. Nevertheless, stress tests indicate that banks and insurers are in a good position to withstand a severe macrofinancial shock, while most investment funds have enough liquid assets to meet redemption shocks. Banks should maintain sound underwriting standards, ensure adequate loan provisions and remain vigilant against potential liquidity risks. Meanwhile, the domestic NBFI sector is expected to pose limited systemic risk to Singapore's financial system but will continue to be closely monitored for evolving risks.

4.1 Banking sector

The Singapore banking system has remained resilient with strong capital and liquidity buffers. Over the past year, credit to non-bank entities resumed an expansion path, supported by increased loans to resident borrowers, alongside the strengthening in the domestic economy. The overall banking FVI level remains benign. Liquidity and maturity risks have remained low, as banks maintained comfortable liquidity positions and healthy loan-to-deposit (LTD) ratios. Meanwhile, leverage vulnerabilities fell further on the back of stronger capital ratios even as loan growth picked up.

Banks in Singapore are well-positioned to manage challenges to the macrofinancial outlook from an uncertain political and economic environment. Asset quality remains healthy despite still-elevated borrowing costs, and banks have adequate precautionary buffers to guard against any credit quality decline. The IWST 2024 exercise affirmed that banks have sufficient capital and liquidity buffers to withstand potential downside risks arising from severe macrofinancial stresses (see Box B "Industry-wide Stress Test of D-SIBs").

Banking sector FVI level	Q3 2023	Q3 2024
Overall Banking FVI		
Resident leverage risk		
Non-resident leverage risk		
Liquidity risk		
Maturity risk		

FVI level quintiles

Lower Vulnerability **Average Vulnerability** **Higher Vulnerability**

Credit growth to non-bank entities was supported by increased loans to residents…

Overall credit growth has recovered from the trough in 2023, peaking in early 2024 before moderating in Q3. These developments were largely driven by interbank lending (see Chart Panel 4A "Banking Sector: Credit Growth Trends", Chart 4A1). Meanwhile, lending to non-bank entities has also registered sustained expansion since March this year, as global interest rates have transitioned towards an easing cycle.

Lending to non-resident, non-bank entities was stable over most of 2024 with banks continuing to intermediate funds to Developed Asia and Europe (see Chart Panel 4B "Banking Sector: Cross-border Lending Trends"). However, lending denominated in USD continued to weigh on loan growth, due to higher US interest rates and the stronger dollar (**Chart 4.1**).

Loans to Singapore residents accounted for most of the recovery in overall lending to non-bank entities. There was a broad-based increase in lending to domestic-oriented sectors, including the property-related and information & communications sectors (Chart Panel 4A, Chart 4A2). While loans to trade-related sectors remained weak, some nascent signs of improvement have appeared as the manufacturing and wholesale trade sectors grew on a sequential basis in September.

… with continued access to financing for resident SMEs

SMEs have continued to have good access to bank financing over the past year with the number of SME borrowers increasing further (see Chart Panel 2A "Small and Medium-sized Enterprise Financing Conditions", Chart 2A2). However, total loan value to SMEs remained weak (Chart Panel 2A, Chart 2A1). The decline in outstanding loans to SMEs was due to a natural runoff of loans under the enhanced Enterprise Singapore (ESG) loan schemes for SMEs introduced during the COVID-19 pandemic, which ceased in Q3 2022.

A slight tightening in credit conditions due to the pass-through of the earlier rise in global interest rates has also contributed to the weakness in SME credit. Despite this, overall credit risk premia remained below the historical average (**Chart 4.2**). Notably, credit risk premia for secured loans declined significantly in 2022 and are presently near zero (**Chart 4.3**).

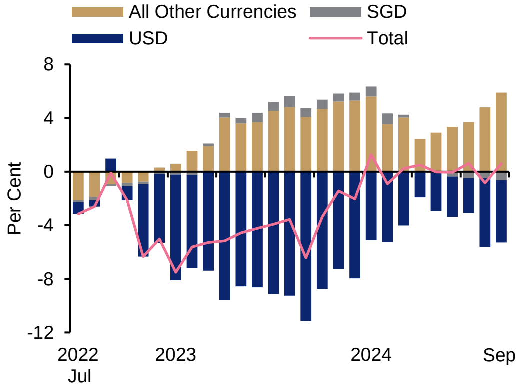
Credit demand is projected to rise while banks maintain prudent underwriting standards

Banks expect credit demand to rise in the near term, underpinned by an increased appetite for corporate capital expenditures and acquisitions in the broader economy as financial conditions ease.[23] Nonetheless, banks plan to maintain current prudent underwriting standards as a safeguard against heightened uncertainty over the economic outlook including from geopolitical tensions.

23 MAS conducted a survey of banks in September 2024 to seek industry views on realised credit conditions over the past six months and expectations for Q4 2024 and Q1 2025.

Chart 4.1 Decline in USD-denominated loans weighed on non-resident loan growth

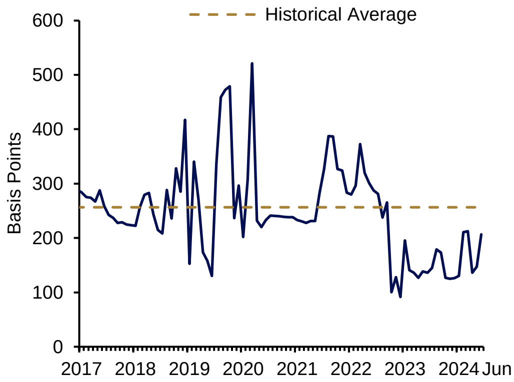
Y-o-y growth of non-resident loans to non-bank entities by currency



Source: MAS

Chart 4.2 Overall credit risk premia remained below historical average

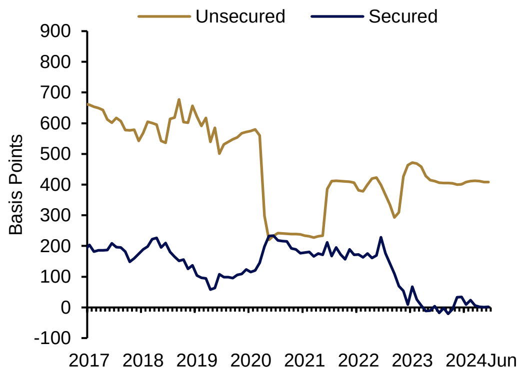
Estimated risk premium of overall new SME loans



Source: MAS survey

Chart 4.3 Credit risk premia for secured loans declined significantly to near zero

Estimated risk premium of new secured and unsecured SME loans



Source: MAS survey

Note: The spread for each unique loan facility extended by the reporting banks to SMEs was estimated by calculating the difference between the interest rate charged at inception for the facility and the 3-month compounded SORA on the date the facility was granted.

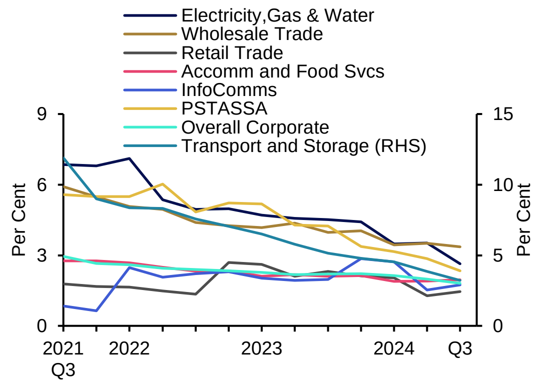
Asset quality remains healthy

Overall asset quality improved over the past year for most sectors. Sectors with relatively higher NPL ratios in the past, such as the transport and storage, electricity, gas & water, and wholesale trade sectors, saw their NPL ratios declining to multi-year lows (**Chart 4.4**).

However, the construction, manufacturing and real property and development of land sectors continue to face headwinds with marginal rises in NPL ratios (**Chart 4.5**), as still-high borrowing costs alongside higher labour and material costs impacted profitability and cash flows.

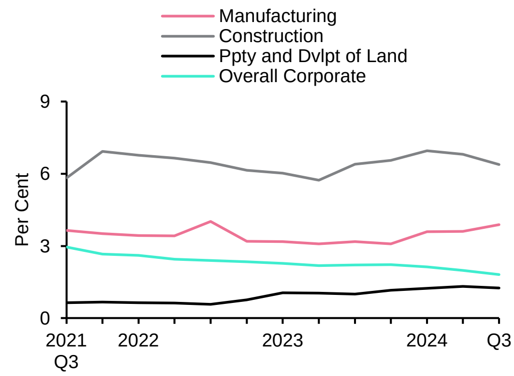
The overall outlook for asset quality is benign with the special mention ratio[24] falling to 2.5% as of Q3 2024 (see Chart Panel 4C "Banking Sector: Asset Quality and Liquidity Indicators", Chart 4C1), driven by corporate loans. Looking ahead, credit quality is expected to remain healthy as corporate earnings are projected to remain favourable in H2 2024 as the Singapore economy continues to expand (see Chapter 2 "Singapore Corporate Sector").

Chart 4.4 Sectors with relatively higher NPL ratios, saw ratios decline to multi-year lows



Source: MAS

Chart 4.5 Several sectors saw a slight rise in NPL ratios, as they continued to face higher input costs



Source: MAS

Banks continue to maintain adequate provisioning and capital buffers, aided by strong profitability

Banks have strengthened precautionary buffers to guard against a potential rise in credit costs. The banking system's total provisioning coverage[25] increased to 119.9% as of Q3 2024, up from 111.1% the same period the year before. Local banking groups maintained healthy provisioning coverage of 252% as of Q3 2024, alongside the improvement in their NPL ratios (see Chart Panel 4D "Banking Sector: Local Banking Groups", Chart 4D4).

Local banking groups have continued to register healthy net profits in 2024, as both net interest income and non-interest income rose. The robust earnings also underpin local banks' strong capital positions, with aggregate common equity tier 1 (CET1) Capital Adequacy Ratios (CAR) well above regulatory requirements, at 16.7%[26] as of Q3 2024. Strong capital buffers, supplemented by ample provisions, will ensure that local banking groups are well placed to withstand any deterioration in credit quality.

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24 Credit facilities that exhibit potential weaknesses but are not yet classified as NPL.

25 The banking system's total provisioning coverage is the sum of general and specific provisions as a share of unsecured non-performing assets (NPA).

26 The local banking groups' CET1 CAR rose from 14.0% as of Q3 2023, largely due to changes associated with the phased adoption of the final Basel III reforms which came into effect from 1 July 2024.

The results of the IWST 2024 exercise affirm that banks in Singapore have adequate capital buffers to weather potential downside risks (see Box B "Industry-wide Stress Test of D-SIBs").

Liquidity positions remain strong

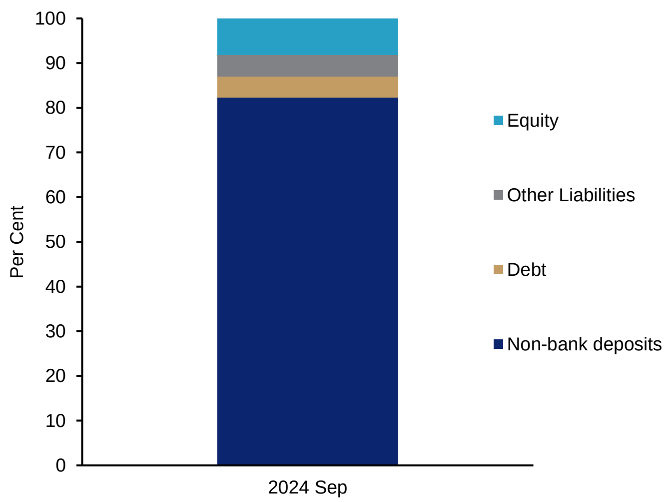
The banking sector's risk from liquidity vulnerabilities have remained low and stable over the past year.

Liquidity buffers of domestic systemically important banks (D-SIBs) are well above all-currency and SGD minimum regulatory Liquidity Coverage Ratio (LCR) requirements (see Chart Panel 4E "Banking Sector: Domestic Systemically Important Banks"). D-SIBs' Net Stable Funding Ratios (NSFRs) have also remained above the regulatory requirement.

The banking system's funding structure continues to be healthy, underpinned by stable non-bank deposits, which constitutes more than 80% of the system's total funding (**Chart 4.6**). Deposits registered robust growth over the past year, largely driven by higher deposits from individuals, supporting the banking sector's capacity to supply credit to the economy. SGD and FCY LTD ratios remain below 100%.

Chart 4.6 Stable non-bank deposits accounted for more than 80% of total banking system funding

Funding structure of banks in Singapore



Source: MAS

Note: As of September 2024, Singapore is a net interbank lender. Thus, interbank funding is not featured in the chart.

Nevertheless, sustained high interest rates, slower global economic growth, escalating trade tensions or geopolitical conflicts could raise credit costs and dampen lending volumes, putting pressure on banks' profitability and capital positions. Banks should thus continue to maintain sound underwriting standards, ensure that their loan provisions are adequate, and remain vigilant against potential liquidity risks.

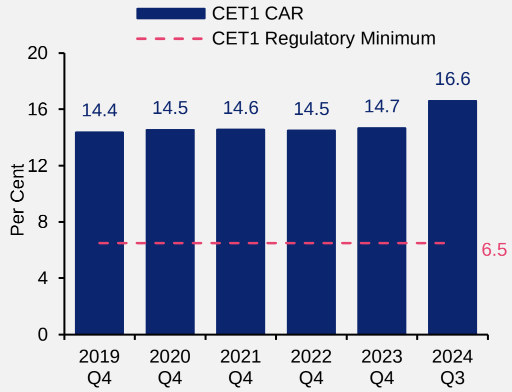
Box B
Industry-wide Stress Test of D-SIBs

D-SIBs continue to be well-capitalised with adequate buffers

D-SIBs have maintained strong capital positions, with their aggregate CET1 CAR, at 16.6%, remaining well above the regulatory minimum in Q3 2024 (**Chart B1**). D-SIBs have also maintained robust credit risk management practices and adequate provisioning buffers to cushion against potential credit losses amid uncertainties in the global macroeconomic outlook (**Chart B2**).

Chart B1 D-SIBs' capital positions have remained well above regulatory requirements

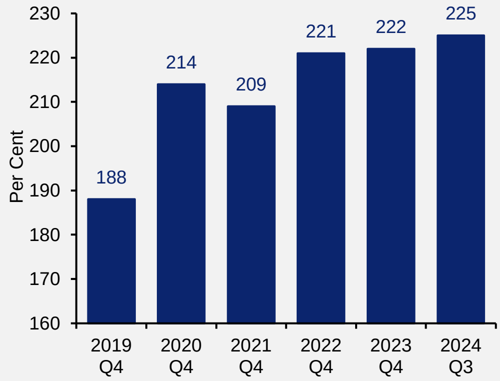
D-SIBs' aggregate CET1 capital ratios[27]



Source: MAS

Chart B2 D-SIBs have allocated adequate provisions to cushion against credit losses

D-SIBs' total provisioning coverage ratio*



Source: MAS

*Total provisions/total unsecured NPA

MAS' IWST 2024 assessed D-SIBs' resilience to renewed inflationary pressures and a global recession

The IWST 2024 exercise required D-SIBs to assess the resilience of their balance sheet and capital positions under two macroeconomic scenarios (Central and Adverse) over a three-year horizon.

The Central scenario assumes that the global economy continues to expand at a steady pace, with the last mile of disinflation supported by continued equilibrating supply-demand conditions. The Singapore economy expands around its potential rate, and inflation eases amid moderating external cost conditions.

In contrast, the Adverse scenario features an intensification of geopolitical conflicts and trade tensions that exacerbates supply chain and trade disruptions. The quick succession of

[27] See footnote 26 in Chapter 4 "Singapore Financial Sector".

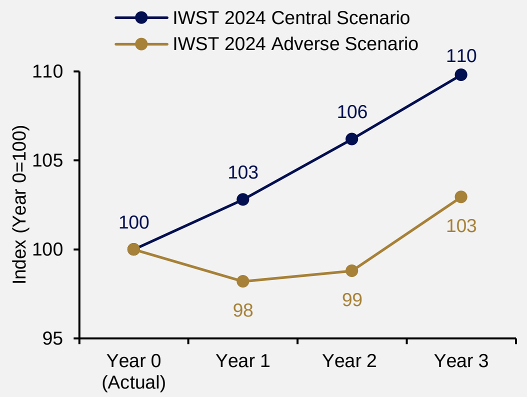
supply shocks reignites inflationary pressures, forcing major central banks to respond with tighter monetary policies. The global economy slips into a deep and protracted recession (**Chart B3**) as higher borrowing costs and volatile financial conditions result in weaker corporate profitability, pullback in investment and depressed consumer sentiments. Rising rates further dampen demand for commercial properties, leading to a deterioration in outlook for the highly-leveraged CRE sector in major economies and significant declines in CRE prices.

At the same time, the Chinese economy faces headwinds from a deepening residential property market downturn that weighs on domestic demand as consumer confidence and household wealth are adversely impacted. Regional economies are impacted extensively due to their close trade ties with China, in addition to the softening global demand conditions.

Under this Adverse scenario, the Singapore economy enters a recession amid weak external demand, rising commodity prices and elevated interest rates (**Chart B4**). The global recession leads to a sharp downturn in the external-facing sectors such as manufacturing, transportation and finance & insurance. Elevated interest rates and rising unemployment put pressure on corporate and household debt servicing abilities, causing banks to incur increased credit losses as defaults rise. Singapore banks also face elevated market and funding liquidity risks via their extensive cross-border interlinkages.

Chart B3 The global economy enters a recession under the Adverse scenario

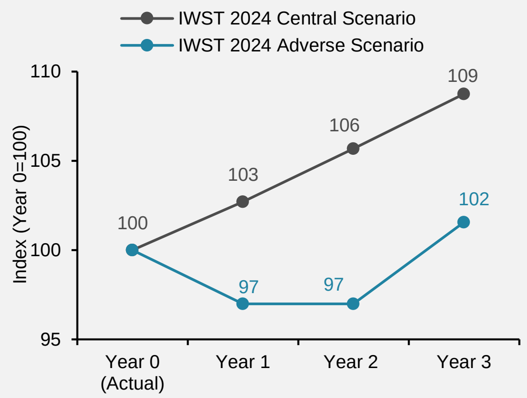
Global real GDP levels



Source: MAS

Chart B4 The Singapore economy contracts under the Adverse scenario amid weak external demand

Singapore's real GDP levels



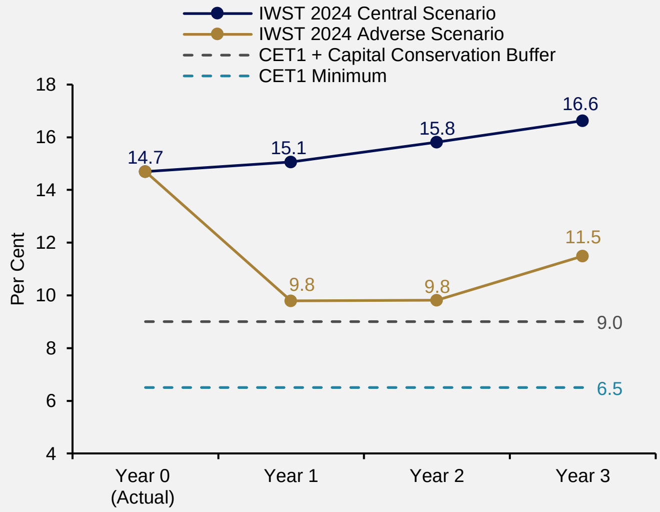
Source: MAS

D-SIBs' aggregate CET1 CAR is projected to remain well above minimum regulatory requirements under the Adverse scenario

Under the Adverse scenario, the decline in D-SIBs' projected aggregate CET1 CAR would level-off at 9.8% in Years 1 and 2, and remain above MAS' combined CET1 and capital conservation buffer (CCB) requirement of 9.0% (**Chart B5**). Each D-SIB's CET1 CAR is also projected to remain well above the minimum CET1 regulatory requirement. Overall, D-SIBs' capital buffers are sufficient to withstand such a severe macrofinancial shock.

Chart B5 D-SIBs' aggregate CET1 CAR remains above regulatory minimum under the Adverse scenario

Projected D-SIBs' aggregate CET1 CAR

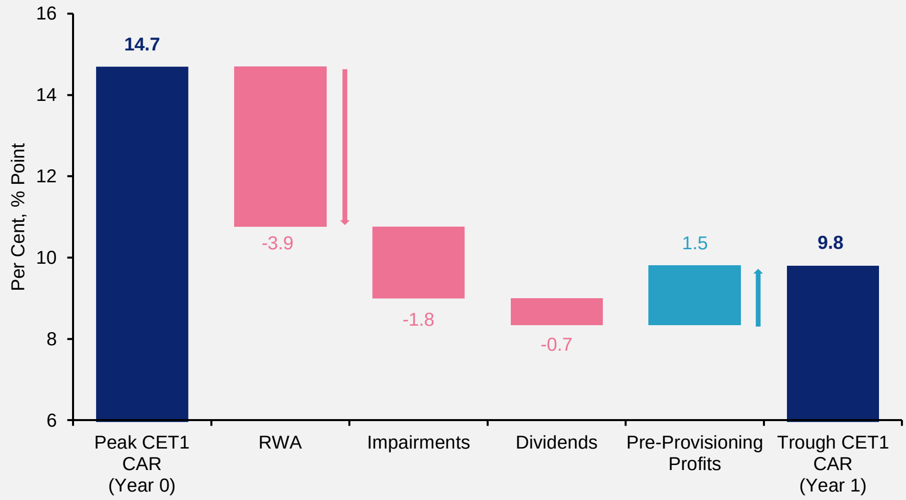


Source: D-SIBs' submissions, MAS estimates

The decline in the aggregate CET1 CAR under the Adverse scenario is primarily driven by the rise in credit risk-weighted assets (RWA) (3.9 percentage points) and credit impairments (1.8 percentage points), reflecting a significant deterioration in the asset quality of D-SIBs' corporate and retail portfolios (**Chart B6**). While D-SIBs' profitability would be adversely impacted by declining non-interest and net trading incomes due to weak macroeconomic and financial market conditions, net interest incomes would increase due to widening net interest margins (NIMs) in the high interest rate environment, with lending rates rising faster than deposit rates.

Chart B6 The decline in D-SIBs' aggregate CET1 CAR is primarily driven by a deterioration in asset quality

Impact of drivers on peak-to-trough decline in aggregate CET1 CAR



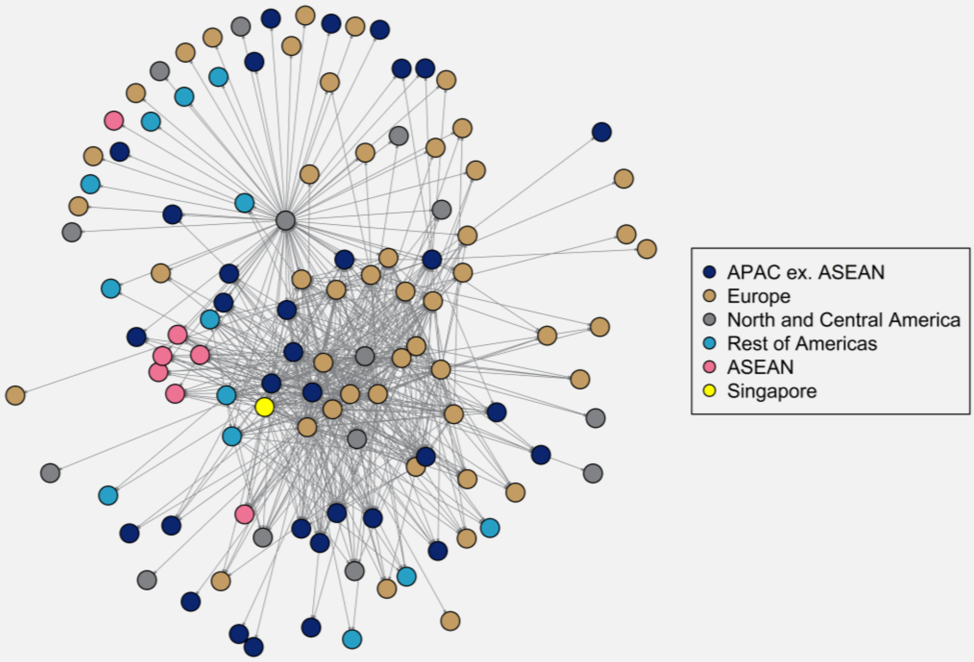
Source: D-SIBs' submissions, MAS estimates

MAS extended its stress testing to assess risk transmissions from cross-border interlinkages

Building on earlier work assessing *domestic* interbank contagion,[28] MAS enhanced its stress testing toolkit to assess first- and second-order *cross-border* interbank contagion impacts.[29] Given the interconnectedness of the domestic banking system with the global financial system, global banks can pose systemic risks to Singapore as adverse shocks can amplify and propagate through the cross-border interbank network (**Chart B7**).

Chart B7 Banking systems are highly interconnected globally

Cross-border network diagram of interbank exposures larger than USD 1 billion (coloured by region)^, as of December 2023



Source: BIS, MAS estimates

^Each node represents a country's banking system, based on available data from the Consolidated Banking Statistics hosted by the BIS, which measures banking activity from a nationality perspective and includes foreign operations of the home country's banks

A stress contagion was simulated, with multiple risk transmission channels…

Cross-border interbank contagion was simulated by assuming a tail-risk scenario of the default of a single country's banking system as the trigger event. The analysis then simulated

28 See the *MAS Financial Stability Review 2022*, "Box B: Industry-Wide Stress Test of D-SIBs".

29 The assessment of contagion does not include the macrofinancial effects that can ensue from a banking system failure in key countries.

the interactions among three risk transmission channels—counterparty country banking system defaults, funding liquidity stresses and fire sales of assets.[30]

- Creditor country banking systems incur first-order capital losses on the non-recoverable portion (i.e. loss given default, LGD) of their direct exposures to the defaulted country's banking system.
- Debtor country banking systems withdraw cross-border lending to address their own funding gaps.
- The sudden pullback of funding forces counterparty country banking systems into asset fire sales to restore liquidity positions. The market losses further deplete the capital buffers of the impacted banking systems, leading to additional capital losses.

Contagion cascades through the global interbank network when counterparty country banking systems deplete their capital buffers and default, triggering further rounds of second-order shocks within the network. The simulation terminates when no new defaults materialise. Multiple simulations were conducted, each triggered in turn by the failure of a different country's banking system.

… and conservative adverse assumptions.

The stress simulations adopted conservative assumptions,[31] where country banking systems:

- were only able to recover 50% (i.e. LGD of 50%) of loans extended to any defaulted country;
- face a 50% pullback of funding that had previously been obtained from any defaulted country's banking system; and
- incur significant liquidation and replacement costs in the form of a 50% haircut on their assets under a fire sale.

Results suggest that the impact on Singapore's banking system remains contained for any single-trigger, tail-risk contagion scenario.

This analysis focused on the 20 counterparty countries' banking system defaults that would have the largest capital impact on Singapore's banking system arising only from the cross-border interbank linkages. It does not take into account macrofinancial spillover impact.

Notably, in a tail-risk scenario with the maximum capital loss, the overall CET1 CAR would decline by an additional 220 bps under the Adverse scenario discussed above, and remain above MAS' minimum regulatory requirements. At the same time, the overall LCR would also stay above regulatory thresholds, as the combined liquidity impact from the funding pullback

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30 The methodology takes reference from the network simulations of credit and funding liquidity shocks by Espionosa-Vega, M.A. and Solé, J (2010) "Cross-Border Financial Surveillance: A Network Perspective", *IMF Working Paper* No. 2010/105.

31 The LGD and haircut assumptions are more severe than the parameters prescribed by the corresponding sections under Basel III and MAS Notices.

and fire sale of assets would not exceed 5% of the banking system's total stock of high-quality liquid assets (HQLA).

Sum-up

D-SIBs have maintained strong capital positions amid the current uncertain environment. The IWST 2024 exercise affirmed that D-SIBs have adequate capital buffers to withstand potential downside risks arising from a resurgence in inflationary pressures, a global recession, and renewed tightening in financial conditions.

MAS also conducted a cross-border interbank network analysis to assess first- and second-order contagion impacts on the Singapore banking system. The simulations show that banks have adequate capital buffers to cushion the additional impact arising from key contagion transmission channels (i.e. counterparty country banking system defaults, funding liquidity stresses, and fire sales of assets). MAS will continue to refine its stress testing capabilities to ensure that stress tests remain relevant as a tool for risk assessment and management.

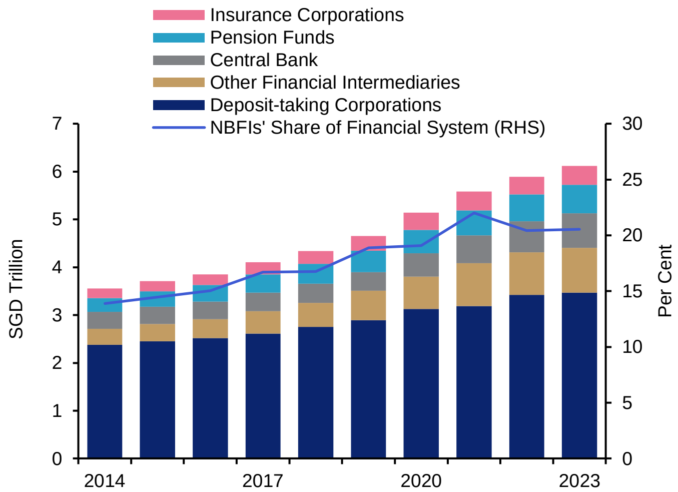
4.2 Non-bank sector

Assets in Singapore's NBFI sector grew by 4% over the past year driven by expansions of insurers and investment funds, amid resilient global economic growth and gains in global capital markets. As a share of the overall Singapore financial system, NBFI assets increased by 0.1 percentage point to reach 20% (**Chart 4.1**). Insurers have fared well and maintained healthy CARs, while investment funds have been able to meet investor redemptions in an orderly manner.

While global markets remain vulnerable to shocks, the NBFI sector in Singapore is not expected to pose systemic risk to Singapore's financial system. Stress tests indicate that insurers can continue to meet regulatory capital and liquidity requirements under an adverse scenario of multiple economic and financial shocks. Similarly, MAS liquidity stress simulations affirm that most investment funds have sufficient liquid assets to meet redemption shocks. Further, the simulations show that in the event of a stress event, potential fire sales conducted by investment funds will not cause significant contagion spillovers to Singapore banks and insurers given the small degree of portfolio overlap. Exposures also appear manageable in the over-the-counter (OTC) derivatives market, where an analysis of crowded trades shows that the majority of such trades tend to be in deep and liquid global markets, limiting contagion risks.

Chart 4.1 NBFI assets grew over the past year, with their share of domestic financial system assets edging up

Assets and NBFIs' share of financial system assets



Source: MAS estimates

Note: "Deposit-taking Corporations" refer to Banks and Finance Companies. "NBFIs" refer to Insurance Corporations and Other Financial Intermediaries.

Investment Funds

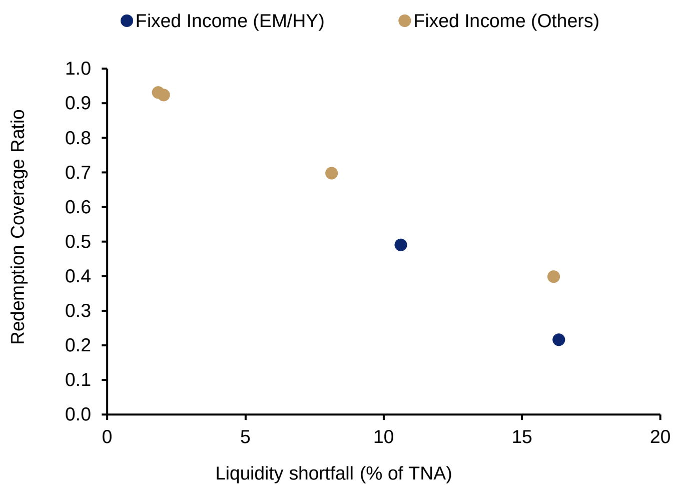
Open-ended funds have managed liquidity risks well. Fund managers indicated that the volume of significant redemptions over the past 12 months has remained stable and similar to past years. Investment funds have been able to meet all significant redemption requests in an orderly manner.

Under a liquidity stress simulation exercise,[32] most funds would have sufficient liquid assets to meet redemption needs under a severe redemption shock. However, a limited number of fixed income funds would face a liquidity shortfall of between 2 and 16% of their total net assets (TNA) under the redemption shocks (**Chart 4.2**). Nevertheless, these funds accounted for a relatively small 7.0% of the TNA of the funds assessed in the exercise.

An additional factor protecting the financial system from contagion is that these funds do not have a large degree of portfolio overlap with banks and insurers, and thus are unlikely to initiate a wave of fire sales. This year's exercise examined common holdings in debt and equity securities between investment funds and Singapore banks and insurers. More details can be found in Box C "Fire Sale Contagion from Investment Funds under Liquidity Stresses".

Chart 4.2 Under a severe stress scenario, some fixed income funds with low redemption coverage ratios may face a liquidity shortfall of up to 16% of TNA

Redemption coverage ratio and liquidity shortfall



Source: MAS estimates

Hedge funds have taken on more gross leverage, although their leverage ratios have remained within global norms. Leveraged Collective Investment Schemes (CIS) managed in Singapore—and especially hedge funds—are subject to regular monitoring of exposures and

32 Funds included in the stress simulation were open-ended funds managed/sub-managed in Singapore, with an investment focus in equity and/or fixed income, and AUM greater than SGD500 million as of end-December 2023.

leverage levels. As of end-2023, the financial leverage for these hedge funds was 0.43 times of their net asset value (NAV) with gross leverage at 8.7 times NAV, within global norms.[33]

Insurers

Resilient global economic growth over the past year has resulted in an increase in new business premiums[34] and growth in asset holdings of insurers in Singapore. The prevailing still-elevated interest rates are also generally favourable to life insurers, given the typically longer duration of their insurance liabilities compared to their assets. Nevertheless, higher interest rates may constrain new business and lead to increased policy redemptions due to heightened competition from alternative investments offering higher returns. Ongoing geopolitical uncertainties could impact global financial markets (for example through falling equity prices and credit spread widening) and insurers' solvency and liquidity positions. Notwithstanding the challenging environment, the CARs of insurers in Singapore remain healthy (see Chart Panel 4F "Insurance Sector", Chart 4F1).

To assess insurers' resilience to unanticipated shocks, MAS conducted an IWST featuring a base scenario and a stress scenario over a three-year horizon. In the stress scenario, which assumed sharp increases in interest rates, falling equity prices and widening credit spreads, insurers were impacted mainly by a drop in asset values arising from the financial shocks. However, in the case of life insurers, higher interest rates also reduced the value of long-term guaranteed liabilities, mitigating some of the negative impact. Insurers were able to continue meeting regulatory capital requirements, in certain cases after undertaking management actions such as reducing bonus allocations and dividend transfers to shareholders, and changing asset allocations.

OTC Derivatives

The total gross notional stock of derivatives outstanding amounted to SGD74.9 trillion as of September 2024 in the Singapore OTC derivatives market, representing an increase of 16% since September 2023. Interest rate (IR) and foreign exchange (FX) OTC derivatives accounted for the bulk of transactions booked and traded in Singapore, constituting 62% and 35% of the total outstanding notional amount respectively (see Chart Panel 4G "Over-the-counter (OTC) Derivatives", Chart 4G1). Foreign banks and dealers were the most prominent counterparties in the market, participating in about 90% of outstanding transactions (Chart Panel 4G, Chart 4G2). The NBFI presence in the OTC derivatives market also increased as a proportion of outstanding transactions, rising from 21% in 2022 to 26% in 2024. The OTC derivatives market comprised relatively few influential counterparties, evidenced by the high centralisation values under a network analysis (Chart Panel 4G, Chart 4G5).

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33 Leverage ratios, derivatives and borrowing exposures are estimated based on funds with NAV greater than or equivalent to USD500 million. This is aligned with reporting thresholds used by the International Organization of Securities Commissions (IOSCO).

34 Refers to weighted new business premiums which measures premiums collected on new policies by taking: (i) 10% of the value of single premium products, (ii) 100% of a year's premiums for annual premium products, and (iii) a pro-rated percentage of premiums for products with premium payment durations of less than ten years.

While synthetic leverage in the OTC derivatives market has risen, most crowded trades were in deep and liquid markets

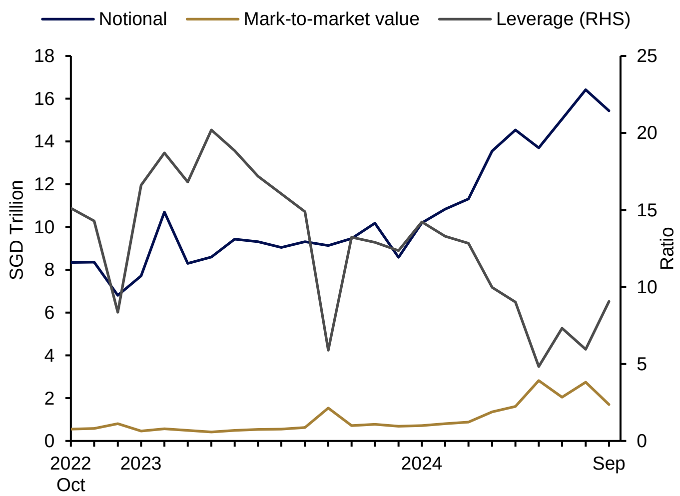
In early August, markets sold off aggressively in reaction to the disappointing US labour market release, following the US Federal Reserve and Bank of Japan policy meetings which were perceived to be hawkish. While the episode was short-lived, the extent of the market movements reflected a heightened sensitivity to macro news, where the unwinding of leveraged trading positions such as FX carry trades, amplified the initial reaction.

In Singapore, synthetic leverage undertaken by NBFIs in the form of OTC derivatives transactions has trended down over the past year but increased slightly after June 2024. Gross notional exposure stood at 9.1 times of gross market value in September 2024 (**Chart 4.3**).

Rising leverage may amplify financial vulnerabilities such as the risk of rapid and correlated trades unwinding during periods of market stress. NBFIs were increasingly crowded in the FX and IR assets, although most of these trades tended to be in deep and liquid markets (see Box D "Crowded Trades and Concentration Risks in the OTC Derivatives Market").

Chart 4.3 NBFIs' leverage and notional exposures increased in recent months

NBFIs' OTC derivatives positions



Source: Depository Trust & Clearing Corporation Data Repository (Singapore) Pte Ltd (DDRS), MAS estimates

Private credit funds have grown rapidly but risk remains limited in Singapore

The assets under management (AUM) of private credit funds in Singapore grew to SGD55 billion or 1.0% of industry AUM as of December 2023, from SGD45 billion the year before. To assess linkages of such funds with the rest of the financial system, MAS conducted a survey of major banks and insurers on their exposures to private credit. It showed that banks and insurers in Singapore have relatively small exposures to private credit and have risk management policies to mitigate risks from these exposures.

Nonetheless, risks to FIs can increase for a variety of reasons, including a higher interest rate environment which would make it challenging for the borrowing firms to service their debts. Vulnerabilities in private credit may also grow as increasing competition and pressure to deploy capital could lead to lower underwriting standards and weaker covenants. MAS will continue to monitor the growth and vulnerabilities arising from private credit to Singapore.

Outlook

Looking ahead, while the domestic NBFI sector continues to grow, it is expected to pose limited systemic impact to the rest of Singapore's financial system. MAS will continue to closely monitor the evolving risks posed by NBFIs.

Box C
Fire Sale Contagion from Investment Funds under Liquidity Stress

This year's liquidity stress simulation for investment funds[35] examined the spillover effects of funds' fire sales[36] on banks and insurers due to their common asset holdings. The analysis referenced a study by Caccioli et al. (2022), which assessed the impact of fire sales on the UK financial system using granular information on portfolio holdings.

Methodology

First, we gauged the extent of portfolio overlap across different types of FIs, adopting the methodology used by Barucca et al. (2021). The **weighted portfolio similarity** between two FIs i and k is calculated as:[37]

$$CosSimilarity_{ik} = \frac{\sum_j w_{ij} w_{kj}}{\sqrt{\sum_j w_{ij}^2} \times \sqrt{\sum_j w_{kj}^2}}$$

where w_{ij} and w_{kj} are the (initial) amounts of asset j held by institutions i and k, respectively. The weighted portfolio similarity is a normalised score ranging between 0 and 1, where 0 indicates no similarity between the portfolios of the FIs and 1 indicates complete similarity.

Next, a redemption shock on investment funds was simulated. Similar to the analysis in the 2023 FSR, redemption shocks were calibrated based on historical fund flows (MAS, 2023). Under a more conservative assumption, assets were liquidated in a pro-rata manner to meet redemptions.

35 The exercise covered authorised and restricted CIS constituted in Singapore, as well as recognised and restricted foreign CIS offered to Singapore investors and managed out of Singapore.

Authorised CIS refer to Singapore-constituted CIS that are authorised pursuant to Section 286 of the Securities and Futures Act 2001 (SFA) and offered to retail investors. Recognised CIS refer to foreign-constituted CIS that are recognised pursuant to Section 287 of the SFA and offered to retail investors. Restricted CIS refer to CIS that have submitted a notification pursuant to Section 305 of the SFA to make offers to accredited investors (or investors with a minimum amount of SGD200,000 per transaction).

36 Fire sales is an indirect channel of contagion where an entity or sector's forced sale of assets at heavily discounted prices can lead to losses across FIs holding the same assets. Fire sale risk can occur, for instance, when open-ended investment funds are forced to liquidate their assets to meet significant liquidity demands arising from large redemptions.

37 To derive a sectoral picture of "similarity", for instance between banks and funds, each bank's portfolio similarity with each fund was first calculated, and the average across these values was then taken.

To capture non-linear effects of fire sales on market prices, the price function from Cont and Wagalath (2016) was adopted. The function is concave—modelling how a fire sale could accelerate—and takes into account market depth:

$$\Psi_j(\Pi_j) = 0.5 \times \left(1 - \exp\left(-\frac{\Pi_j}{0.5 \times \delta_j}\right)\right)$$

where Ψ_j is the percentage decline in the price of asset j, $\Pi_j = \sum_i \pi_{ij}$ is the total amount of asset j sold across all institutions, π_{ij} is the amount of asset j sold by institution i, and δ_j is the market depth[38] of asset j. Peak-to-trough losses were capped at 50%, based on maximum historical stress losses observed in equity markets.

The **aggregate fire sale loss** arising from the redemption shock is a function of the price impact and the remaining amount of assets held by all FIs:

$$R^{firesales} = \sum_i \sum_j (w_{ij} - \pi_{ij}) \times \Psi_j(\Pi_j)$$

Note: For this analysis, $\pi_{ij} = 0$ for banks and insurers since they are not the receivers of the initial shock.

Portfolio composition varied across different sectors

Data on equity and fixed income holdings at the security level, identified by the International Securities Identification Number (ISIN) where available, was used for the analysis.[39] Fixed income products constituted the largest share of banks' and insurers' portfolio holdings (**Table C1**). By contrast, investment funds' holdings were concentrated in equities, although about a quarter of funds by value specialised in fixed income.

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38 Market depth was computed following the methodology used by Cont and Schanning (2017). For more details, see "Box D: Liquidity Stress Simulation of Investment Funds in Singapore" in MAS' 2023 *Financial Stability Review*.

Where the trading volume of an individual security was unavailable, it was proxied by average trading volume of the corresponding asset class for a similar jurisdiction or region, with a similar credit rating. This was then scaled by the outstanding amounts of the security. Daily volatility was estimated using the daily returns of equity and bond market indices.

39 For banks, data was collected from a subset of banks participating in the 2024 IWST, which collectively accounted for 70% of Singapore's banking system debt and equity exposures. For insurers, regulatory data collected under MAS Notice 122 was used. For funds, data was collected through a survey of open-ended investment funds managed/sub-managed in Singapore with an investment focus in equity and/or fixed income, and AUM greater than SGD500 million as of end-December 2023.

Table C1 Singapore banks' and insurers' holdings were largely in debt securities, while most funds were equity funds

Gross mark-to-market (MTM) values in SGD billions, as of end-December 2023

	Equities[a]	Corporate bonds	Sovereign bonds	Funds	Others[b]	Total
Banks	10.0	203.9	311.5	3.3	30.3	559.0
Insurers	33.2	112.8	62.6	52.6	6.4	267.6
Funds	201.4	47.1	28.0	16.3	1.0	293.7
Equity	192.9	-	0.1	8.0	-	200.9
Fixed income	0.1	41.8	23.6	0.8	0.9	67.2
Balanced	8.4	5.3	4.3	7.5	0.1	25.6
Total	244.6	363.8	402.1	72.2	37.7	1,120.3

(a) "Equities" include listed equities, ETFs, REITs and other listed funds.

(b) "Others" include securitised products, structured products, private equity and private credit.

Source: MAS

While larger portfolio similarities were seen within the same sectors, especially for banks and equity funds (**Table C2**), portfolio similarities across different sectors appeared limited. In particular, there was very little similarity (<0.01) between banks' and insurers' portfolios, and those of the investment fund sector. The small area of overlap comprised largely common holdings of Singapore equity and sovereign debt.

Table C2 Portfolio overlap across different sectors was mostly limited

Average weighted portfolio similarity across sectors (normalised scores)

	Banks	Insurers	Funds (All)	Funds (Equity)	Funds (Fixed income)	Funds (Balanced)
Banks	0.16	0.05	<0.01	<0.01	0.02	<0.01
Insurers		0.04	<0.01	<0.01	<0.01	<0.01
Funds (All)			0.06			
Funds (Equity)				0.14	<0.01	0.04
Funds (Fixed income)					0.02	<0.01
Funds (Balanced)						0.02

Source: MAS estimates

Fire sale losses arising from funds' asset sales were relatively small

Under this year's liquidity stress simulation scenario (see Chapter 4 "Singapore Financial Sector", Section 4.2 "Non-bank sector"), 22.5% of funds' total net assets (or SGD68.0 billion) would need to be liquidated by investment funds to meet redemption needs. Since they did not share a large degree of portfolio overlap, the impact of funds' fire sales on banks and insurers was small, compared to the initial redemption shock (**Table C3**). However, the spillover impact on insurers was somewhat larger as insurers held more equities than banks.

Table C3 Resulting losses on banks' and insurers' portfolios was relatively small

Asset sales and fire sale losses in SGD billions, and as a percentage of (initial) total net assets

	Banks	Insurers	Funds
Asset sales (redemption shock)			68.0 (22.5%)
Fire sale losses	2.3 (0.4%)	4.7 (1.8%)	17.9 (5.9%)

Source: Bloomberg, Refinitiv Eikon, MAS estimates

Sum-up

Investment funds in Singapore are unlikely to pose significant contagion risks to Singapore banks and insurers via fire sales in bond and equity markets. This reflects the low degree of portfolio overlap between investment funds, and banks and insurers. MAS will continue to improve its understanding of the linkages between NBFIs and the rest of the financial system, as part of its ongoing commitment to monitoring risks related to the operations of NBFIs in Singapore.

References

Barucca, P, Mahmood, T and Silvestri, L (2021), "Common asset holdings and systemic vulnerability across multiple types of financial institution", *Journal of Financial Stability*, Volume 52, Article 100810.

Caccioli, F, Ferrara, G and Ramadiah, A (2022), "Modelling fire sale contagion across banks and non-banks", *Bank of England (BoE) Staff Working Paper*, No. 878.

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Cont, R and Wagalath, L (2016), "Fire sales forensics: measuring endogenous risk", *Mathematical Finance*, 26(4), 835-866.

Monetary Authority of Singapore (2023), "Box D: Liquidity Stress Simulation of Investment Funds in Singapore", *Financial Stability Review*.

Box D
Crowded Trades and Concentration Risks in the OTC Derivatives Market

Motivation

This box outlines how MAS is augmenting its surveillance toolkit for OTC derivatives markets by monitoring crowded trades and concentrated positions of NBFIs.

Crowded trades occur when a significant number of non-bank market participants take up similar positions in a security, investment theme or strategy. They are of concern because of the potential for procyclical fire sales when a shock causes many participants to unwind their common positions at the same time. This dynamic was evident during the unwinding of "yen carry trade" positions in August 2024, as well as the UK gilts crisis in 2022.[40] The risk posed by a crowded trade tends to rise as the share of the crowded position in market participants' portfolios increases.[41]

Scope

Although most analyses of crowded trades had focused on equities (EQ),[42] the scope of analysis was expanded in this study to include FX and IR derivatives. The latter two are the largest asset classes traded in Singapore (see Chart Panel 4G "Over-the-counter (OTC) Derivatives", Chart 4G1).

For this analysis, the focus was on linear products.[43] An assessment of crowdedness in options and other complex derivatives exhibiting non-linear payoffs requires enhanced valuation and aggregation techniques, which will be addressed in future work. **Table D1** provides a detailed breakdown of the types of trades within the scope of the current monitoring focus.

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40 See Pinter, G, Siriwardane, E and Walker, D (2024), "Fire sales of safe assets", *Bank of England Staff Working Paper* No.1089.

41 See Cont, R and Schaanning, E (2017), "Fire Sales, Indirect Contagion and Systemic Stress Testing", available at SSRN: http://dx.doi.org/10.2139/ssrn.2541114.

42 See Hong Kong Monetary Authority (HKMA)'s crowded trades indicator by Cheng, K, Liu, Z, Silvia, P and Yu, L (2023), "Building an integrated surveillance framework for highly leveraged NBFIs – lessons from the HKMA", *BIS Papers* No. 137.

43 Linear products provide a linear payoff, whereby a rise or decline in the value of the underlying asset yields a corresponding change in returns. Examples of linear products include futures, forwards and swaps.

Table D1 The study included linear FX and IR products, as well as EQ swaps and CFDs

Scope of study

Asset class	Underlying asset	Crowded trades scope	Coverage by the DDRS
FX	Currency pairs	Forwards and FX swaps	74% of total gross notional outstanding FX derivatives
IR	Currencies (or corresponding benchmarks)	Single-currency fixed-float interest rate swaps (IRS) and overnight-index swaps (OIS)	80% of total gross notional outstanding IR derivatives
EQ	• Individual securities • Investment themes derived by categorising individual securities into geographies and sectors	Single name equity swaps and contract for differences (CFDs)	31% of total gross notional outstanding EQ derivatives

Source: DDRS, MAS estimates

Methodology

There are typically two aspects to assessing crowdedness—when many entities are exposed to a common security or strategy; and when aggregated positions are too large to be liquidated.[44]

To gauge the level of crowdedness in linear FX, IR and EQ products traded or booked in Singapore, a candidate list of securities forming a significant share of any NBFI's portfolio was created. The share of underlying assets in each NBFI's portfolio was calculated based on the absolute values of net notional outstanding positions for the FX[45] and IR asset classes, and gross notional outstanding positions for EQ.

We then applied the following conditions to obtain a list of trades that were assessed to be susceptible to crowding. If any of the three following conditions applied, a trade from the candidate list was deemed crowded:

- Interconnectedness: A trade was deemed crowded when it formed a significant share of many NBFIs' portfolios. By considering both concentration and commonality of exposures of NBFIs, this condition reflects the extent and severity of impact during sharp changes to asset prices.

44 See Brown, G, Howard, P and Lundblad, C (2021), "Crowded Trades and Tail Risk", *The Review of Financial Studies*, Vol.35(7), pp. 3231–3271, and Chincarini , L, Lazo-Paz, R and Moneta, F (2024), "Crowded Spaces and Anomalies", available at SSRN: http://dx.doi.org/10.2139/ssrn.4618248.

45 For FX swaps, the far leg was used to determine its net position.

- One-sidedness: Crowded trades were also identified based on the size of NBFIs' overall net positions.[46] This condition considers that NBFIs crowding on the same side of a trade may further exacerbate any impact from asset devaluation (i.e. one-sided markets).

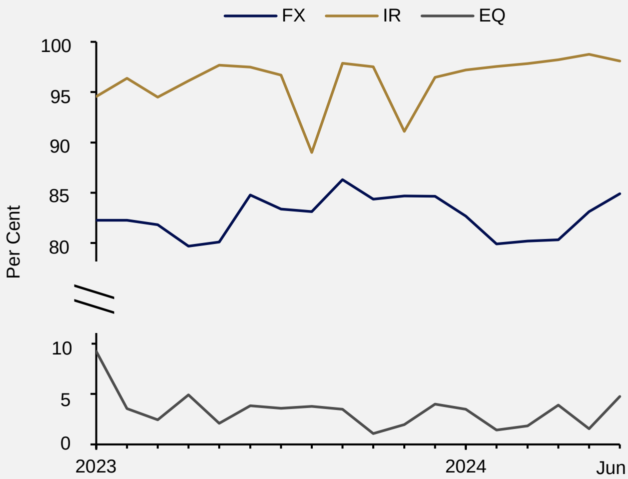
- Relative size and liquidity: Crowdedness was proxied by the number of days required for NBFIs to collectively and fully exit their outstanding net positions given the average market daily trading volume (ADV).[47] This condition compares the size of NBFIs' holdings to the volume and liquidity of the market.

NBFIs' portfolios became more crowded in FX and IR derivatives

Crowded trades have constituted a higher share of NBFIs' portfolios in FX and IR since January 2023 (**Chart D1**). As crowded trades comprised nearly all of NBFIs' IR exposures, these institutions appear to have largely similar portfolios in interest rates. Conversely, the share of crowded trades in NBFIs' equity exposures was much lower, which suggests a higher degree of variation among NBFIs' equity portfolios.

Chart D1 The shares of crowded trades increased slightly over the past year

NBFIs' share of portfolio in crowded trades by asset class



Source: DDRS, MAS estimates

NBFIs crowded more into liquid trades in FX and IR, and less in EQ

Within FX and IR, NBFIs mostly crowded into trades that have deep and liquid global markets (**Charts D2 and D3**). The concentration of such trades indicates that global shocks

[46] The one-sided condition was not applied to the EQ asset class due to insufficient information to determine the direction of EQ contracts.

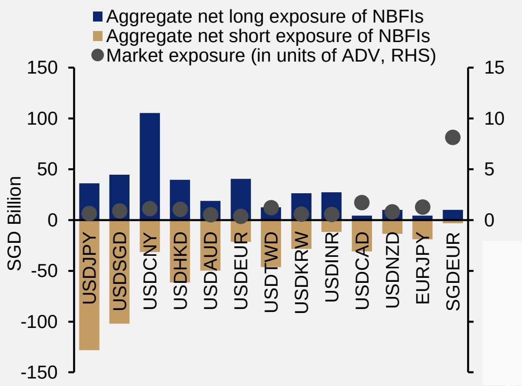
[47] The ADVs of FX and IR currencies in the Singapore market were proxied by the average of daily new trades reported to DDRS. Commercial data was used to determine the ADV of the underlying EQ single name securities to reflect leverage providers' ability to liquidate their positions.

can exert a significant impact on Singapore NBFIs. Conversely, shocks originating from NBFIs in Singapore are less likely to move these global markets.

Crowded trades identified for poor liquidity were typically traded by fewer NBFIs (**Chart D3**). However, many NBFIs shared common exposures to slightly longer-dated interest rate swaps in Asian currencies. Scarce liquidity in these products may lead to large negative returns if, for example, NBFIs attempt to exit their positions when there is a sudden unexpected increase in the underlying interest rate.

Chart D2 Crowded FX trades were mostly in deep and liquid markets

Crowded FX currency pairs as of 30 June 2024

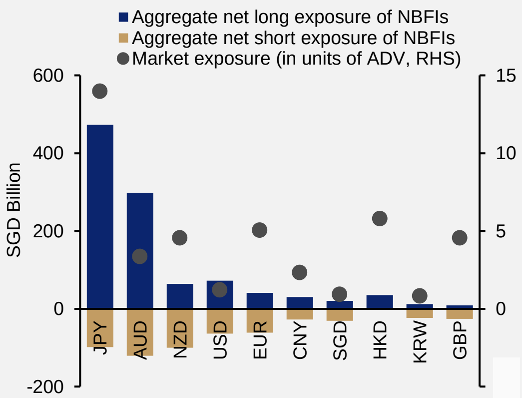


Source: DDRS, MAS estimates

Note: An entity that is long (short) a currency pair (USDJPY) will buy (sell) the base currency (USD) and sell (buy) the quote currency (JPY) forward. Each NBFI's net positions are aggregated to arrive at an overall net long/short exposure. Market exposure is defined as the maximum of the absolute values of aggregate net long exposure and aggregate net short exposure across all NBFIs, divided by ADV to approximate the likelihood of fire sales should positions be unwound.

Chart D3 Larger IR trades tended to be in more liquid markets

Crowded IRS as of 30 June 2024



Source: DDRS, MAS estimates

Note: An entity that is long (short) a fixed-float IRS is a net receiver (payer) of the floating rate. Each NBFI's net positions are aggregated to arrive at an overall net long/short exposure. Market exposure is defined as the maximum of the absolute values of aggregate net long exposure and aggregate net short exposure across all NBFIs, divided by ADV to approximate the likelihood of fire sales should positions be unwound.

As previously indicated, NBFIs have been generally diversified in equities, with little sign of crowding in individual securities. However, there were some pockets of crowdedness. For example, many NBFIs held significant positions in the Taiwanese information technology sector and in the Asian industrials sector as of end-June 2024.

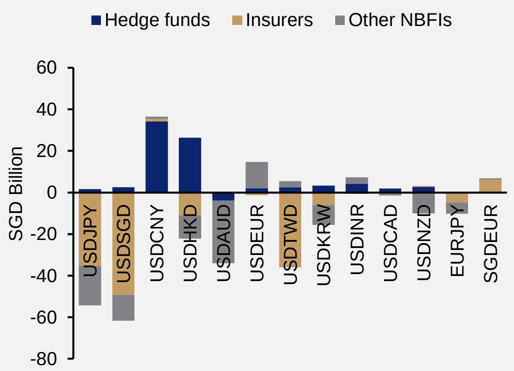
Hedge funds and insurers were the main contributors to crowdedness

As leveraged investors, hedge funds are particularly likely to take on larger (and, in some cases, shorter-term) one-sided bets. For example, hedge funds were net long USD against both the CNY and HKD in June 2024 (**Chart D4**), while in the IR market they were long on JPY and AUD interest rate products (**Chart D5**).

By contrast, insurers are more likely to enter into derivative contracts to hedge their investments. This can be seen from the net long positions in the JPY, SGD and TWD held by Japanese, domestic and Taiwanese insurers respectively (**Chart D4**). Although domestic insurers were also net short SGD IRS, these exposures were small as the insurers do not offer defined benefit schemes akin to the UK pension funds.

Chart D4 Insurers crowded into specific currencies to meet hedging needs

Net notional outstanding of crowded FX currency pairs by NBFI types as of June 2024

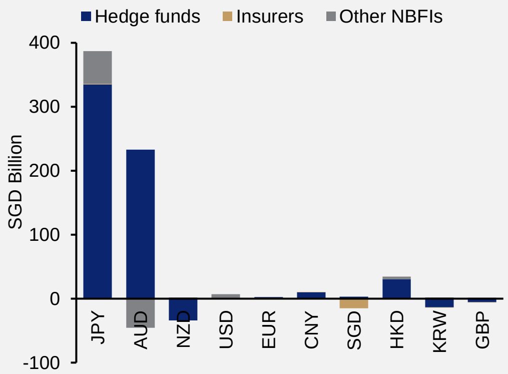


Source: DDRS, MAS estimates

Note: Positive values refer to long positions. A sector that is net long (short) a currency pair (USDJPY) will buy (sell) the base currency (USD) and sell (buy) the quote currency (JPY) forward. Outstanding notional amounts are netted across each NBFI sub-sector, e.g. the hedge fund sub-sector is net long USDCNY.

Chart D5 Hedge funds took on large one-sided positions as directional bets

Net notional outstanding of crowded IRS by NBFI types as of June 2024



Source: DDRS, MAS estimates

Note: Positive values refer to long positions. A sector that is long (short) a fixed-float IRS is a net receiver (payer) of the floating rate. Outstanding notional amounts are netted across each NBFI sub-sector, e.g. the hedge fund sub-sector is net long JPY trades.

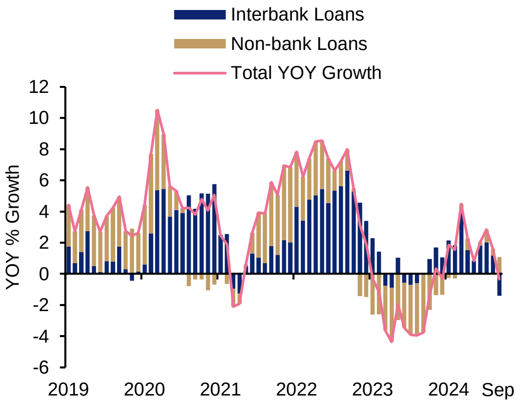
Sum-up

Detection of crowded trades is an important addition to the surveillance toolkit in the ongoing monitoring of Singapore's OTC derivatives markets. The analysis of NBFIs' portfolio concentration in crowded trades can also be integrated as part of entity-based monitoring, to inform further investigation and supervisory follow-up where necessary. MAS will continue to refine the methodology and enhance its surveillance capabilities in monitoring concentration risks that may arise.

Chart Panel 4A Banking Sector: Credit Growth Trends

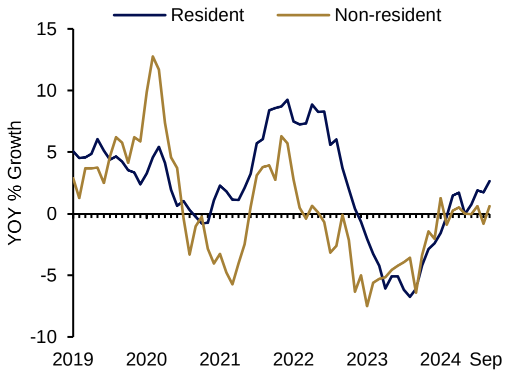
Overall credit was supported by non-bank lending, which has been expanding since March 2024.

Chart 4A1 Overall loan growth



Source: MAS

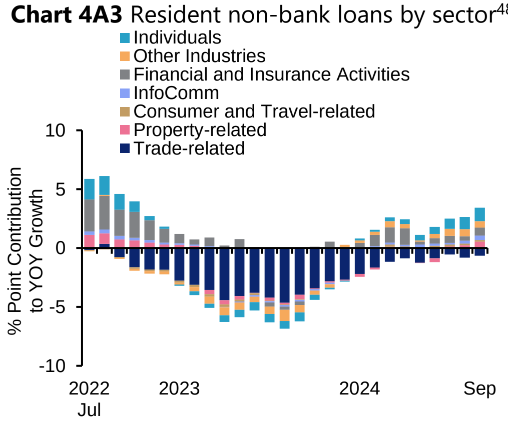
Chart 4A2 Non-bank loan growth by residency



Source: MAS

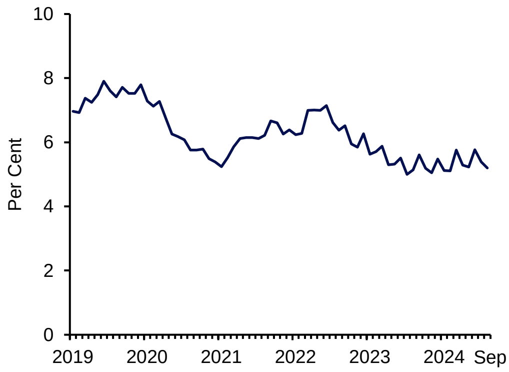
Resident non-bank credit growth improved on a y-o-y basis across most sectors.

Chart 4A3 Resident non-bank loans by sector[48]



Source: MAS

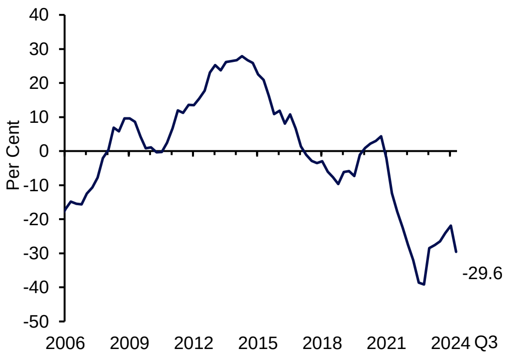
Chart 4A4 Share of trade financing in non-bank loans



Source: MAS

The credit-to-GDP gap for Singapore remained negative at −29.6% as of Q3 2024.

Chart 4A5 Credit-to-GDP gap



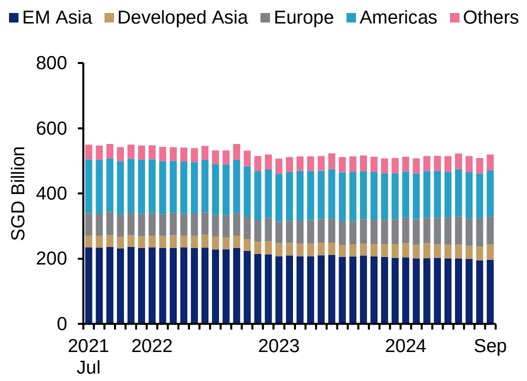
Source: MAS

48 Consumer and Travel-related refers to Retail Trade and Accommodation and Food Services Activities. Property-related refers to Construction and Property and Development of Land. Trade-related refers to Manufacturing, Wholesale Trade, and Transport and Storage.

Chart Panel 4B Banking Sector: Cross-border Lending Trends[49]

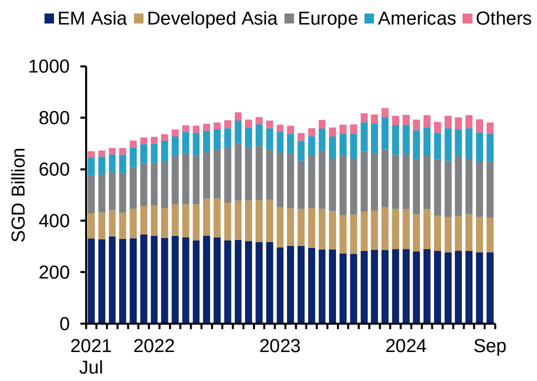
While non-bank loans to EM Asia and Americas declined slightly on a y-o-y basis in September 2024, EM Asia continues to account for the largest share of cross-border non-bank loans.

Chart 4B1 Cross-border non-bank loans by region



Source: MAS

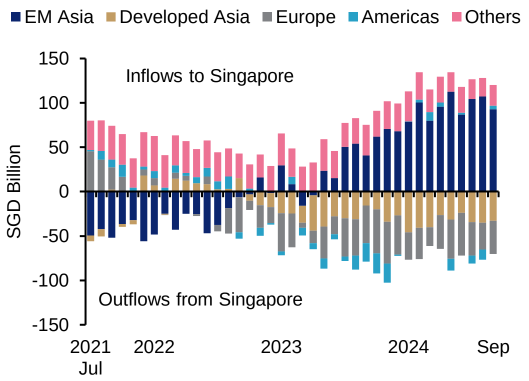
Chart 4B2 Cross-border interbank loans by region



Source: MAS

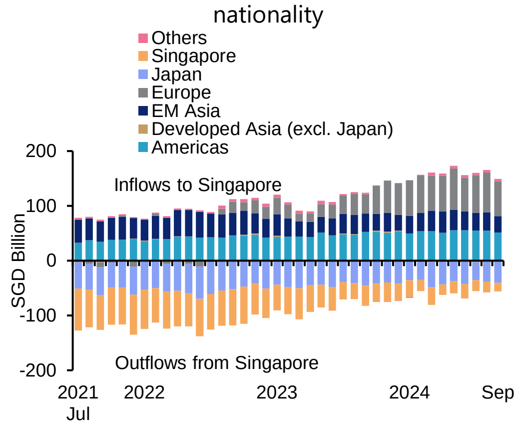
Banks in Singapore have been net providers of credit to Developed Asia and Europe since end-2022. Local and Japanese banks have continued lending to EM Asia.

Chart 4B3 Net lending positions by region



Source: MAS

Chart 4B4 Net lending positions to EM Asia by bank nationality



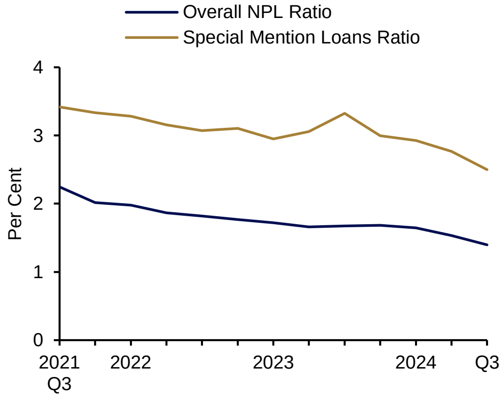
Source: MAS

49 Under the new MAS Notice 610/1003 reporting requirements implemented in July 2021, some data are not comparable with historical periods (pre-July 2021) due to a change in data definitions. Please refer to the Data Annex in MAS' 2021 *Financial Stability Review* for more details.

Chart Panel 4C Banking Sector: Asset Quality and Liquidity Indicators

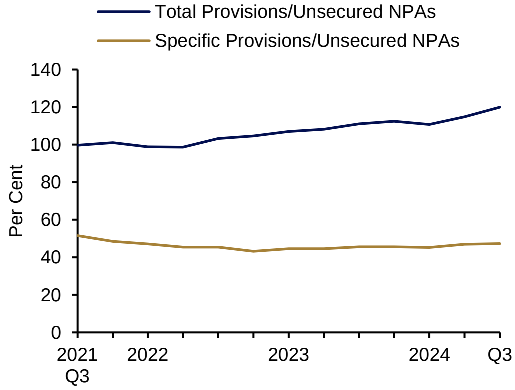
The overall NPL ratio improved to 1.4% as of Q3 2024. Total provisioning coverage increased to 119.9% as of Q3 2024.

Chart 4C1 Banking system's asset quality



Source: MAS

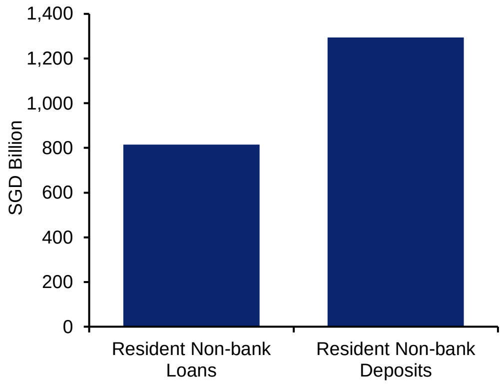
Chart 4C2 Banking system's provisioning coverage



Source: MAS

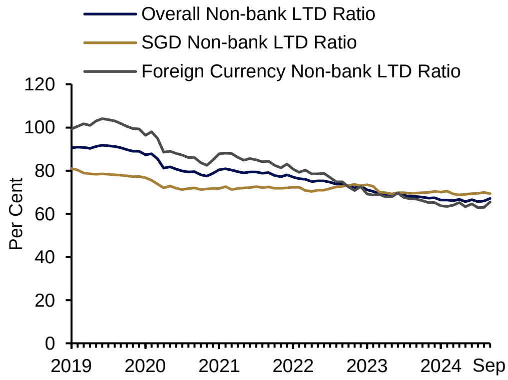
Banks' liquidity positions are strong. Resident deposits are more than sufficient to fund resident loans. All LTD ratios remained below 100% as of September 2024.

Chart 4C3 Banking system's resident non-bank loans and deposits (as of September 2024)



Source: MAS

Chart 4C4 Banking system's non-bank LTD ratios

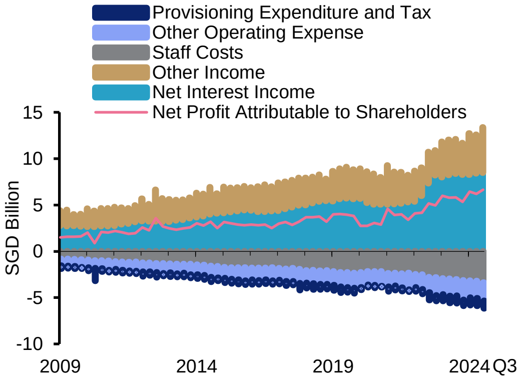


Source: MAS

Chart Panel 4D Banking Sector: Local Banking Groups

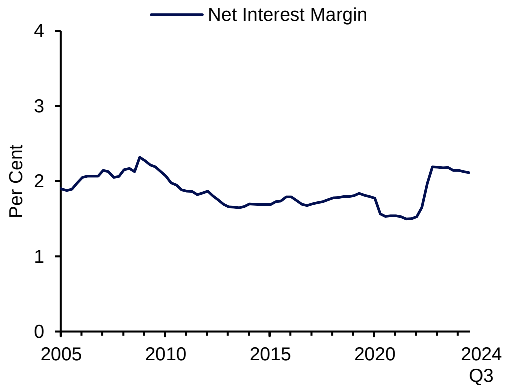
Local banking groups' net profits increased as of Q3 2024, supported by growth in net interest income and non-interest income. NIMs have stabilised at 2.1% as of Q3 2024.

Chart 4D1 Local banking groups' profit components



Source: Local banking groups' financial statements, MAS

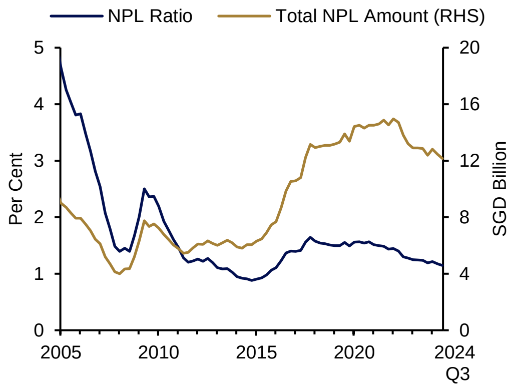
Chart 4D2 Local banking groups' NIMs



Source: Local banking groups' financial statements, MAS

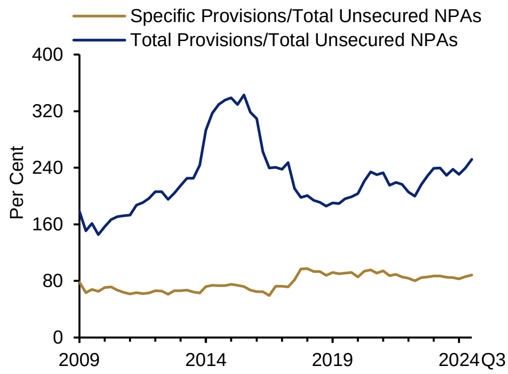
Asset quality remained stable, with a slight improvement in NPL ratio to 1.14% as of Q3 2024. Local banking groups maintained a healthy total provisioning coverage of 252% as of Q3 2024.

Chart 4D3 Local banking groups' NPLs



Source: Local banking groups' financial statements, MAS

Chart 4D4 Local banking groups' provisioning coverage

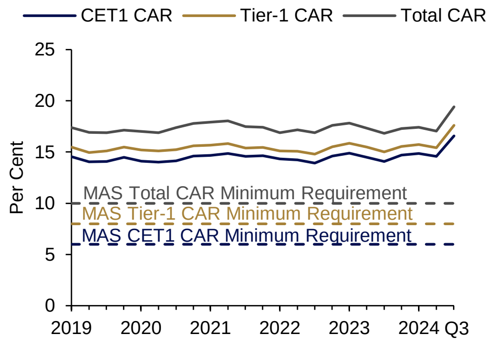


Source: Local banking groups' financial statements, MAS

Chart Panel 4E Banking Sector: Domestic Systemically Important Banks

D-SIBs maintained robust capital positions, with aggregate CARs well above regulatory requirements.

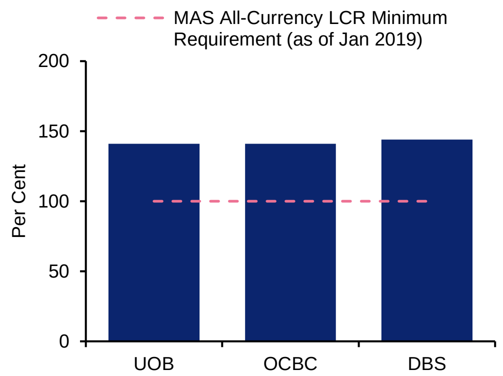
Chart 4E1 D-SIBs' CARs[50]



Source: D-SIBs' financial statements, MAS

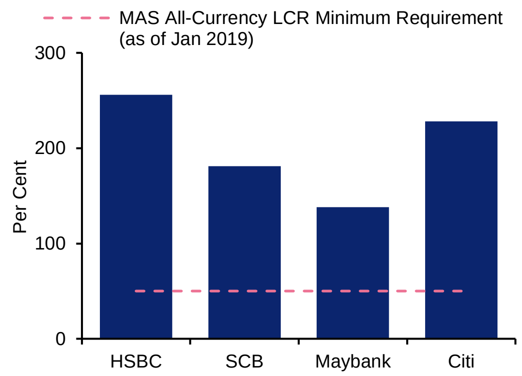
D-SIBs maintained strong liquidity positions that are well above regulatory requirements.

Chart 4E2 Local banking groups' all-currency LCRs (Q3 2024)



Source: Local banking groups' financial statements, MAS

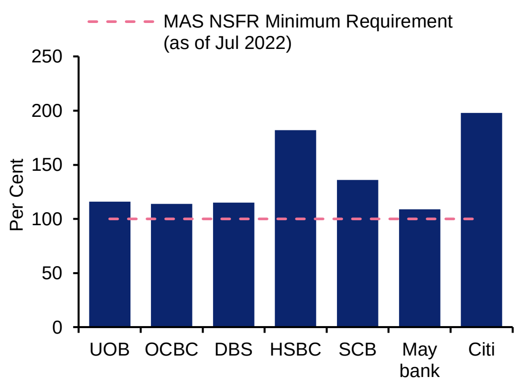
Chart 4E3 Other D-SIBs' all-currency LCRs (Q3 2024)



Source: Other D-SIBs' financial statements, MAS

D-SIBs' net stable funding ratios have exceeded the MAS regulatory minimum.

Chart 4E4 D-SIBs' NSFRs (as of Q3 2024)



Source: D-SIBs' financial statements, MAS

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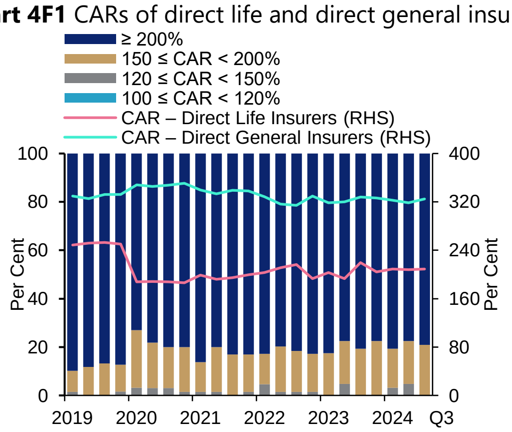
50 See footnote 26 in Chapter 4 "Singapore Financial Sector".

Chart Panel 4F Insurance Sector

The insurance industry in Singapore remains well-capitalised. The average CAR for direct life and general insurance firms is well above regulatory requirements.[51]

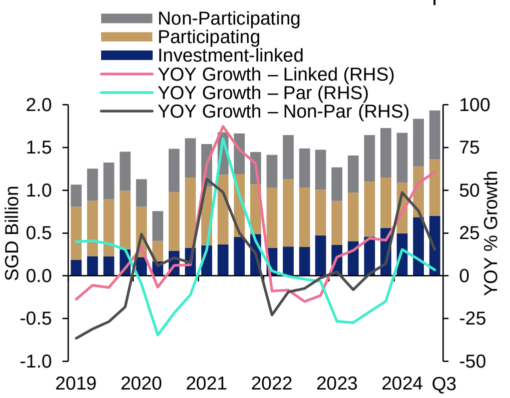
Chart 4F1 CARs of direct life and direct general insurers



Source: MAS estimates

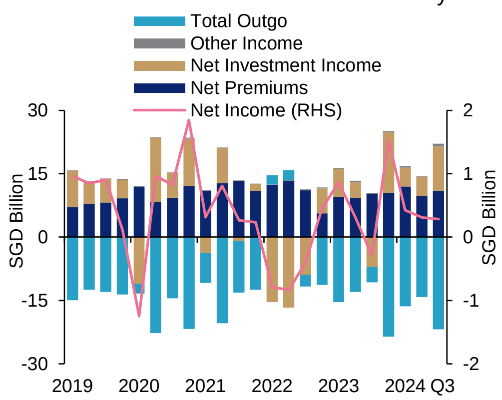
New business premiums of the direct life insurance industry increased across all product types. The industry reported an increase in net income due to unrealised investment gains.

Chart 4F2 Direct life insurers: New business premiums



Source: MAS estimates

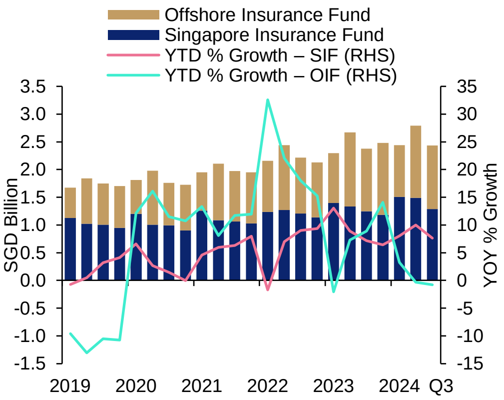
Chart 4F3 Direct life insurers' net income by source



Source: MAS estimates

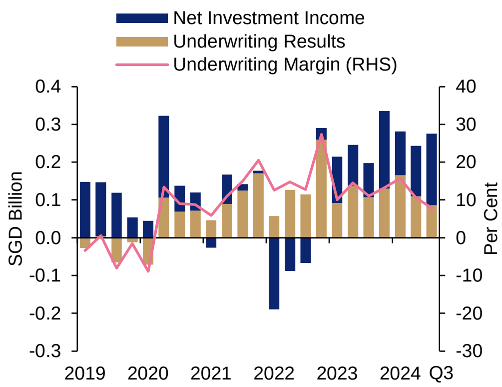
Gross premiums of the direct general insurance industry increased, primarily driven by growth in Singapore Insurance Fund business. The industry also reported positive underwriting and unrealised investment gains in 2024.

Chart 4F4 Direct general insurers: Gross premiums



Source: MAS estimates

Chart 4F5 Direct general insurers: Operating results



Source: MAS estimates

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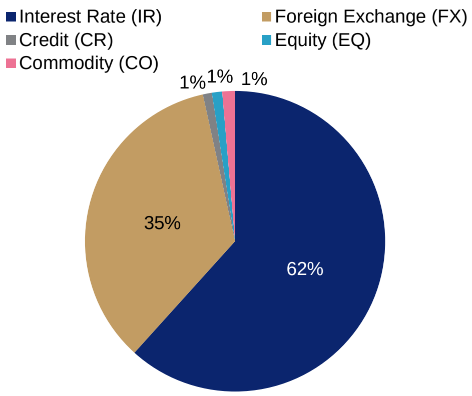
51 With effect from Q1 2020, the valuation and capital framework for insurers in Singapore has been enhanced (revised Risk Based Capital framework, RBC 2). The CAR of the two regimes are thus not directly comparable.

Chart Panel 4G Over-the-counter (OTC) Derivatives

IR and FX formed the largest asset classes in the OTC derivatives market, based on outstanding notional amounts. Foreign banks and dealers remained key participants in the market.

Chart 4G1 OTC derivatives market in Singapore by asset class and notional outstanding (end-Sep 2024)



Source: DDRS, MAS estimates

Chart 4G2 Cross-sectoral breakdown of notional outstanding (end-Sep 2024) – All asset classes

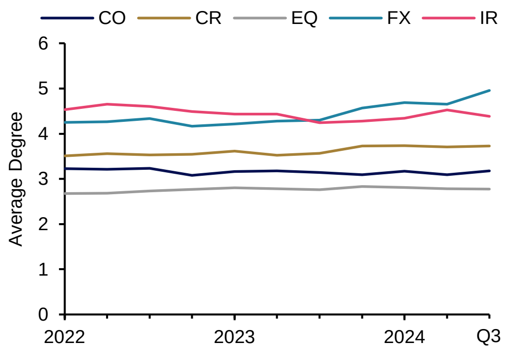
%	LCL BK	FGN B/D	NBFI	C/O	CCP	INDIV
LCL BK	0.4	3.5	0.4	0.2	2.4	0.3
FGN B/D		35.7	22.0	1.3	28.4	1.2
NBFI			0.8	0.1	2.4	<0.1
C/O				0.8	<0.1	

Source: DDRS, MAS estimates

Note: LCL BK = Locally incorporated bank; FGN B/D = Foreign bank or dealer; C/O = corporate or other non-bank entity

Average degree per counterparty for FX increased over the past year. Concentration amongst counterparties fell for CR trades, while that for EQ and IR trades saw a small uptick.

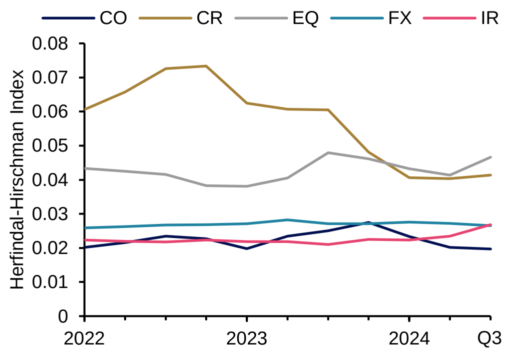
Chart 4G3 Average degree



Source: DDRS, MAS estimates

Note: Degree refers to the number of connections a counterparty has. Connections are counted based on outstanding positions.

Chart 4G4 Herfindahl-Hirschman Index (HHI)

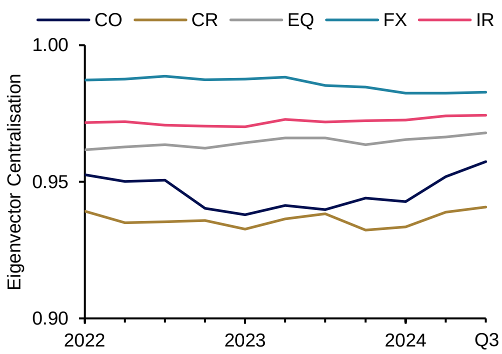


Source: DDRS, MAS estimates

Note: HHI is calculated based on the sum of the squares of the proportion of notional outstanding of each counterparty. Intragroup trades and trades with CCPs have been excluded.

High centralisation values show that the OTC derivatives network in Singapore has comprised relatively few influential counterparties.

Chart 4G5 Eigenvector centralisation



Source: DDRS, MAS estimates

Special Features on Financial Stability

Special Feature 1

Asian Capital Flows and the Global Financial Cycle

Executive Summary

Emerging economies in Asia have taken steps to reduce their vulnerability to shifts in capital flows and avoid the negative effects of externally driven boom-and-bust cycles. While acknowledging the influence of a "global financial cycle" that can limit monetary policy effectiveness, most attempt to operate within the framework of the traditional international macroeconomic policy trilemma by combining open capital accounts with complementary measures to preserve policy autonomy. Flexible exchange rate regimes—now employed by many Asian central banks—should allow economies to adjust to external shocks while setting interest rates based on domestic conditions. Where currency flexibility alone cannot fully defray shocks, strategic currency intervention can ease difficult policy trade-offs. Macroprudential measures (MPMs)—implemented with increasing frequency over the past 15 years—can help discourage buildups of potential flight capital.

To assess if these measures have worked, this Special Feature undertakes an empirical study of the impact of external shocks, external policy changes, and the global financial cycle (GFCy) on capital flows to emerging market economies in Asia over the past 30 years. It finds a statistically significant decline in the impact of the GFCy on capital flows to Asia since the 2008–10 Global Financial Crisis (GFC). However, while capital flows have become less responsive to the GFCy, the impact of the GFCy and G7 monetary policy on domestic prices and activity, including interest rates, economic activity, and even policy rates, remains large and undiminished by most measures. This leads to the conjecture that greater attention by policymakers to traditional capital flows—including through the use of MPMs and capital flow management measures (CFMs)—may have pushed cross-border financial activity into other instruments such as derivatives. Further, trade flow linkages may have strengthened during a period of increased common global shocks. At the same time, evidence of continued dependence of domestic policy rates on the GFCy may indicate that policymakers have become more anticipatory of cross-border interest rate differentials.

Introduction

Asian emerging economies have weathered the recent round of post-pandemic interest rate hikes surprisingly well, avoiding the capital surges and sudden stops that characterised previous external monetary shocks. During the 1997–99 Asian Financial Crisis (AFC), the 2008–10 GFC, and, to a lesser extent, the Taper Tantrum episode of 2013, capital outflows aggravated deteriorating financial conditions in many emerging market economies, often forcing policymakers into adopting procyclical policies. The frequency and severity of such episodes lent support to the thesis associated with Hélène Rey, that the well-known monetary trilemma of international economics has collapsed into a dilemma. To the extent that this is the case, countries would have to choose between having a closed capital account or being at the mercy of the GFCy, and domestic monetary policy would be ineffective in the face of global capital flows. But the recent experience in Asia and elsewhere suggests that either domestic policy is more effective than previously thought, or that conditions have changed.

To test these ideas, the present study explores the influence of global monetary conditions or a GFCy on a subset of Asian economies (subsequently referred to as AEE),[52] looking first at the impact on capital flows directly, and then analysing the exposure of domestic prices and the business cycle to global factors. Employing standard statistical methods from the literature, it investigates whether the causal relationship has changed over time. The study concludes with a discussion of possible explanations for these developments.

Outline

The paper begins with a literature survey describing recent research on capital flows and the influence of the GFCy on emerging market economies. This is followed by a set of stylised facts on capital flows, exchange rates and interest rates of countries in the region.

The initial empirical section examines the determinants of capital flows to AEE in light of recent research. Following the literature, global factors are obtained through the use of factor analysis of an array of macroeconomic and financial variables pertaining to the US and other advanced economies. A series of regressions under alternate specifications is then used to determine the extent to which global factors can explain changes in capital flows—defined either as net flows on the financial account ex foreign direct investments (FDI) and reserves, or as net acquisition by foreigners of domestic debt instruments (bonds and bank deposits)—and how this relationship has changed over the past thirty years.

The second empirical section extends this approach to domestic monetary and financial conditions. Dependent variables include domestic policy rates, short-term (90-day) interest rates and proxies for the business cycle. As with capital flows, these[53] are regressed on various measures of advanced economy policy stances and the GFCy to determine how correlations have changed over time.

Possible explanations for patterns identified in the empirical work are then considered, including the role played by increased currency flexibility, MPMs, increased savings (fiscal and external) surpluses, and strategic intervention by national authorities. The increased use of unconventional tools, including spot and forward intervention, CFMs and MPMs, is also discussed.

52 The economies studied are Indonesia, Malaysia, the Philippines, South Korea, and Thailand—the countries most affected during the 1997–99 Asian financial crisis.

53 Dependent variables with a unit root—notably interest rates—are analysed in differenced form.

A concluding section suggests possible policy measures to dampen the impact of the GFCy on AEE economies.

Literature Survey

Several recent studies have explored the impact of GFCys on capital flows and domestic variables. A notable point of reference is Hélène Rey's frequently cited 2014 Mundell-Fleming lecture and paper (2016), which highlights the power of the GFCy and the limits it imposes on the ability of policymakers in noncore countries to influence domestic monetary and credit conditions. Obstfeld (2015) accepts Rey's general thesis that global market conditions and US Federal Reserve policy decisions affect financial and credit conditions across borders, but argues that this does not invalidate the policy trilemma. Focusing on capital flows—specifically fixed income flows into bank deposits and bonds—rather than price changes, Forbes and Warnock (2020) conclude that the capital flow waves seen in the 1980's and 1990's have in recent years diminished to ripples.

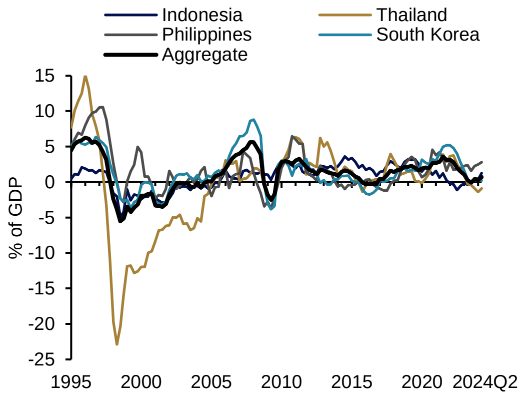
Employing a conventional panel regression on widely cited indicators of the global cycle, including the VIX volatility index and US policy rate, Cerutti et al. (2019) find that the GFCy explains less than 10 percent of capital flow variation (R^2 less than 0.1). In a subsequent study, Cerutti and Claessens (2024) find that the GFCy explains a much higher share of interest rate variation than it does the variation in capital flows. Most recently, noting that EMEs have been quite resilient to the recent round of central bank tightening, the BIS (2024) makes the case that susceptibility to external shocks and correlations with the GFCy have diminished since the GFC.

External Account Trends

By some measures, capital flows to emerging economies in Asia have moderated since the GFC of 2008–10. Notably, net and gross inflows as a share of GDP have diminished, as have gross inflows to debt instruments (bonds and bank deposits) (**Chart S1.1**). In addition, the variance of capital flows into fixed income instruments, including bank deposits and portfolio debt (both domestic and international bonds), has declined by more than half as a share of GDP over the past decade (**Chart S1.2**).

Chart S1.1 Capital flows have diminished in magnitude...

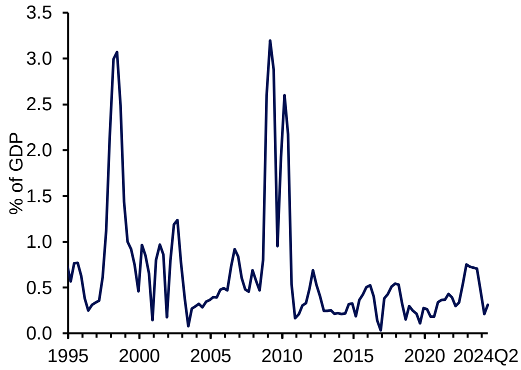
Debt flows as a share of GDP



Source: IMF, Haver Analytics, MAS estimates

Chart S1.2 ... and volatility has declined

Volatility of aggregate debt flows

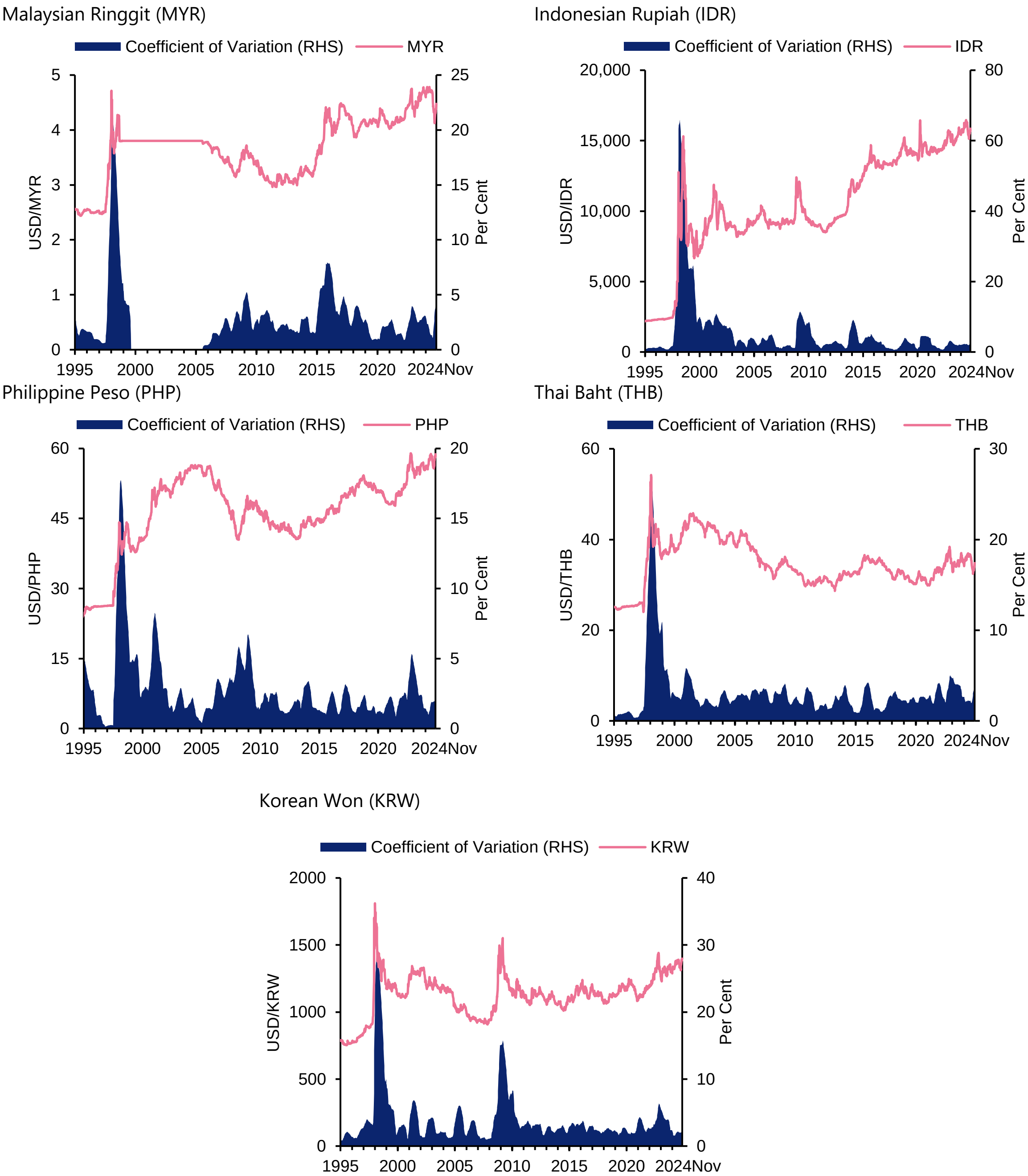


Source: IMF, Haver Analytics, MAS estimates

Note: Volatility is measured as the standard deviation of the quarterly flows as a percentage of GDP using 4-quarter rolling windows.

Other immediate comparisons are less suggestive of a structural change. Nominal exchange rate volatility has not shifted decisively, even as many countries have moved to adopt flexible exchange rate inflation targeting regimes. Some currencies (MYR, IDR, PHP) have registered trend nominal declines against the USD while others (THB, KRW) have fluctuated around a stationary mean (**Chart S1.3**). Since 2000, these five currencies have depreciated by an average of 1.5% annually against the USD, while annual inflation has averaged 3.3% in the five economies and 2.6% in the US.

Chart S1.3 Exchange Rates against the USD and Exchange Rate Volatilities



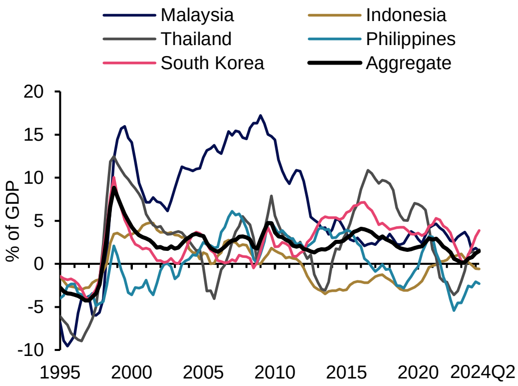
Source: Bloomberg, MAS estimates

Note: Exchange rate volatility is measured as the coefficient of variation (defined as standard deviation over the mean) for the weekly bilateral exchange rate against the USD using 52-week rolling windows.

Current accounts strengthened after the AFC, rising from average deficits of 4% of GDP before the crisis to surpluses of 2% by 2005, but since 2022, average surpluses have slipped to 1% (**Chart S1.4**). During this period, the average share of government bonds held by foreigners has declined by more than half from a high of 59% in 1997 to 23% in 2023 (**Chart S1.5**), reflecting the reduced rate of inflows into interest-bearing instruments.

Chart S1.4 Current accounts have generally been in surplus since the AFC

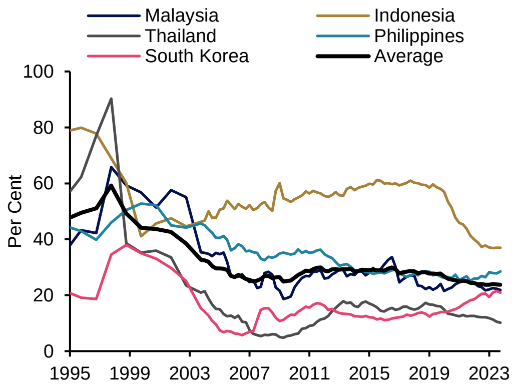
Current account balance as a share of GDP



Source: IMF, Haver Analytics, MAS estimates

Chart S1.5 The share of foreign investors in government debt markets has fallen

Share of general government gross debt held by foreign investors



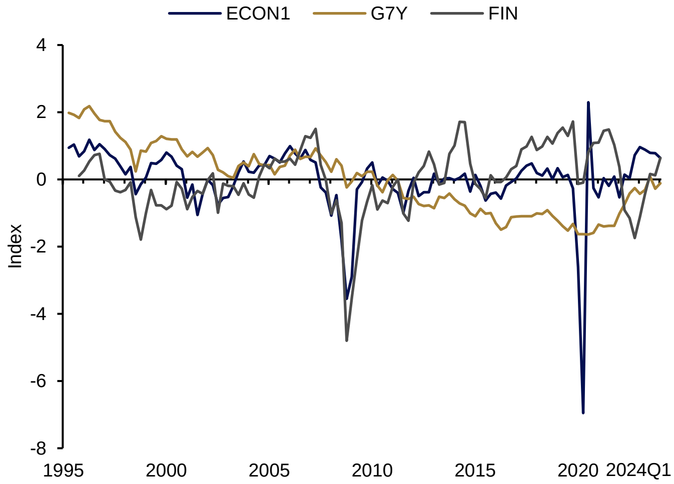
Source: IMF, Haver Analytics, MAS estimates

Estimating the Global Financial Cycle

Following Mirinda-Agrippino and Rey, Obstfeld and others, the global cycle is proxied by extracting the first two principal components (ECON1 and ECON2) from a vector of 25 cyclical economic and policy variables, including quarterly GDP, M2, real policy rates, and real exchange rates in core economies (US, Europe, Japan). An additional factor (FIN) is computed as the first principal component of a vector of 14 financial variables, including the VIX index of implied US equity volatility, the MOVE implied bond volatility index, and several gauges of realised volatilities and credit spreads. Factors for medium- and long-term yields in G7 economies (G7Y) and for a weighted combination of yields across the U.S. yield curve (YIELD) are also included. All data are quarterly, covering the period 1995–2024, and each variable is normalised to zero mean and a standard deviation of one. As shown in **Chart S1.6**, ECON1 tracks global output growth, falling sharply with recessions in 2008 and 2020. FIN is calculated as the inverse of the principal component of implied volatilities; it is thus negatively correlated with the VIX index and may be interpreted as an index of risk appetite. G7Y is effectively a weighted average of long-term yields in G7 countries.

Reflecting the downward trend in long-term yields prior to the COVID-19 pandemic, the hypothesis that YIELD, G7Y, and some domestic interest rate series have unit roots cannot be rejected. Accordingly, regressions involving these variables are performed in both levels and first differences; results from both are reported when informative. ECON1, ECON2, and FIN are stationary (I(0)), as are capital flows.

Chart S1.6 Global Financial Cycle Principal Components



Source: MAS estimates

Testing for the External Drivers of Asian Capital Flows

In the first battery of regressions, capital flows[54] to major Asian emerging market economies[55] are regressed on contemporaneous values of ECON1, ECON2, FIN, YIELD and G7Y using ordinary least squares (OLS), following the approach taken in other studies of the GFCy. For the sake of robustness, two types of capital flows are measured: the overall capital/financial account excluding change in reserves, and inward debt flows computed as changes in banking and portfolio debt liabilities, with the latter being the measure employed in the Forbes/Warnock and Cerutti et al studies. Capital flows are derived from quarterly balance of payments accounts from the IMF. Regressions for country i have the form:

$$CF_{i,t} = \alpha + \beta_{i,1}F_{1,t} + \cdots + \beta_{i,k}F_{k,t} + \epsilon_t$$

where $CF_{i,t}$ denotes capital flows (either definition) to country i in period t, and $F_{j,t}$ is the value of one of the global factors (e.g., ECON1, ECON2, etc) in period t. In some cases, lagged values of global factors were included, although this made little difference to the results. The focus, and the primary measure of the dependence of a variable on the GFCy, is the share of variance explained by the independent variables—that is, R^2 in the case of OLS. This approach is selected over the alternative, employed by several authors, of regressing capital flows on a set of global and domestic variables and basing the analysis on the estimated coefficient and associated t-statistic for the global variable(s) (e.g. ECON 1). A limitation of the standard approach, as distinguished from that used here, is that t-statistics do not show the share of variation explained by global factors (which may be quite low). Moreover, an increase in the coefficient or t-statistic associated with a specific variable or variables does not necessarily imply that the share of variation explained by global factors has increased. If domestic variables are included as regressors, endogeneity may be an issue as well. As a check,

[54] Calculated as changes in banking and portfolio debt liabilities.

[55] Indonesia, Malaysia, the Philippines, South Korea, and Thailand.

estimated coefficients and standard deviations are reported for some regressions, such as the comprehensive panel regressions in **Table S1.3.**

Regressions are done for an initial 15-year period (Period 1: 1995q1–2010q4)[56] running through the GFC, then repeated for the following period (Period 2: 2011q1–2024q2), with the R^2 measure of fit, as well as p-values, in Period 1 compared with the same measure in Period 2 for each country (**Table S1.1**). The choice of end-2010 as the dividing point for the sample corresponds to the end of the GFC, and keeps the two test periods at approximately equal lengths. Also provided, in parentheses, are the probability values from an F-test of the null hypothesis that the independent variables explain none of the variation in capital flows. Given a standard 95% confidence interval, a p-value below 0.05 implies confirmation that global factors influence capital flows in the period in question.

Table S1.1 Adjusted R^2 Values with P-values of F-Statistics in Parentheses

	Bank and Portfolio Liabilities		Total Capital Flow	
	Period 1	Period 2	Period 1	Period 2
Indonesia	0.33 (0.00)	0.25 (0.00)	0.20 (0.01)	0.24 (0.00)
South Korea	0.52 (0.00)	0.03 (0.29)	0.35 (0.00)	0.14 (0.04)
Malaysia	0.41 (0.00)	0.27 (0.00)	0.38 (0.00)	−0.04 (0.63)
Philippines	0.27 (0.00)	0.02 (0.29)	0.14 (0.01)	0.03 (0.03)
Thailand	0.39 (0.00)	0.15 (0.04)	0.37 (0.00)	0.06 (0.20)

Source: MAS estimates

As a control, the exercise is repeated (**Table S1.2**) with the inclusion of a lagged term for the differential between foreign (US) and domestic policy interest rates:

$$CF_{i,t} = \alpha + \beta_{i,1}F_{1,t} + \cdots + \beta_{i,k}F_{k,t} + \beta_{k+1}UIP_{t-1} + \epsilon_t$$

As bond flows are likely to respond to (exchange-rate adjusted) interest differentials, leaving this term out risks omitted variable bias. However, inclusion of the rate differential may induce some endogeneity, as domestic rates may be set in part in response to capital flows. In the event, R^2 results do not change markedly, although they are somewhat higher for Malaysia and the Philippines in Period 1 and Indonesia in Period 2.

56 Starting dates vary somewhat depending on data availability.

Table S1.2 Adjusted R^2 Values with P-values, IR Differential Included as Regressor

	Bank and Portfolio Liabilities		Total Capital Flow	
	Period 1	Period 2	Period 1	Period 2
Indonesia	0.33 (0.00)	0.34 (0.00)	0.31 (0.00)	0.36 (0.00)
South Korea	0.56 (0.00)	0.06 (0.19)	0.37 (0.00)	0.12 (0.07)
Malaysia	0.59 (0.00)	0.28 (0.00)	0.40 (0.00)	−0.06 (0.73)
Philippines	0.45 (0.02)	−0.03 (0.61)	0.24 (0.08)	0.10 (0.39)
Thailand	0.30 (0.00)	0.13 (0.06)	0.25 (0.02)	0.06 (0.21)

Source: MAS estimates

In both sets of regressions, every country (**Tables S1.1 and S1.2**)—with the partial exception of Indonesia—shows a large drop from the earlier to the later period. Coefficients in Period 1 generally have the expected signs—capital flows are positively correlated with the business cycle and negatively correlated with interest rates. While t-statistics for individual coefficients are typically under 1.96, F-tests in Period 1 indicate, for each of the five countries evaluated, strong rejection of the null hypothesis that capital flows are independent of global factors. Results for Period 2 are less consistent across countries, but for each country, global factors explain a lower share of capital flow variation in Period 2 than in Period 1.

As a comprehensive test of the hypothesis that the relationship between global factors and capital flows has shifted over time, data from across the five countries are aggregated in fixed-effects panel regressions for each of the two periods. A Chow test for structural break strongly confirms a structural shift in the panel regression from Period 1 to Period 2. Estimated coefficients, reported in **Table S1.3**, generally correspond to theoretical priors in Period 1 (with the exception of the sign on G7Y), with t-statistics indicating significance at the 1 percent level. T-statistics weaken in Period 2, with some coefficients displaying the wrong signs.

Table S1.3 Results from Panel Regression of Portfolio and Bank Liabilities of Asian Economies

	Period 1	Period 2
Constant Term	1.007 (0.598)*	3.351 (0.905)***
ECON1	2.511 (0.923)***	−1.396 (0.453)***
FIN	1.678 (0.493)***	1.419 (0.552)**
YIELD	−5.447 (1.23)***	1.426 (1.31)
ECON2	−1.551 (0.788)*	1.922 (1.027)*
G7Y	2.892 (1.093)***	1.524 (1.096)
Adjusted R^2	0.237	0.110

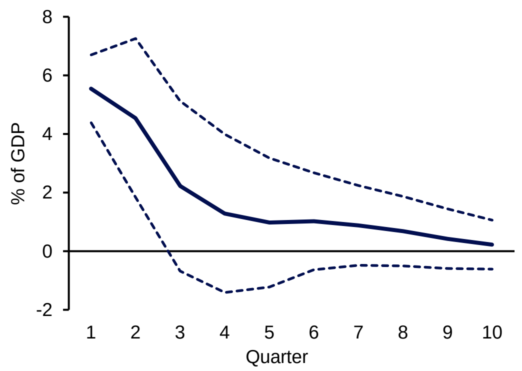
Source: MAS estimates

*, ** and *** indicate statistical significance at 10%, 5% and 1% level respectively.

Note: Chow Breakpoint Test confirms the presence of a breakpoint (F(6,555) statistic of 13.15 against the 1% critical value of 3.06) between periods 1 and 2.

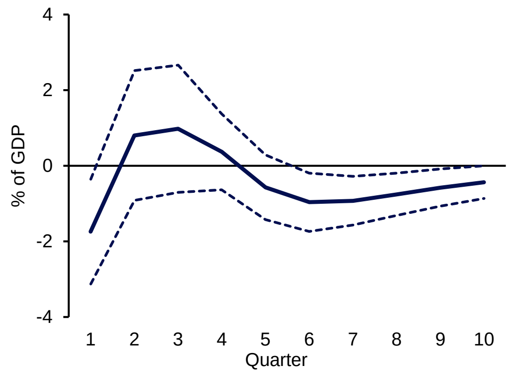
While not the main focus of the analysis, impulse response functions generated from vector autoregressions (VARs) of Korean capital flows on ECON1 illustrate the difference between Period 1 and Period 2 for a subset of the data (**Charts S1.7 and S1.8**). In the first period, the global economic cycle (ECON1) is shown to have a statistically significant and persistent impact on capital inflows, while in the second period the impact is negligible.

Chart S1.7 Impulse Response of Korean Capital Flows to ECON1 (Period 1)



Source: MAS estimates

Chart S1.8 Impulse Response of Korean Capital Flows to ECON1 (Period 2)



Source: MAS estimates

Taken together, these estimations indicate that the exposure of Asian capital flows to the GFCy has diminished. However, they do little to identify the underlying causes of this development, nor do they establish whether the reduced dependence of capital flows on the GFCy leaves AEE central banks with more policy space to respond to country-specific shocks.

Domestic Interest Rates and Activity

If Asian emerging economies have become more independent of the GFCy, then domestic business cycles and other domestic variables such as interest rates should be less correlated with GFCy markers. This proposition is tested for GDP growth, credit growth, and domestic interest rates across the sample group of five Asian markets. While the strategy is similar to that employed with capital flows, the presence or possible presence of unit roots in some of the dependent variables prompts a greater emphasis on the use of equations in first differences (notably for interest rates) in this battery of tests. When tests are run in first differences, a lagged "uncovered interest parity (UIP) divergence" term is added to the regression in order to capture the arbitrage effect of departures from UIP. This term is defined as: *US interest rate*$_{t-1}$ – *Domestic interest rate*$_{t-1}$,[57] and is effectively equivalent to the interest rate differential term employed in the previous section.

Similar to the capital flows tests, regressions for GDP and credit growth for country i have the form:

[57] In theory, the UIP divergence term should equal *(1+US interest rate*$_{t-1}$*) – (1+Domestic interest rate*$_{t-1}$*)*(Expected depreciation)-1*. If the expected depreciation rate is a constant k, then this is approximately equal to $(i_{US} - i_D - k)$, and a regression with one term for the interest rate differential and at least one constant would provide the same explanatory power (R^2) as a regression incorporating the full UIP divergence term.

$$G_{i,t} = \alpha + \beta_{i,1}F_{1,t} + \cdots + \beta_{i,k}F_{k,t} + \epsilon_t$$

These generally show little explanatory power in either period, with R^2 below or in the range of those related to capital flows (**Table S1.4**). However, in contrast to the results for capital flows, there is no tendency for R^2 to fall from Period 1 to Period 2. Unsurprisingly, the global variable most tightly correlated with GDP growth is ECON1, which closely tracks advanced economy GDP growth.

Table S1.4 Results of Regression of Credit Growth and GDP Growth on Global Conditions

	Adjusted R^2			
	Credit Growth		Q-O-Q GDP Growth	
	Period 1	Period 2	Period 1	Period 2
Indonesia	0.20	0.10	0.03	0.34
South Korea	0.04	0.27	0.17	0.10
Malaysia	0.01	0.16	0.29	0.50
Philippines	-	-	0.16	0.57
Thailand	0.01	0.30	0.15	0.31

Source: MAS estimates

Note: Estimates make use of data on Adjusted Credit to the Nonfinancial Sector from BIS, for which data on the Philippines is not available.

By contrast, regressions with domestic interest rates as the dependent variable indicate continued dependence on global financial conditions. R^2 for domestic interest rates generally rise from Period 1 to Period 2. Evaluated in levels and measured against global factors, the average adjusted R^2 for policy rates rises from .55 to .63, while the corresponding figure for 90-day rates increase from .41 to .57 and that for 10-year rates shifts up from .39 to .44.

Given some indication of unit roots in the dependent variables,[58] these regressions are also performed in first differences with a UIP divergence term included in **Table S1.5**, as described above.

The general specification for one country (dropping the country subscript) has the form:

$$\Delta R_t = \alpha + \sum_{i=1}^{n} \beta_i F_{i,t} + \sum_{j=1}^{m} \beta_j \Delta F_{i,t} + \beta_{n+m+1} UIP_{t-1} + \epsilon_t$$

where F_i are global factors and ΔF_j are either first differences of yield-based global factors or first differences of individual yields such as 10-year US Treasury yields. In these regressions in first differences, the average R^2 rises from 0.27 in Period 1 to 0.40 in Period 2 for the policy rate, and from 0.31 to 0.45 for the 10-year yield. The explanatory variables most closely linked with domestic interest rates are foreign long-term rates. Similar trends are obtained when

58 Durbin-Watson statistics range from 0.23 to 1.30, which fall outside the standard 1.50–2.50 reference range.

actual differenced US, German, and Japanese 10-year yields are substituted for the differenced global factors G7Y and YIELD in the baseline regressions.

Table S1.5 Results of Regression of Policy Rate and 10-Year Bond Rate Change on Global Conditions

	Adjusted R^2			
	Policy Rate (1st Difference)		10-Year Bond Rate (1st Difference)	
	Period 1	Period 2	Period 1	Period 2
Indonesia	0.27	0.19	0.58	0.42
South Korea	0.45	0.60	0.39	0.64
Malaysia	0.38	0.30	0.21	0.60
Philippines	0.11	0.59	−0.07	0.21
Thailand	0.13	0.29	0.44	0.38
Average	**0.27**	**0.40**	**0.31**	**0.45**

Source: MAS estimates

As a robustness check to establish that the exclusion of other domestic variables (e.g. credit growth) from the above domestic interest rate regressions did not bias the results, VARs were conducted with domestic interest rates (R), foreign interest rates G7Y (F), and domestic quarterly growth (ΔY) as endogenous variables. For the vector $\zeta_t = [\Delta R_t \ \ \Delta F_t \ \ \Delta Y_t]'$ the following relationship was estimated:

$$\zeta_t = \alpha + \beta_1 \zeta_{t-1} + \beta_2 \zeta_{t-2} + \epsilon_t$$

Decomposition of the sources of variance in domestic 10-year interest rates over the long run showed the following shares of the variance of domestic interest rates (out of 100) attributable to foreign interest rate shocks:

Table S1.6a Share of Variance in Domestic 10-Year Rate Attributable to Foreign Interest Rate Shocks from Variance Decomposition

	Total	Period 1	Period 2
Indonesia	11.54	25.81	22.83
South Korea	45.18	35.01	60.57
Malaysia	30.81	37.31	24.01
Philippines	8.54	11.33	15.44
Thailand	44.23	55.89	37.14
Average	**28.06**	**33.07**	**32.00**

Source: MAS estimates

Variance decompositions for the vector $\zeta_t = [\Delta R_t \ ECON1_t \ \Delta F_t \ \ \Delta Y_t]'$ (i.e. incorporating the global economic cycle) yielded similar variance shares for the combination of the two external variables:

Table S1.6b Share of Variance in Domestic 10-Year Rate Attributable to Total Foreign Shocks from Variance Decomposition

	Total	Period 1	Period 2
Indonesia	18.11	41.73	36.03
South Korea	45.30	31.17	64.54
Malaysia	29.82	41.72	34.81
Philippines	9.47	13.72	26.34
Thailand	43.93	58.11	36.66
Average	**29.33**	**37.29**	**39.67**

Source: MAS estimates

Calculations of the share of variance in domestic rates attributable to foreign shocks in the VAR are generally in line with those from the OLS regressions. For all VARs, the domestic business cycle accounted for a small share of the variance in domestic interest rate (less than 7 percent on average).

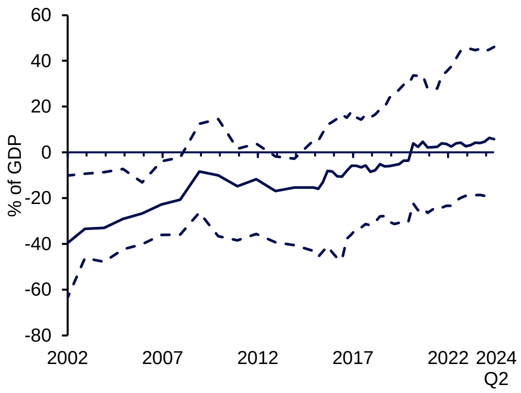
The continued strong dependence of AEE policy rates on global variables is surprising in light of the results from the previous section. Given the reduced dependence of AEE capital flows on the GFCy demonstrated there, it might be supposed that AEE policy rates have become more independent of both the GFCy and policy decisions of advanced economy central banks. However, R^2 for policy rates follow those for longer term rates in showing little sign of falling from Period 1 to Period 2. This suggests that the countercyclical policy space available to AEE central banks may not in fact have increased, despite the decline in the dependence of capital flows on the GFCy. This and alternative hypotheses are considered in the next section.

What has changed?

Capital flows to AEE have declined in volume and volatility over time, while becoming apparently less sensitive to global factors and the GFCy. One possible explanation is that AEE economies have become generally less reliant on foreign capital, preferring to self-finance through higher savings, and thus less likely to suffer large capital outflows during crisis periods. Some evidence does point in this direction; net international investment positions have improved somewhat over the past two decades while the average ratio of gross foreign reserves to GDP has risen from 16% in the late 1990s to 30% in 2020 (**Charts S1.9 and S1.10**). However, even now, far from registering large recurrent surpluses, current accounts as a share of GDP are generally close to balance.

Chart S1.9 Net international investment positions have gradually improved over the past two decades

Average net international investment position as a share of GDP

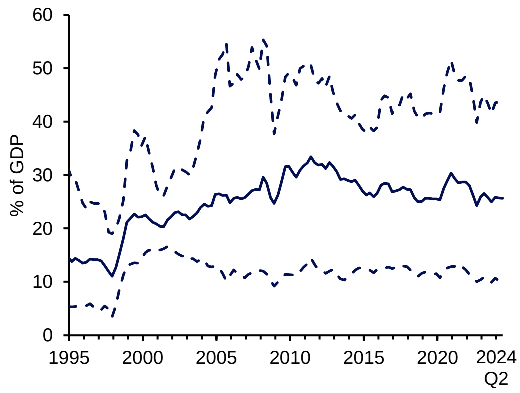


Source: IMF, Haver Analytics, MAS estimates

Note: Dashed lines indicate the min/max ranges.

Chart S1.10 Foreign reserve holdings have increased post-AFC

Average gross foreign reserves as a share of GDP



Source: IMF, Haver Analytics, MAS estimates

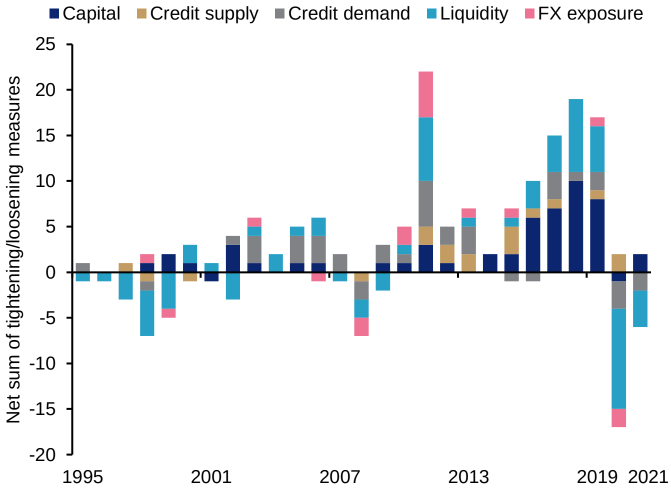
Note: Dashed lines indicate the min/max ranges.

Other recent policy developments may have also contributed to the reduced sensitivity of capital flows to global factors. With some qualifications, central banks in the region have moved in the direction of greater exchange rate flexibility and the use of inflation targeting frameworks. In principle, by shifting more of the risk of cross-border and cross-currency investment to foreign investors, exchange rate flexibility may work to dampen some inflows. At the same time, there has been a substantial increase in the use of MPMs to discourage the buildup of financial imbalances and asset bubbles (**Chart S1.11**). Even when not aimed explicitly at cross-border flows, such measures can help to discourage speculative inflows. Further factors that may play a role in moderating recorded capital inflows, and/or reducing the influence of global factors on net flows, could include strategic deployment of currency intervention, as has been advocated by some recent researchers.[59] Relatedly, the increased use by central banks of forward intervention and swaps may have discouraged or diverted some types of capital flows such as portfolio flows.

[59] See, for example, Adrian et al. (2021).

Chart S1.11 The use of MPMs has risen since the GFC

Annual macroprudential tightening (+1) and loosening (−1) actions in AEE



Source: IMF, MAS estimates

Note: The chart shows the sum of tightening (+1) and loosening (−1) actions for 17 different types of MPMs across the five economies (MY, ID, TH, PH, KR) on an annual basis. These have been grouped into five categories based on the main focus of the measure in question.

In this context, the observed continued strong dependence of interest rates and domestic activity on global factors appears incongruous. The contrast with the reduced dependence of Asian capital flows on external factors calls for further analysis. Among the hypotheses that could account for this paradox is the possibility that, during the period under study (Period 2), AEEs and advanced economies were responding to common shocks, such as the COVID-19 pandemic and subsequent inflationary supply squeeze, which are not directly reflected in the global factors used in this study. Other economic links, such as trade channels, may also have strengthened over the periods under study. Complementarily, it may be that the transmission channels for external shocks have shifted, from the banking and portfolio flows that were prominent in the Asian crisis and subsequent events, to derivatives, swaps and structured products that transfer risk but are not fully captured and more difficult to track through the balance of payment statistics. Some support for this view may be found in the fact that trade in financial derivatives for the five economies grew rapidly from 2010 to 2022, rising by 154% from USD92 billion to USD233 billion, ahead of the global increase of 89% over the same period.[60] In either case, or if there is some further unexplained factor, this may mean that despite the apparent respite from capital flows, central banks have not managed to gain greater autonomy from the global cycle.

An alternative possibility is that, even if policy space has increased in principle, central banks have become more aware of the power of the GFCy and less willing to move against it. Based on past experience, some AEE central banks may be more inclined to anticipate and minimise cross-border interest rate differentials, even if the consequences of not doing so are less dire than in the past. Regressions of changes in AEE domestic policy rates on changes in the Fed Funds rate since the Federal Reserve moved away from its zero interest rate policy

60 Based on the results of the BIS Triennial Central Bank Survey of foreign exchange and OTC derivatives markets in 2022. Foreign exchange derivative activity data are not available for the MYR, IDR, THB, and PHP. Total foreign exchange activity (spot and OTC derivatives) is used as a proxy for FX derivative activity.

(ZIRP) in 2016 (**Table S1.7**) suggest that this may have been the case since 2017. To accommodate recent and anticipated changes in the Fed funds rate, the estimated relationship is:

$$\Delta PR_t = a + \beta_1 \Delta FF_t + \beta_2 \Delta FF_{t-1} + \beta_3 \Delta FF_{t+1} + \beta_4 UIP_{t-1} + \epsilon_t$$

As indicated in **Table S1.7**, for each country in the sample, there is a substantial increase in the extent to which the policy rate moves with the Federal Funds rate in the 2018–24 period.

Table S1.7 Dependence of Change in Domestic Policy Rate on Change in Fed Funds Rate: Full Sample and post-ZIRP Period 2018–24

	Adjusted R^2	
	Full Sample	**2018Q1–2024Q1**
Indonesia	0.08	0.51
South Korea	0.31	0.87
Malaysia	0.15	0.57
Philippines	0.40	0.73
Thailand	0.20	0.65

Source: MAS estimates

Conclusion

While the increased independence of capital flows from the GFCy is a promising sign for the development of policy autonomy and countercyclical policy space, it is clear that global factors are still important determinants of domestic prices and activity in Asian emerging market economies. This does not entail that central banks are powerless, that the trilemma does not apply, or that flexible exchange rates are not able to cushion external shocks. Rather, it implies that global shocks, including shocks from G7 policy decisions, continue to have strong influence on domestic economic and financial conditions. In these circumstances, exchange rate flexibility remains an appropriate policy to respond to external shocks, supported in some cases by MPMs, foreign exchange intervention, or, in more extreme and temporary circumstances, capital flow measures.

Further research will be needed to expand upon the empirical results presented here. If derivatives or other markets now substitute to some extent for portfolio flows, this may be shown in the net exposures entailed by the derivatives positions of large investors. It should also be possible to test for the importance of common shocks in shared monetary policy decisions. Similarly, additional tests for the independence of EME central bank policy decisions may be constructed. Finally, new-generation Dynamic Stochastic General Equilibrium (DSGE) models incorporating divergences from UIP and appropriate market imperfections may address the question of transmission of shocks independent of capital flows.

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Special Feature 2

Expanding the Toolkit for Macroprudential Surveillance

MAS is introducing a new Financial Stress Index (FSI) for Singapore to complement the Financial Conditions (FCI) and Financial Vulnerability Indices (FVI), which have also been enhanced. Together, the three indicators are designed to improve monitoring of systemic financial stability risks, in combination with a broad range of other financial information employed in MAS' regular surveillance and policy deliberations. This Special Feature briefly describes the construction of the FSI, the revised methodologies for the FCI and FVI, and how they are used for conjunctural assessments.

Monitoring FCI, FSI and FVI through the financial cycle

The FCI, FSI, and FVI help to identify distinct aspects of financial stability risks across the financial cycle:

- The FCI gauges the ease of financing for the real economy. A higher FCI points to tighter financing conditions that weigh on credit intermediation and economic activity, while a lower FCI indicates easier financing conditions that facilitate borrowing and support economic growth.

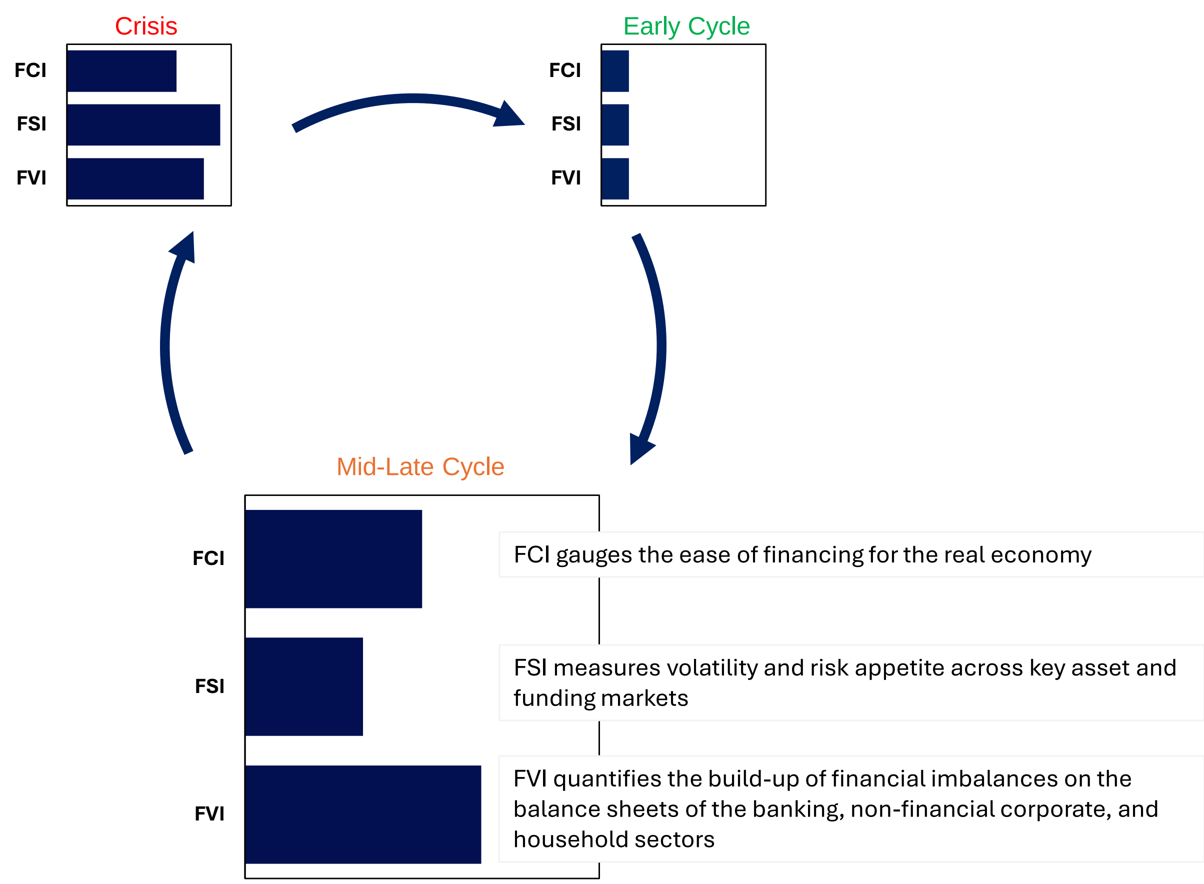
- The FSI measures volatility and risk appetite across key asset and funding markets. A higher FSI is associated with elevated and systemic financial market distress, while a lower FSI points to localised dislocations at lower levels of intensity.

- The FVI quantifies the build-up of financial imbalances on the balance sheets of the banking, non-financial corporate, and household sectors. A higher FVI suggests greater susceptibility to shocks.

While the FCI, FSI and FVI can be used independently to assess market and financial conditions, they convey complementary information for the surveillance of risks (**Figure S2.1**). For instance, while all three indicators tend to be at low levels during the early stages of a financial cycle, the FVI may begin to rise if accommodative financing conditions incentivise risk-seeking behaviour. This could lead to a build-up of financial vulnerabilities in financial institutions, households, and corporates. Even as leverage continues to rise, tighter monetary conditions may cause the FCI to increase. Subsequently, negative shocks to the financial system—reflected in spikes in the FSI—can produce destabilising dynamics such as asset fire sales and credit crunches, particularly where vulnerabilities have accumulated. If the financial system's intermediation capacity is compromised, these initial shocks could further tighten financial conditions (e.g. higher credit costs, shortfalls in credit supply), causing the FCI to move even higher.

Figure S2.1 The FCI, FVI and FSI each measure distinct aspects of the financial cycle, and together they convey complementary information about the evolution of the financial cycle

Evolution of FCI, FVI and FSI across a stylised financial cycle



Source: MAS

Financial Stress Index

The FSI gauges risk appetite and systemic stress in key asset and funding markets in Singapore. It is inspired by the European Central Bank (ECB)'s Composite Indicator of Systemic Stress (CISS) (Hollo et al., 2012), with some notable differences in data sources and the treatment of cross-correlation between indicators.[61] Its strength lies in synthesising information across different markets, summarising market stress levels and flagging potential contagion risks.

Methodology

By design, the FSI covers a broad range of indicators drawn from five key market segments: the financial intermediaries sector, money markets, equity markets, bond markets, and foreign exchange markets. Indicators are detrended, ranked against their own histories

[61] Other related approaches include the St. Louis Fed (Mendez-Carbajo, 2020) and Kansas City Fed's (Hakkio and Keeton, 2009) financial stress indices, which are available at lower frequencies (only weekly or monthly) and focus on the US financial system.

on a rolling basis, and assigned a score of between zero and one. This scoring system lends the FSI a natural interpretation: it is a measure of the level of stress in a market relative to its own history. A score of 1 means that distress in the market concerned is at the most severe level.

FSI's for each market are combined into an aggregate FSI through a novel approach that identifies both the average level of distress by sector, and the effect of contagion across sectors.[62] The 'contagion factor' measures the extent to which co-movement between sectors increases during periods of market stress or dysfunction.

Key Findings

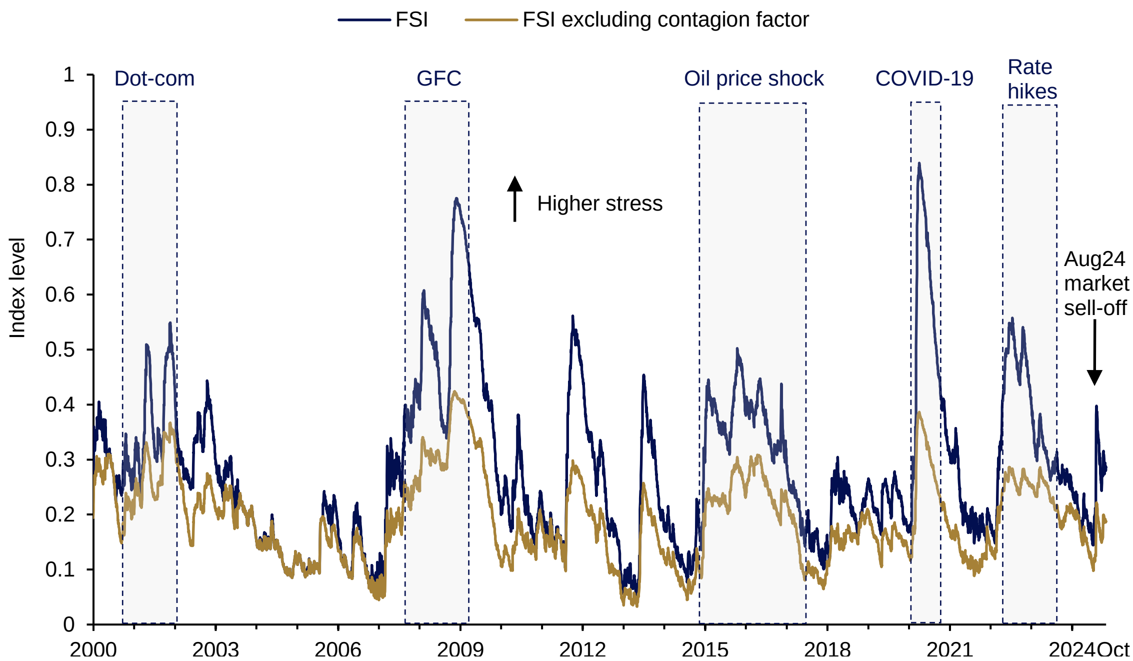
The FSI shows that Singapore's financial system has weathered a series of systemic stress events since 2000, all originating from external financial shocks (**Chart S2.1**). Of these, the COVID-19 shock stands out as the most severe, despite its relatively short duration. It is due to the contagion factor that the COVID-19 shock appears so strong—when measured without it, the COVID-19 shock was comparable in severity to the 2016 oil price shock or the 2022 inflation shock.

At the current conjuncture, domestic markets has continued to function well, as indicated by the FSI. During the global equity market sell-off in early August, risk aversion increased moderately in the domestic government bond and equity markets, and in the financial intermediaries segment. Other key markets, notably the foreign exchange and money markets, functioned normally over this period. The FSI has since eased considerably with the recent decline in global market implied volatility. MAS will continue to monitor the FSI, and its components, closely given the increased sensitivity of markets to economic data surprises, diverging monetary policy stances, and geopolitical developments.

[62] The most important common factor driving the range of indicators in each market segment, i.e., the first principal component, is extracted. These common factors are readily interpreted as indices for the level of dysfunction in each market segment. The aggregate FSI is then calculated as the root mean square of common factors from all market segments. The loadings on each common factor are determined by a formula related to the pairwise correlations between them. All things equal, the higher the correlation, the higher the level of the FSI.

Chart S2.1 COVID-19 shock was the most severe of the significant systemic stress events since 2000

Financial Stress Index (FSI)



Source: Bloomberg, MAS estimates

Financial Conditions Index

The Singapore FCI was introduced in the 2019 FSR (MAS, 2019) as a summary measure of the cost and quantity of financing in Singapore. Many other central banks employ similar tools.[63] With the introduction of the FSI, the FCI construction methodology has been updated to allow for a more intuitive reading of the impact of the FCI on Singapore's economic activity. For example, market functioning indicators and the Singapore dollar exchange rate are now excluded.[64]

Methodology

There are two key changes to the FCI methodology. First, the FCI has been updated with a revised list of indicators that are strongly connected to financing conditions in Singapore. These fall under four categories: global market indicators, interest rates, asset prices, and money and credit growth. New indicators include valuation of CRE (which serves as collateral

63 Many jurisdictions and institutions have developed their own FCI, with the Bank of Canada (Gauthier et al., 2004) as one of the early adopters. Others include the BoE (BoE, 2019), the IMF (IMF, 2017) and the Organisation for Economic Co-operation and Development (OECD) (Davis et al., 2016). These FCIs differ in terms of variables considered and the methodology used. Principal components (e.g. IMF) and impulse response functions or regressions (e.g. BoE, OECD) were the common methodologies for constructing FCIs.

64 The Singapore dollar exchange rate is not included in the FCI as the effects of Singapore's exchange rate policy are transmitted through the tradeable sectors, notably the terms-of-trade channel. It is also important to note the partial equilibrium nature of the impact of FCI in economic activity, which does not incorporate the aggregate demand impact of exchange rate policy.

for business borrowing) and corporate and household loan growth. Some indicators related to market functioning have been removed and incorporated in the FSI.

Second, to better explain how financial conditions impact Singapore's economy, the revised index adopts a Principal Component Regression (PCR) approach (Stock and Watson, 2002). This essentially entails performing linear regression with the principal components to summarise their impact on the output gap.[65] The FCI thus indicates the extent to which market conditions restrain or support economic activity.

Key Findings

The updated FCI shows that tightening financial conditions had significant impact on growth deviation during past crises—dot-com bubble crash, global financial crisis, and COVID-19 pandemic (**Chart S2.2**). More modest tightening was also observed during the 2014–2016 oil price shock, and more recently with the rise in global interest rates during 2022–2023. Across the earlier events, asset prices had been the main drivers of the FCI, which suggested the importance of risk sentiment and collateral value in influencing financing conditions in Singapore.

In recent years, money and credit indicators have featured more prominently as key drivers of the FCI. The tightening of financial conditions during the early phase of the COVID-19 pandemic was largely due to a retrenchment in credit, which reflected lender concerns over corporate viability. After the implementation of pandemic-support credit measures, conditions eased discernibly. The subsequent global monetary policy tightening in 2022–2023 led to a contraction in money and credit indicators as households and corporates deleveraged amid rising interest rates and economic uncertainty. Banks observed that corporates preferred to draw down cash buffers first, instead of borrowing at higher rates.[66]

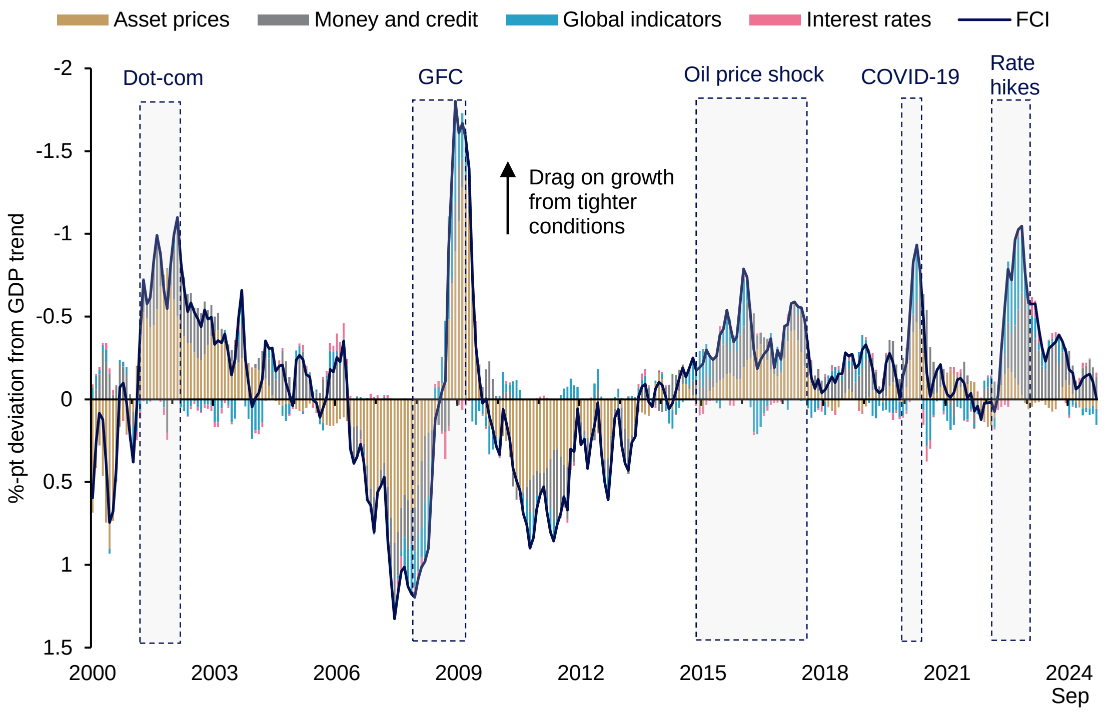
Since H2 2023, financing conditions in Singapore have become less restrictive as global interest rates plateaued and then started to ease. Notably, money and credit conditions have also improved. Recent moderate increases in financial stress did not translate into any discernible retrenchment in the availability of credit.

[65] Stock and Watson (2002) introduced the concept of PCR. Our construction entails summarising the indicators into a few common factors and estimating their impact on economic activity via a linear regression against the output gap. Combining the two common FCI methodologies (i.e. Principal Component Analysis (PCA) and regression), the PCR approach benefits from both the dimension and noise reduction abilities of PCA as well as interpretability and linkages to economic activity through the regression. Deutsche Bank's FCI used this approach.

[66] MAS conducted a survey of banks in September 2023 to seek industry views on realised credit conditions over the past six months and expectations for Q4 2023 and Q1 2024. For more details, see Chapter 4 "Singapore Financial Sector" of the MAS' Financial Stability Review 2023.

Chart S2.2 Asset prices tended to drive variations in the FCI, but money and credit drove FCI trends in the most recent rate hike cycle

Financial Conditions Index (FCI) by category



Source: Bloomberg, DOS, URA, US FRED, MAS estimates

Financial Vulnerabilities Index

MAS introduced the FVI in 2019 to measure the extent of financial vulnerabilities in Singapore. Adapted from the IMF Global Financial Stability Report, the FVI quantifies the financial imbalances of the banking, non-financial corporate, and household sectors using risk metrics such as leverage, funding, and FX/liquidity mismatches. A higher FVI suggests greater susceptibility of the sector to shocks, while a lower FVI implies higher resilience.

Methodology

MAS has revised the household sector FVI risk categories to ensure continued relevance:

- Leverage risk: Indicators on debt servicing have been included to assess mortgage repayment risk.
- Liquidity risk: This has been enhanced to reflect the vulnerability of households to short-term debt obligations. such as credit card borrowings. It now incorporates the availability of liquidity buffers to mitigate such risks.
- Asset price volatility risk: This is a new category to assess household exposures to market risks in the property sector.

Another key change is the presentation of the FVI in terms of levels, instead of year-on-year changes, for a more direct assessment of vulnerabilities in the system.

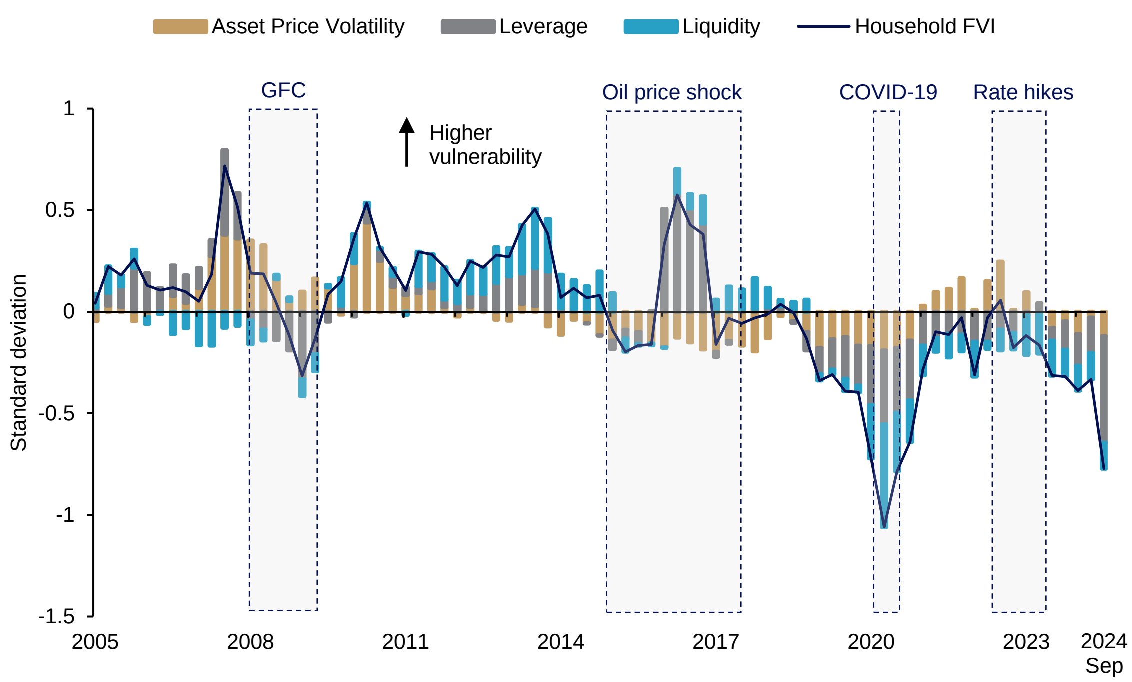
Key Findings

The revised household FVI better captures risks to households and breaks down the risk drivers more distinctly across time (**Chart S2.3**). It shows a build-up of vulnerabilities prior to the GFC, as expressed by the positive values (i.e. standard deviations from historical average), stemming from increases in asset price volatility and heightened leverage. Similarly, asset price volatility and further leverage, reflecting higher property prices amid the low interest rate environment, induced a build-up of household vulnerabilities after the GFC.

Following the COVID-19 pandemic, household vulnerabilities rose from historically low levels, as property prices advanced again while liquidity buffers moderated from higher levels accumulated during COVID. To mitigate the build-up of these risks, the interagency taskforce tightened property measures such as tax and credit-related tools (e.g. Additional Buyer's Stamp Duty (ABSD), Total Debt Servicing Ratio (TDSR)) from 2021 to 2023. More recently, the overall level of vulnerabilities has remained relatively contained, and has even showed signs of improvement, supported by the recent macroprudential measures.

Chart S2.3 The revised household FVI better captures salient risks to households and breaks down the risk drivers more distinctly

Household Financial Vulnerabilities Index (FVI), in standard deviations from historical average



Source: DOS, URA, MAS estimates

Sum-up

The three indicators reviewed in this Special Feature will be used in conjunction with other existing tools and financial information to assess financial risks and vulnerabilities in Singapore. The revised FCI indicates the potential impact of the current cost and quantity of financing in Singapore on economic activity. The FVIs summarise the level of vulnerabilities in the corporate, household and banking sectors, supplemented by an existing broad range of indicators for surveillance of sector-specific risks. The new FSI measures risk appetite in key asset and funding markets in Singapore, including potential contagion risks. Going forward, MAS will continue to regularly review and explore refinements to the indicators to ensure that they remain relevant in the evolving macrofinancial environment.

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